



RTL Design Sherpa

RAPIDS Micro-Architecture Specification 0.1

September 30, 2026

Table of Contents

1 Rapids Mas Index.....	32
2 Document Information.....	32
2.1 Revision History.....	32
2.2 References.....	32
2.2.1 Related Documents.....	32
2.3 Terminology.....	33
2.4 Document Conventions.....	34
3 Architecture Overview.....	34
3.1 Overview.....	34
3.1.1 Key Features.....	35
3.1.2 What Is Shared with RAPIDS Beats.....	35
3.2 Block Diagram.....	35
3.2.1 Figure 1.1.1: RAPIDS Architecture Block Diagram.....	35
3.3 Data Flow Overview.....	37
3.3.1 Sink Path (Network to Memory).....	37
3.3.2 Figure 1.1.2: Sink Path Data Flow.....	37
3.3.3 Source Path (Memory to Network).....	38
3.3.4 Figure 1.1.3: Source Path Data Flow.....	38
3.4 Byte-Granular Design Points.....	38
3.5 Chapter Map.....	39
4 Top-Level Port List.....	40
4.1 Parameters.....	40

4.2 Port List (324 ports).....	42
4.2.1 Clock and Reset (shared - single clock/reset for the whole wrapper)....	42
4.3 SOURCE HALF (u_src) - shared-infrastructure ports (src_ prefixed).....	42
4.3.1 APB Programming Interface.....	42
4.3.2 Per-channel configuration.....	42
4.3.3 Scheduler Configuration (global).....	43
4.3.4 Descriptor Engine Configuration (global).....	43
4.3.5 Control Engine Configuration (Phase 2, global).....	44
4.3.6 Descriptor AXI Monitor Configuration.....	44
4.3.7 AXIS data-path monitor-lite configuration (rapids TASK-015).....	45
4.3.8 Status.....	46
4.3.9 Descriptor AXI Monitor Status.....	46
4.3.10 AXIS data-path monitor-lite status (rapids TASK-015).....	46
4.3.11 Descriptor AXI Monitor perf window (feeds SRC_MON.DAXMON_PERF_*).....	47
4.3.12 Descriptor Fetch AXI Master.....	48
4.3.13 Control Read AXI Master (32-bit) [Phase 2].....	49
4.3.14 Control Write AXI Master (32-bit) [Phase 2].....	50
4.4 SOURCE HALF (u_src) - direction-unique ports (no prefix).....	51
4.4.1 AXI Transfer Configuration (source-only).....	51
4.4.2 Source Path - AXIS Master Interface (SRAM -> Network); tid = channel id	51
4.4.3 AXI4 Master - Data Read (Memory -> Source SRAM).....	52
4.5 SINK HALF (u_snk) - shared-infrastructure ports (snk_ prefixed).....	52

4.5.1 APB Programming Interface.....	52
4.5.2 Per-channel configuration.....	53
4.5.3 Scheduler Configuration (global).....	53
4.5.4 Descriptor Engine Configuration (global).....	53
4.5.5 Control Engine Configuration (Phase 2, global).....	54
4.5.6 Descriptor AXI Monitor Configuration.....	54
4.5.7 AXIS data-path monitor-lite configuration (rapids TASK-015).....	55
4.5.8 Status.....	56
4.5.9 Descriptor AXI Monitor Status.....	56
4.5.10 AXIS data-path monitor-lite status (rapids TASK-015).....	57
4.5.11 Descriptor AXI Monitor perf window (feeds SNK_MON.DAXMON_PERF_*).....	57
4.5.12 Descriptor Fetch AXI Master.....	58
4.5.13 Control Read AXI Master (32-bit) [Phase 2].....	59
4.5.14 Control Write AXI Master (32-bit) [Phase 2].....	60
4.6 Monitor Bus (SINGLE aggregated stream for the whole core).....	61
4.6.1 Monitor Bus.....	61
4.7 SINK HALF (u_snk) - direction-unique ports (no prefix).....	61
4.7.1 AXI Transfer Configuration (sink-only).....	61
4.7.2 Sink Path - AXIS Slave Interface (Network -> SRAM); tid = channel id...62	
4.7.3 AXI4 Master - Data Write (Sink SRAM -> Memory).....	62
4.8 Debug Interface (per half, src_dbg_ / snk_dbg_ prefixed).....	63
4.8.1 Source half debug.....	63
4.8.2 Sink half debug.....	64

4.8.3 Active-channel sideband for per-channel bus instrumentation.....	64
5 Clocks and Reset.....	65
5.1 Clock Domain.....	65
5.1.1 Clock Specifications.....	65
5.1.2 Clock Distribution.....	65
5.2 Reset Specification.....	66
5.2.1 Reset Signal.....	66
5.2.2 Reset Timing Diagram.....	66
5.2.3 Figure 1.3.1: Reset Timing.....	66
5.3 Reset Behavior.....	67
5.3.1 On Reset Assert.....	67
5.3.2 Reset Sequence.....	67
5.4 Configuration During Reset.....	68
5.5 Reset Macros.....	68
5.6 Power-On Reset Considerations.....	68
6 Scheduler.....	69
6.1 Overview.....	69
6.1.1 Key Features.....	69
6.1.2 Block Diagram.....	70
6.1.3 Figure 2.1.1: Scheduler Byte Math.....	70
6.2 Parameters.....	71
6.3 Port List.....	72
6.3.1 Clock, Reset and Configuration.....	72
6.3.2 Status.....	72

6.3.3 Descriptor Engine Interface.....	73
6.3.4 Data Read Interface (to AXI Read Engine).....	73
6.3.5 Data Write Interface (to AXI Write Engine).....	73
6.3.6 Completion Interface.....	74
6.3.7 Control Engine Interface.....	75
6.3.8 Monitor Bus Interface.....	75
6.4 Descriptor Format.....	76
6.5 Byte-to-Beat Math.....	76
6.6 Address Advance.....	78
6.7 Packet Records.....	78
6.8 Finite State Machine.....	78
6.9 Operation.....	80
6.9.1 Data Request Outputs.....	80
6.9.2 Write Completion.....	80
6.9.3 Control Descriptors.....	80
6.10 Error Handling.....	80
6.11 Recoverable Write-Progress Timeout.....	81
6.12 MonBus Events.....	81
6.13 Waveforms.....	82
7 Descriptor Engine Specification.....	82
7.1 Overview.....	82
7.1.1 Key Features.....	82
7.1.2 Block Diagram.....	83
7.1.3 Figure 2.2.1: Descriptor Engine Block Diagram.....	83

7.2 Parameters.....	83
7.3 Port List.....	84
7.3.1 Clock and Reset.....	84
7.3.2 APB Programming Interface.....	84
7.3.3 Configuration Interface.....	84
7.3.4 AXI4 Read Master Interface.....	85
7.3.5 Scheduler Interface.....	86
7.4 FSM States.....	86
7.4.1 Figure 2.2.2: Descriptor Engine FSM.....	86
7.5 Operation.....	88
7.5.1 Descriptor Chain Processing.....	88
7.5.2 Figure 2.2.3: Descriptor Chain Timing.....	88
7.6 Address Range Checking.....	88
7.7 Prefetch and Autonomous Chaining.....	89
7.7.1 Chaining Decision.....	89
7.7.2 Deferred (Throttle-Blocked) Chain Fetch.....	89
8 AXI Read Engine.....	90
8.1 Overview.....	90
8.1.1 Key Features.....	90
8.1.2 Block Diagram.....	91
8.1.3 Figure 2.3.1: AXI Read Engine Block Diagram.....	91
8.2 Parameters.....	91
8.2.1 Derived Constants.....	92
8.3 Port List.....	93

8.3.1 Clock, Reset and Configuration.....	93
8.3.2 Scheduler Interface.....	93
8.3.3 SRAM Allocation and Fill Interface.....	94
8.3.4 AXI4 Read Master Interface.....	94
8.3.5 Debug.....	95
8.4 Operation.....	95
8.4.1 Burst Length Cap (rapids BUG-009).....	95
8.4.2 Byte-Granular Burst Sizing.....	95
8.4.3 Streaming Pipeline.....	96
8.4.4 Figure 2.3.2: Burst Segmentation with a Mid-Beat Start.....	96
8.4.5 Done Strobe.....	97
8.5 Error Handling.....	97
8.6 Differences from RAPIDS Beats.....	97
9 AXI Write Engine.....	98
9.1 Overview.....	98
9.1.1 Key Features.....	98
9.1.2 Block Diagram.....	99
9.1.3 Figure 2.4.1: AXI Write Engine Block Diagram.....	99
9.2 Parameters.....	99
9.2.1 Derived Constants.....	100
9.3 Port List.....	100
9.3.1 Clock, Reset and Configuration.....	100
9.3.2 Scheduler Interface.....	101
9.3.3 SRAM Reservation and Drain Interface.....	102

9.3.4 AXI4 Write Master Interface.....	102
9.3.5 Debug and Status.....	103
9.4 Operation.....	104
9.4.1 Burst Length Cap (rapids BUG-009).....	104
9.4.2 Byte-Granular Burst Sizing.....	104
9.4.3 Write Strobes.....	104
9.4.4 Streaming Pipeline.....	105
9.4.5 Two Completion Strobes.....	105
9.5 Error Handling.....	105
9.6 Differences from RAPIDS Beats.....	106
10 Beats Alloc Control Specification.....	106
10.1 Overview.....	107
10.1.1 Key Features.....	107
10.1.2 Block Diagram.....	107
10.1.3 Figure 2.5.1: Beats Alloc Control Block Diagram.....	107
10.2 Concept: Virtual FIFO.....	107
10.2.1 Use Case: Pre-Allocation for AXI Reads.....	108
10.3 Parameters.....	108
10.4 Port List.....	108
10.4.1 Clock and Reset.....	108
10.4.2 Write Interface (Allocation Requests).....	108
10.4.3 Read Interface (Data Written).....	109
10.4.4 Status Outputs.....	109
10.5 Operation.....	110

10.5.1 Pointer Arithmetic.....	110
10.5.2 Timing Diagram.....	110
10.5.3 Figure 2.5.2: Allocation and Release Timing.....	110
10.6 Integration Example.....	110
11 Beats Drain Control Specification.....	111
11.1 Overview.....	112
11.1.1 Key Features.....	112
11.1.2 Block Diagram.....	112
11.1.3 Figure 2.6.1: Beats Drain Control Block Diagram.....	112
11.2 Concept: Data Availability Tracking.....	112
11.2.1 Complementary Roles.....	112
11.3 Parameters.....	113
11.4 Port List.....	113
11.4.1 Clock and Reset.....	113
11.4.2 Write Interface (Data Written to SRAM).....	113
11.4.3 Read Interface (Drain Requests).....	114
11.4.4 Status Outputs.....	114
11.5 Operation.....	114
11.5.1 Pointer Arithmetic.....	114
11.5.2 Timing Diagram.....	115
11.5.3 Figure 2.6.2: Data Arrival and Drain Timing.....	115
11.6 Integration Example.....	115
11.7 Relationship: alloc_ctrl vs drain_ctrl.....	116
12 Latency Bridge (beats) Specification.....	116

12.1 Overview.....	117
12.1.1 Key Features.....	117
12.1.2 Block Diagram.....	117
12.1.3 Figure 2.7.1: Latency Bridge Block Diagram.....	117
12.2 Concept: bridging a registered read.....	118
12.3 Parameters.....	118
12.4 Port List.....	118
12.4.1 Clock and Reset.....	118
12.4.2 Upstream Interface (from the registered FIFO).....	119
12.4.3 Downstream Interface (to the consumer).....	119
12.4.4 Status and Debug.....	119
12.5 Operation.....	120
12.5.1 Figure 2.7.2: Latency Bridge Timing (consumer always ready).....	120
12.6 Integration Context.....	121
12.7 Design Considerations.....	121
13 Control-Read Engine Specification.....	122
13.1 Overview.....	122
13.1.1 Key Features.....	122
13.1.2 Block Diagram.....	123
13.1.3 Figure 2.8.1: Control-Read Engine Block Diagram.....	123
13.2 Parameters.....	123
13.3 Port List.....	124
13.3.1 Clock and Reset.....	124
13.3.2 Scheduler Interface.....	124

13.3.3 Configuration Interface.....	125
13.3.4 AXI4 Read Master Interface.....	125
13.3.5 MonBus Interface.....	126
13.4 FSM States.....	127
13.4.1 Figure 2.8.2: Control-Read Engine FSM.....	127
13.5 Operation.....	128
13.5.1 Masked Compare.....	128
13.5.2 Retry Budget and Pacing.....	128
13.5.3 Null Address Fast-Path.....	129
13.5.4 AXI Read Transaction.....	129
13.5.5 Channel Reset.....	129
13.6 MonBus Reporting.....	129
13.7 Descriptor Field Mapping.....	130
14 Control-Write Engine Specification.....	131
14.1 Overview.....	131
14.1.1 Key Features.....	131
14.1.2 Block Diagram.....	131
14.1.3 Figure 2.9.1: Control-Write Engine Block Diagram.....	131
14.2 Parameters.....	132
14.3 Port List.....	132
14.3.1 Clock and Reset.....	132
14.3.2 Configuration Interface.....	133
14.3.3 Scheduler Interface.....	133
14.3.4 AXI4 Write Master Interface.....	133

14.3.5 MonBus Interface.....	134
14.4 FSM States.....	135
14.4.1 Figure 2.9.2: Control-Write Engine FSM.....	135
14.5 Operation.....	137
14.5.1 Doorbell Write Sequence.....	137
14.5.2 Figure 2.9.3: Control-Write Doorbell Timing.....	137
14.5.3 Request Handshake and Skid Buffer.....	138
14.5.4 Address Validation.....	138
14.5.5 Error Reporting.....	138
14.5.6 Channel Reset.....	138
14.6 Contrast with the Control-Read Engine.....	139
15 Scheduler Group.....	140
15.1 Overview.....	140
15.1.1 Key Features.....	140
15.1.2 Block Diagram.....	140
15.1.3 Figure 3.1.1: Scheduler Group Block Diagram.....	140
15.2 Parameters.....	141
15.3 Port List.....	142
15.3.1 Clock, Reset and Kick-Off.....	142
15.3.2 Configuration Interface.....	143
15.3.3 Status.....	144
15.3.4 Descriptor AXI Master Interface.....	144
15.3.5 Control Read and Control Write AXI Interfaces.....	145
15.3.6 Scheduler Data Interface.....	145

15.3.7 Packet Record Interface.....	146
15.3.8 MonBus Interface.....	147
15.4 Internal Architecture.....	147
15.4.1 Descriptor Path.....	147
15.4.2 Control Engines.....	147
15.4.3 MonBus Aggregation.....	147
15.5 Differences from RAPIDS Beats.....	148
16 Scheduler Group Array.....	148
16.1 Overview.....	148
16.1.1 Key Features.....	149
16.1.2 Block Diagram.....	149
16.1.3 Figure 3.2.1: Scheduler Group Array Block Diagram.....	149
16.2 Parameters.....	150
16.3 Port List.....	151
16.3.1 Clock, Reset and Kick-Off.....	151
16.3.2 Global Scheduler and Descriptor Engine Configuration.....	152
16.3.3 Descriptor Monitor Configuration and Status.....	153
16.3.4 Descriptor-AXI Perf Window.....	154
16.3.5 Status.....	155
16.3.6 Shared AXI Masters.....	155
16.3.7 Per-Channel Scheduler Data Interface.....	155
16.3.8 Per-Channel Packet Record Interface.....	156
16.3.9 MonBus Interface.....	157
16.4 Shared Master Arbitration.....	157

16.4.1 Figure 3.2.2: Shared Master Arbitration.....	157
16.5 MonBus Aggregation.....	158
16.6 Differences from RAPIDS Beats.....	158
17 Sink Data Path.....	159
17.1 Overview.....	159
17.1.1 Key Features.....	159
17.1.2 Figure 3.3.1: Sink Data Path Block Diagram.....	159
17.2 Parameters.....	160
17.3 Interfaces.....	160
17.3.1 Fill Interface.....	160
17.3.2 Configuration.....	161
17.3.3 Scheduler Interface.....	161
17.3.4 AXI4 Write Master.....	162
17.4 Byte Enable Path.....	162
17.5 Differences from RAPIDS Beats.....	162
18 Sink Data Path with AXIS (Ingress).....	163
18.1 Overview.....	163
18.1.1 Key Features.....	164
18.1.2 Figure 3.4.1: Ingress Shifter.....	164
18.2 Parameters.....	164
18.3 Interfaces.....	165
18.3.1 AXI-Stream Slave.....	165
18.3.2 Scheduler Interface.....	166
18.3.3 Other Ports.....	166

18.4 Ingress Operation.....	166
18.4.1 Packet Records.....	166
18.4.2 Placement.....	166
18.4.3 Flush and Completion.....	167
18.4.4 Length Check.....	167
18.4.5 Allocation.....	167
18.5 Differences from RAPIDS Beats.....	167
19 Sink SRAM Controller Specification.....	168
19.1 Overview.....	168
19.1.1 Key Features.....	169
19.1.2 Block Diagram.....	169
19.1.3 Figure 3.5.1: Sink SRAM Controller Block Diagram.....	169
19.2 Parameters.....	170
19.3 Port List.....	170
19.3.1 Clock and Reset.....	170
19.3.2 Fill Interface (ID-Selected).....	171
19.3.3 Arbitrated Drain Interface (to AXI Write Engine).....	171
19.3.4 Debug Interface.....	172
19.4 Drain Selection.....	172
19.4.1 Figure 3.5.2: Drain Selection by the Consumer.....	172
19.5 Integration with AXI Write Engine.....	173
20 Source Data Path Specification.....	173
20.1 Overview.....	174
20.1.1 Key Features.....	174

20.1.2 Block Diagram.....	174
20.1.3 Figure 3.6.1: Source Data Path Block Diagram.....	174
20.2 Parameters.....	175
20.3 Data Flow.....	175
20.3.1 Figure 3.6.2: Source Path Data Flow Sequence.....	175
20.4 Port List.....	176
20.4.1 Clock and Reset.....	176
20.4.2 Configuration.....	176
20.4.3 Scheduler Interface.....	176
20.4.4 AXI Read Master Interface.....	177
20.4.5 Drain Interface (Output to Network).....	177
20.5 Timing Diagram.....	178
20.5.1 Figure 3.6.3: Source Path Transfer Timing.....	178
20.6 Integration Example.....	179
21 Source Data Path with AXIS (Egress).....	180
21.1 Overview.....	180
21.1.1 Key Features.....	180
21.1.2 Figure 3.7.1: Egress Shifter.....	180
21.2 Parameters.....	181
21.3 Interfaces.....	181
21.3.1 Configuration.....	181
21.3.2 Scheduler Interface.....	182
21.3.3 AXI-Stream Master.....	182
21.3.4 Other Ports.....	183

21.4 Egress Operation.....	183
21.4.1 Reservation and Drain.....	183
21.4.2 Pop.....	183
21.4.3 Shift and Hold.....	183
21.4.4 Strobe and Last.....	184
21.4.5 Waveform 3.7: Egress Re-pack.....	184
21.5 Differences from RAPIDS Beats.....	185
22 Source SRAM Controller Specification.....	185
22.1 Overview.....	186
22.1.1 Key Features.....	186
22.1.2 Block Diagram.....	186
22.1.3 Figure 3.8.1: Source SRAM Controller Block Diagram.....	186
22.2 Parameters.....	187
22.3 Port List.....	188
22.3.1 Clock and Reset.....	188
22.3.2 Fill Interface (ID-Selected, from AXI Read Engine).....	188
22.3.3 Arbitrated Drain Interface (to Network).....	188
22.3.4 Debug Interface.....	189
22.4 Drain Selection.....	189
22.4.1 Figure 3.8.2: Drain Selection by the Consumer.....	189
22.5 Comparison: Sink vs Source SRAM Controller.....	190
22.6 Integration Example.....	190
23 RAPIDS Core, Sink Half and Source Half.....	191
23.1 Overview.....	191

23.1.1 Figure 3.9.1: Core Block Diagram.....	191
23.2 Byte-Granular Content.....	192
23.2.1 Packet Record Nets.....	192
23.2.2 Port Renames.....	193
23.2.3 Data Ports.....	193
23.3 Parameters.....	194
23.4 Differences from RAPIDS Beats.....	195
24 RAPIDS Registers Specification.....	195
24.1 Overview.....	195
24.2 Base Register Map (0x100-0x3FF).....	196
24.2.1 Key Register Fields.....	198
24.3 Monitor Register Map (rapids_mon_regs @ 0x1000).....	198
25 RAPIDS Config Block Specification.....	200
25.1 Overview.....	200
25.2 Mapping Groups.....	200
25.2.1 Base Configuration.....	200
25.2.2 Monitor Configuration (hwif_out.MON.*).	201
25.2.3 Performance and Observation.....	202
26 RAPIDS Top.....	202
26.1 Overview.....	202
26.1.1 Figure 3.12.1: Top Hierarchy.....	202
26.2 APB Address Map.....	203
26.3 Parameters.....	204
26.4 Differences from RAPIDS Beats.....	205

27 AXI4 Interface Specification.....	205
27.1 Overview.....	205
27.2 Byte Address to Beat Mapping.....	206
27.3 Burst Rules.....	206
27.4 Write Strobes.....	207
27.5 Limitations.....	207
28 AXI-Stream Interface Specification.....	207
28.1 Overview.....	208
28.2 Packet Contract.....	208
28.3 Source Master Signals.....	208
28.4 Sink Slave Signals.....	209
28.5 Length Check.....	210
29 MonBus Interface Specification.....	210
29.1 Overview.....	210
29.2 MonBus Signal Interface.....	210
29.2.1 Signal Table.....	210
29.2.2 Handshaking Protocol.....	211
29.2.3 Timing Diagram.....	211
29.2.4 Figure 4.3.1: MonBus Packet Transfer Timing.....	211
29.3 MonBus Packet Format.....	211
29.3.1 64-bit Packet Structure.....	211
29.3.2 Figure 4.3.2: MonBus Packet Format.....	212
29.4 Packet Types.....	212
29.4.1 Type Code Definitions.....	212

29.4.2 Payload Formats by Type.....	213
29.5 MonBus Source Aggregation.....	214
29.5.1 RAPIDS Core MonBus Sources.....	214
29.5.2 Aggregation Hierarchy.....	214
29.5.3 Figure 4.3.3: MonBus Aggregation Tree.....	214
29.6 Integration Guidelines.....	215
29.6.1 Consumer Requirements.....	215
29.6.2 Recommended Consumer Patterns.....	215
29.7 Timing Requirements.....	216
30 Packet Record Interface Specification.....	216
30.1 Overview.....	216
30.1.1 Figure 4.4.1: Scheduler Byte Math.....	217
30.2 Signals.....	217
30.3 Protocol.....	218
30.4 Unused Direction.....	219
30.5 Consumer Behavior.....	219
31 RAPIDS Beats Architecture MAS.....	219
31.1 Document Organization.....	220
31.1.1 Front Matter.....	220
31.1.2 Chapter 1: Overview.....	220
31.1.3 Chapter 2: FUB (Functional Unit Blocks).....	220
31.1.4 Chapter 3: Macro (Integration Blocks).....	220
31.1.5 Chapter 4: Interfaces.....	220
31.2 Quick Reference.....	221

31.2.1 FUB Modules (fub_beats/)	221
31.2.2 Macro Modules (macro_beats/)	222
31.3 Architecture Comparison: RAPIDS vs STREAM	223
31.4 Clock and Reset Summary	224
31.4.1 Clock Domains	224
31.4.2 Reset Signals	224
31.5 Interface Summary	224
31.5.1 External Interfaces	224
31.5.2 Internal Buses	225
31.6 Related Documentation	225
31.7 Specification Conventions	225
31.7.1 Signal Naming	225
31.7.2 Parameter Naming	226
31.7.3 Figure and Table Numbering	226

List of Figures

Figure 1.1.1: RAPIDS Architecture Block Diagram.....	35
Figure 1.1.2: Sink Path Data Flow.....	37
Figure 1.1.3: Source Path Data Flow.....	38
Figure 1.3.1: Reset Timing.....	66
Figure 2.1.1: Scheduler Byte Math.....	70
Figure 2.2.1: Descriptor Engine Block Diagram.....	83
Figure 2.2.2: Descriptor Engine FSM.....	86
Figure 2.2.3: Descriptor Chain Timing.....	88
Figure 2.3.1: AXI Read Engine Block Diagram.....	91
Figure 2.3.2: Burst Segmentation with a Mid-Beat Start.....	96
Figure 2.4.1: AXI Write Engine Block Diagram.....	99
Figure 2.5.1: Beats Alloc Control Block Diagram.....	107
Figure 2.5.2: Allocation and Release Timing.....	110
Figure 2.6.1: Beats Drain Control Block Diagram.....	112
Figure 2.6.2: Data Arrival and Drain Timing.....	115
Figure 2.7.1: Latency Bridge Block Diagram.....	117
Figure 2.7.2: Latency Bridge Timing (consumer always ready).....	120
Figure 2.8.1: Control-Read Engine Block Diagram.....	123
Figure 2.8.2: Control-Read Engine FSM.....	127
Figure 2.9.1: Control-Write Engine Block Diagram.....	131
Figure 2.9.2: Control-Write Engine FSM.....	135
Figure 2.9.3: Control-Write Doorbell Timing.....	137
Figure 3.1.1: Scheduler Group Block Diagram.....	140
Figure 3.2.1: Scheduler Group Array Block Diagram.....	149
Figure 3.2.2: Shared Master Arbitration.....	157
Figure 3.3.1: Sink Data Path Block Diagram.....	159
Figure 3.4.1: Ingress Shifter.....	164
Figure 3.5.1: Sink SRAM Controller Block Diagram.....	169
Figure 3.5.2: Drain Selection by the Consumer.....	172
Figure 3.6.1: Source Data Path Block Diagram.....	174
Figure 3.6.2: Source Path Data Flow Sequence.....	175
Figure 3.6.3: Source Path Transfer Timing.....	178
Figure 3.7.1: Egress Shifter.....	180
Figure 3.8.1: Source SRAM Controller Block Diagram.....	186
Figure 3.8.2: Drain Selection by the Consumer.....	189
Figure 3.9.1: Core Block Diagram.....	191

Figure 3.12.1: Top Hierarchy.....	202
Figure 4.3.1: MonBus Packet Transfer Timing.....	211
Figure 4.3.2: MonBus Packet Format.....	212
Figure 4.3.3: MonBus Aggregation Tree.....	214
Figure 4.4.1: Scheduler Byte Math.....	217

List of Tables

Table 1: Table 0.1: Revision History.....	32
Table 2: Table 0.2: Related Documents and Specifications.....	32
Table 3: Table 1.1.1: Block Roles.....	37
Table 4: Table 1.1.2: Design Points.....	38
Table 5: Table 1.2.1: Parameters (from rapids_core_beats.sv).....	40
Table 6: Table 1.2.2: Clock and Reset (shared - single clock/reset for the whole wrapper).....	42
Table 7: Table 1.2.3: APB Programming Interface.....	42
Table 8: Table 1.2.4: Per-channel configuration.....	42
Table 9: Table 1.2.5: Scheduler Configuration (global).....	43
Table 10: Table 1.2.6: Descriptor Engine Configuration (global).....	43
Table 11: Table 1.2.7: Control Engine Configuration (Phase 2, global).....	44
Table 12: Table 1.2.8: Descriptor AXI Monitor Configuration.....	44
Table 13: Table 1.2.9: AXIS data-path monitor-lite configuration (rapids TASK-015).....	45
Table 14: Table 1.2.10: Status.....	46
Table 15: Table 1.2.11: Descriptor AXI Monitor Status.....	46
Table 16: Table 1.2.12: AXIS data-path monitor-lite status (rapids TASK-015).....	46
Table 17: Table 1.2.13: Descriptor AXI Monitor perf window (feeds SRC_MON.DAXMON_PERF_*).....	47
Table 18: Table 1.2.14: Descriptor Fetch AXI Master.....	48
Table 19: Table 1.2.15: Control Read AXI Master (32-bit) [Phase 2].....	49
Table 20: Table 1.2.16: Control Write AXI Master (32-bit) [Phase 2].....	50
Table 21: Table 1.2.17: AXI Transfer Configuration (source-only).....	51
Table 22: Table 1.2.18: Source Path - AXIS Master Interface (SRAM -> Network); tid = channel id.....	51
Table 23: Table 1.2.19: AXI4 Master - Data Read (Memory -> Source SRAM).....	52
Table 24: Table 1.2.20: APB Programming Interface.....	52
Table 25: Table 1.2.21: Per-channel configuration.....	53
Table 26: Table 1.2.22: Scheduler Configuration (global).....	53
Table 27: Table 1.2.23: Descriptor Engine Configuration (global).....	53
Table 28: Table 1.2.24: Control Engine Configuration (Phase 2, global).....	54
Table 29: Table 1.2.25: Descriptor AXI Monitor Configuration.....	54
Table 30: Table 1.2.26: AXIS data-path monitor-lite configuration (rapids TASK-015).....	55
Table 31: Table 1.2.27: Status.....	56

Table 32: Table 1.2.28: Descriptor AXI Monitor Status.....	56
Table 33: Table 1.2.29: AXIS data-path monitor-lite status (rapids TASK-015).....	57
Table 34: Table 1.2.30: Descriptor AXI Monitor perf window (feeds SNK_MON.DAXMON_PERF_*).....	57
Table 35: Table 1.2.31: Descriptor Fetch AXI Master.....	58
Table 36: Table 1.2.32: Control Read AXI Master (32-bit) [Phase 2].....	59
Table 37: Table 1.2.33: Control Write AXI Master (32-bit) [Phase 2].....	60
Table 38: Table 1.2.34: Monitor Bus.....	61
Table 39: Table 1.2.35: AXI Transfer Configuration (sink-only).....	61
Table 40: Table 1.2.36: Sink Path - AXIS Slave Interface (Network -> SRAM); tid = channel id.....	62
Table 41: Table 1.2.37: AXI4 Master - Data Write (Sink SRAM -> Memory).....	62
Table 42: Table 1.2.38: Source half debug.....	63
Table 43: Table 1.2.39: Sink half debug.....	64
Table 44: Table 1.2.40: Active-channel sideband for per-channel bus instrumentation.....	64
Table 45: Table 1.3.1: Clock Specifications.....	65
Table 46: Table 1.3.2: Reset Specifications.....	66
Table 47: Table 1.3.3: Reset States.....	67
Table 48: Table 1.3.4: Configuration Timing Requirements.....	68
Table 49: Table 2.1.1: Scheduler Parameters.....	71
Table 50: Table 2.1.2: Clock, Reset and Configuration Ports.....	72
Table 51: Table 2.1.3: Status Ports.....	72
Table 52: Table 2.1.4: Descriptor Engine Interface.....	73
Table 53: Table 2.1.5: Data Read Interface.....	73
Table 54: Table 2.1.6: Data Write Interface.....	73
Table 55: Table 2.1.7: Completion and Error Interface.....	74
Table 56: Table 2.1.8: Control Engine Interface.....	75
Table 57: Table 2.1.9: Monitor Bus Interface.....	75
Table 58: Table 2.1.10: Descriptor Layout.....	76
Table 59: Table 2.1.11: Beat Totals for a 32-byte Beat.....	77
Table 60: Table 2.1.12: State Encoding.....	79
Table 61: Table 2.1.13: Transitions.....	79
Table 62: Table 2.1.14: MonBus Events.....	81
Table 63: Table 2.2.2: Clock and Reset.....	84
Table 64: Table 2.2.3: APB Programming Interface.....	84
Table 65: Table 2.2.4: Configuration Interface.....	84
Table 66: Table 2.2.5: AXI4 Read Master Interface.....	85

Table 67: Table 2.2.6: Scheduler Interface.....	86
Table 68: Table 2.3.1: AXI Read Engine Parameters.....	91
Table 69: Table 2.3.2: AXI Read Engine Derived Constants.....	92
Table 70: Table 2.3.3: Clock, Reset and Configuration.....	93
Table 71: Table 2.3.4: Scheduler Interface.....	93
Table 72: Table 2.3.5: SRAM Allocation and Fill Interface.....	94
Table 73: Table 2.3.6: AXI4 Read Master Interface.....	94
Table 74: Table 2.3.7: Debug Outputs.....	95
Table 75: Table 2.3.8: Burst Segmentation Example.....	96
Table 76: Table 2.3.9: Error Handling.....	97
Table 77: Table 2.3.10: Read Engine Delta.....	97
Table 78: Table 2.4.1: AXI Write Engine Parameters.....	99
Table 79: Table 2.4.2: AXI Write Engine Derived Constants.....	100
Table 80: Table 2.4.3: Clock, Reset and Configuration.....	100
Table 81: Table 2.4.4: Scheduler Interface.....	101
Table 82: Table 2.4.5: SRAM Drain Interface.....	102
Table 83: Table 2.4.6: AXI4 Write Master Interface.....	102
Table 84: Table 2.4.7: Debug and Status Outputs.....	103
Table 85: Table 2.4.8: Completion Strokes.....	105
Table 86: Table 2.4.9: Error Handling.....	105
Table 87: Table 2.4.10: Write Engine Delta.....	106
Table 88: Table 2.5.2: Clock and Reset.....	108
Table 89: Table 2.5.3: Write Interface.....	108
Table 90: Table 2.5.4: Read Interface.....	109
Table 91: Table 2.5.5: Status Outputs.....	109
Table 92: Table 2.6.2: Clock and Reset.....	113
Table 93: Table 2.6.3: Write Interface.....	113
Table 94: Table 2.6.4: Read Interface.....	114
Table 95: Table 2.6.5: Status Outputs.....	114
Table 96: Table 2.6.6: alloc_ctrl vs drain_ctrl Comparison.....	116
Table 97: Table 2.7.2: Clock and Reset.....	118
Table 98: Table 2.7.3: Upstream Interface.....	119
Table 99: Table 2.7.4: Downstream Interface.....	119
Table 100: Table 2.7.5: Status and Debug.....	119
Table 101: Table 2.7.6: Skid Depth Selection.....	121
Table 102: Table 2.8.2: Clock and Reset.....	124
Table 103: Table 2.8.3: Scheduler Interface.....	124
Table 104: Table 2.8.4: Configuration Interface.....	125

Table 105: Table 2.8.5: AXI4 Read Master Interface.....	125
Table 106: Table 2.8.6: MonBus Interface.....	126
Table 107: Table 2.8.7: MonBus Events.....	129
Table 108: Table 2.8.8: CTRL_READ Descriptor Field Mapping.....	130
Table 109: Table 2.9.2: Clock and Reset.....	132
Table 110: Table 2.9.3: Configuration Interface.....	133
Table 111: Table 2.9.4: Scheduler Interface.....	133
Table 112: Table 2.9.5: AXI4 Write Master Interface.....	133
Table 113: Table 2.9.6: MonBus Interface.....	134
Table 114: Table 3.1.1: Scheduler Group Parameters.....	141
Table 115: Table 3.1.2: Clock, Reset and Kick-Off.....	142
Table 116: Table 3.1.3: Configuration Interface.....	143
Table 117: Table 3.1.4: Status Interface.....	144
Table 118: Table 3.1.5: Descriptor AXI Master Interface.....	144
Table 119: Table 3.1.6: Scheduler Data Interface.....	145
Table 120: Table 3.1.7: Packet Record Interface.....	146
Table 121: Table 3.1.8: MonBus Interface.....	147
Table 122: Table 3.1.9: Scheduler Group Delta.....	148
Table 123: Table 3.2.1: Scheduler Group Array Parameters.....	150
Table 124: Table 3.2.2: Clock, Reset and Kick-Off.....	151
Table 125: Table 3.2.3: Global Configuration.....	152
Table 126: Table 3.2.4: Descriptor Monitor Configuration and Status.....	153
Table 127: Table 3.2.5: Descriptor-AXI Perf Window.....	154
Table 128: Table 3.2.6: Status.....	155
Table 129: Table 3.2.7: Per-Channel Scheduler Data Interface.....	155
Table 130: Table 3.2.8: Per-Channel Packet Record Interface.....	156
Table 131: Table 3.2.9: MonBus Interface.....	157
Table 132: Table 3.2.10: MonBus Source Assignment.....	158
Table 133: Table 3.2.11: Array Delta.....	158
Table 134: Table 3.3.1: Sink Data Path Parameters.....	160
Table 135: Table 3.3.2: Fill Interface.....	160
Table 136: Table 3.3.3: Configuration.....	161
Table 137: Table 3.3.4: Scheduler Interface.....	161
Table 138: Table 3.3.5: Sink Data Path Delta.....	162
Table 139: Table 3.4.1: Additional Parameters.....	164
Table 140: Table 3.4.2: AXI-Stream Slave.....	165
Table 141: Table 3.4.3: Packet Record Ports.....	166
Table 142: Table 3.4.4: Ingress Delta.....	167

Table 143: Table 3.5.2: Clock and Reset.....	170
Table 144: Table 3.5.3: Fill Interface (ID-Selected).....	171
Table 145: Table 3.5.4: Arbitrated Drain Interface.....	171
Table 146: Table 3.5.5: Debug Interface.....	172
Table 147: Table 3.6.2: Clock and Reset.....	176
Table 148: Table 3.6.3: Configuration.....	176
Table 149: Table 3.6.4: Scheduler Interface.....	176
Table 150: Table 3.6.5: AXI Read Master Interface.....	177
Table 151: Table 3.6.6: Drain Interface.....	177
Table 152: Table 3.7.1: Parameters.....	181
Table 153: Table 3.7.2: Configuration.....	181
Table 154: Table 3.7.3: Scheduler Interface.....	182
Table 155: Table 3.7.4: AXI-Stream Master.....	182
Table 156: Table 3.7.5: Egress Shift Rules.....	184
Table 157: Table 3.7.6: Egress Delta.....	185
Table 158: Table 3.8.2: Clock and Reset.....	188
Table 159: Table 3.8.3: Fill Interface (ID-Selected).....	188
Table 160: Table 3.8.4: Arbitrated Drain Interface.....	188
Table 161: Table 3.8.5: Debug Interface.....	189
Table 162: Table 3.8.6: Sink vs Source Controller Comparison.....	190
Table 163: Table 3.9.1: Packet Record Nets per Half.....	192
Table 164: Table 3.9.2: Core Port Renames.....	193
Table 165: Table 3.9.3: Data Ports.....	193
Table 166: Table 3.9.4: Core Parameters.....	194
Table 167: Table 3.9.5: Core Delta.....	195
Table 168: Table 3.10.1: Base Register Map.....	196
Table 169: Table 3.10.2: Monitor Register Map (base 0x1000).....	198
Table 170: Table 3.11.1: Monitor Register-to-Config Mapping.....	201
Table 171: Table 3.12.1: APB Address Map.....	203
Table 172: Table 3.12.2: Top Parameters.....	203
Table 173: Table 3.12.3: Top Delta.....	204
Table 174: Table 4.1.1: Address Split.....	206
Table 175: Table 4.1.2: Burst Rules.....	206
Table 176: Table 4.1.3: Byte-Granular Write Signal.....	207
Table 177: Table 4.2.1: Packet Contract.....	208
Table 178: Table 4.2.2: Source Master Signals.....	208
Table 179: Table 4.2.3: Sink Slave Signals.....	209
Table 180: Table 4.3.1: MonBus Interface Signals.....	210

Table 181: Table 4.3.2: MonBus Packet Fields.....	212
Table 182: Table 4.3.3: MonBus Packet Types.....	212
Table 183: Table 4.3.4: Descriptor Packet Payload.....	213
Table 184: Table 4.3.5: Scheduler Packet Payload.....	213
Table 185: Table 4.3.6: AXI Event Packet Payload.....	213
Table 186: Table 4.3.7: Error Packet Payload.....	214
Table 187: Table 4.3.8: MonBus Source Assignment.....	214
Table 188: Table 4.3.9: MonBus Timing Parameters.....	216
Table 189: Table 4.4.1: Packet Record Signals.....	218
Table 190: Table 4.4.2: Record Rules.....	218
Table 191: FUB Module Summary.....	221
Table 192: Macro Module Summary.....	222
Table 193: RAPIDS Beats vs STREAM Comparison.....	223
Table 194: Clock Domains.....	224
Table 195: Reset Signals.....	224
Table 196: External Interfaces.....	224
Table 197: Internal Buses.....	225

List of Waveforms

Waveform 2.1: Byte Descriptor Start.....	82
Waveform 3.7: Egress Re-pack.....	184

1 Rapids Mas Index

Generated: 2026-09-30

RTL Design Sherpa · Learning Hardware Design Through Practice GitHub · Documentation Index · MIT License

2 Document Information

This document describes RAPIDS, the byte-granular Rapid AXI Programmable In-band Descriptor System. RAPIDS provides network-to-memory and memory-to-network data transfer using descriptor-based DMA with 8 channels. A descriptor length is a byte count, source and destination addresses are byte addresses, and the sink write master drives byte enables. The earlier “Beats” architecture, which counted everything in beats, was the stepping stone to this design and is documented separately in the RAPIDS Beats MAS.

2.1 Revision History

Table 0.1: Revision History

Version	Date	Author	Description
0.1	2026-09-30	RTL Design Sherpa	First byte-granular MAS. Rewrites the chapters that changed from RAPIDS Beats and links the rest.

2.2 References

2.2.1 Related Documents

Table 0.2: Related Documents and Specifications

Source	Title	Version
RTL Design Sherpa	RAPIDS Beats MAS (shared chapters)	0.7
RTL Design Sherpa	RAPIDS HAS	0.1

Source	Title	Version
RTL Design Sherpa	RAPIDS Product Requirements Document	1.0
ARM	AMBA AXI and ACE Protocol Specification	IHI0022H
ARM	AMBA AXI-Stream Protocol Specification	IHI0051A

2.3 Terminology

Beat

One data transfer of an AXI burst or an AXI-Stream transfer. One beat is `BYTE_LANES` bytes, where `BYTE_LANES = DATA_WIDTH/8`. The board design uses 256 bits, so a beat is 32 bytes.

Byte offset

The low `OFF_W` bits of a byte address, where `OFF_W = $clog2(BYTE_LANES)`. It is the lane at which a transfer begins inside its first memory beat.

Memory beat

A beat as it appears on the AXI side, aligned to `BYTE_LANES`. A packet that starts at a non-zero offset occupies one more memory beat than a stream of the same length would.

Stream beat

A beat as it appears on AXI-Stream, packed from lane 0 with no offset.

Packet record

The pair `{bytes, offset}` that the scheduler issues once per DATA descriptor, so the AXIS shifter knows how the memory-side and stream-side layouts relate.

Spill

The bytes of a stream beat that did not fit in the current memory beat during ingress shifting. They are held for the next memory beat.

Channel

One of 8 independent DMA channels. Each has its own descriptor chain, scheduler and SRAM allocation.

Descriptor

A 256-bit structure holding the source address, destination address, length in bytes, next pointer, and control flags.

TYPE=EXT

The descriptor type that describes a row/column transfer. It stays beat-aligned by design: addresses are aligned and lengths are multiples of the beat size.

DMA

Direct Memory Access. Data movement without processor involvement.

MonBus

The 64-bit monitor bus that carries event packets out of the design.

2.4 Document Conventions

- Shared chapters are linked from the RAPIDS Beats MAS and marked “(shared with RAPIDS Beats)” in the index. Module names in those chapters carry a `_beats` suffix that this design drops.
- Requirement priorities follow the RAPIDS PRD.
- No emojis appear in this document.

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

3 Architecture Overview

Module: `rapids_core.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro/`
Status: Implemented **Last Updated:** 2026-09-30

3.1 Overview

RAPIDS moves data between AXI-Stream and system memory in units of bytes. A descriptor’s length is a byte count in the existing 32-bit field, and its source and destination are byte addresses. A linear descriptor may start at any byte on either side. The design keeps the streaming, no-FSM

data path of RAPIDS Beats and adds three things around it: byte-to-beat math in the scheduler, byte enables through the sink SRAM to the write master, and a shifter at each AXI-Stream edge that converts between packed stream beats and memory-aligned beats.

3.1.1 Key Features

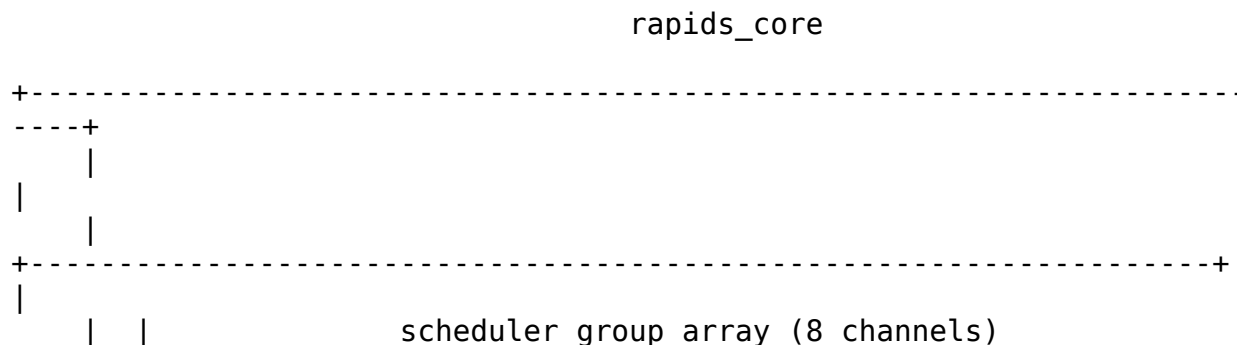
- **8 Independent Channels:** Each channel has its own descriptor chain and SRAM allocation.
- **Separate Data Paths:** Sink (network to memory) and source (memory to network).
- **Byte Granularity:** Length in bytes, byte addresses, any starting offset for linear descriptors.
- **Byte Enables:** The sink SRAM entry is {strb, data}; the write engine drivesWSTRB from it.
- **Packet Records:** The scheduler issues one {bytes, offset} record per DATA descriptor to the AXIS shifter.
- **One AXIS Packet per Descriptor:** The source path emits exactly one packet, with TLAST on the final byte.
- **Safe Bursts:** AXI bursts never cross a 4 KB boundary; AxADDR is beat-aligned.
- **MonBus Integration:** Same monitor packets as RAPIDS Beats.

3.1.2 What Is Shared with RAPIDS Beats

The descriptor engine, the control-read and control-write engines, the SRAM controllers, the register block and config block, and the MonBus specification are unchanged. Those chapters are linked from the RAPIDS Beats MAS.

3.2 Block Diagram

3.2.1 Figure 1.1.1: RAPIDS Architecture Block Diagram



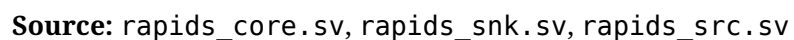


Table 1.1.1: Block Roles

Block	Role
scheduler	Loads a descriptor, computes beats from offset and length, issues the packet record, tracks completion
scheduler_group_array	Eight scheduler groups sharing one descriptor AXI master; per-channel packet-record ports
snk_data_path_axis	Converts packed AXIS beats into memory-aligned beats with byte enables; checks the packet length
snk_data_path	SRAM of {strb, data} entries plus the AXI write engine
src_data_path	SRAM of data entries plus the AXI read engine; unchanged from Beats
src_data_path_axis	Converts memory-aligned beats into packed stream beats with a partial last beat

3.3 Data Flow Overview

3.3.1 Sink Path (Network to Memory)

3.3.2 Figure 1.1.2: Sink Path Data Flow

- Scheduler loads a descriptor, computes beats_total, pulses sched_wr_pkt_valid
 | (record {bytes, offset} enters the per-channel record queue)
 v
- s_axis packet arrives; tready is held low until the channel's record is known
 |
 v
- Ingress shifter: stream beat << (offset*8), OR the spill of the previous beat
 | low DATA_WIDTH bits = this memory beat, high bits become the new spill
 v
- Fill logic writes {strb, data} into the sink SRAM

(snk_sram_controller)



5. axi_write_engine: aligned AWADDR, capped burst,WSTRB from the SRAM entry



6. System memory; B response commits the beats and completes the descriptor

3.3.3 Source Path (Memory to Network)

3.3.4 Figure 1.1.3: Source Path Data Flow

1. Scheduler loads a descriptor, computes beats_total, pulses sched_rd_pkt_valid



2. axi_read_engine: aligned ARADDR, capped burst, beats into the source SRAM



3. src_data_path_axis reserves drain blocks (round-robin, cfg_drain_size beats)



4. Egress shifter: memory beat >> (offset*8), OR the next beat's low lanes

partial | the first pop only primes the hold; the last beat is



5. m_axis: TDATA packed from lane 0, TSTRB contiguous, TLAST on the final byte

3.4 Byte-Granular Design Points

Table 1.1.2: Design Points

Point	Rule
Length unit	Bytes, in the existing 32-bit descriptor field
Beats moved	(offset + length + BYTE_LANES - 1) >> OFF_W, zero when length is zero
First burst	May begin mid-beat; the address it

Point	Rule
	presents is beat-aligned and the offset is carried by the record
Later bursts	Beat-aligned
4 KB rule	A burst is truncated at the next 4 KB boundary
Sink lanes not written	WSTRB bit is 0
Source last beat	Partial; TSTRB marks the valid low lanes, TLAST set
TYPE=EXT	Beat-aligned by design (permanent): aligned addresses, beat-multiple lengths

The 4 KB rule and the byte math are specified in [Scheduler](#) and [AXI Read Engine](#).

3.5 Chapter Map

Topic	Chapter
Scheduler byte math and record issue	Scheduler
Record queue interface	Packet Record Interface
Ingress shifter	Sink Data Path AXIS
Egress shifter	Source Data Path AXIS
Everything else	Linked from the RAPIDS Beats MAS via the index

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

4 Top-Level Port List

Module: `rapids_core_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

4.1 Parameters

Every parameter of `rapids_core_beats`, as declared. The short aliases at the end are what the port widths below are written in (NC, AW, DW, IW, SCW, CIW, SW).

Table 1.2.1: Parameters (from `rapids_core_beats.sv`)

Parameter	Type	Default	Notes
NUM_CHANNELS	int	8	-
CHAN_WIDTH	int	$\$clog2(\text{NUM_CHANN}$ $\text{ELS})$	-
ADDR_WIDTH	int	64	-
DATA_WIDTH	int	512	-
AXI_ID_WIDTH	int	8	-
SRAM_DEPTH	int	512	-
SEG_COUNT_WIDTH	int	$\$clog2(\text{SRAM_DEPT}$ $\text{H}) + 1$	-
PIPELINE	int	1	write-engine pipelining; 1 is the perf design point (see <code>rapids_beats_top</code>)
AR_MAX_OUTSTANDI NG	int	8	-
AW_MAX_OUTSTANDI NG	int	8	-
W_PHASE_FIFO_DEP TH	int	64	-
B_PHASE_FIFO_DEP TH	int	16	-
AXIS_ID_WIDTH	int	8	-
AXIS_DEST_WIDTH	int	4	-

Parameter	Type	Default	Notes
AXIS_USER_WIDTH	int	1	-
DESC_MON_BASE_AGE ENT_ID	int	16	0x10 - Descriptor Engines (16-23)
SCHED_MON_BASE_A GENT_ID	int	48	0x30 - Schedulers (48-55)
DESC_AXI_MON_AGE NT_ID	int	8	0x08 - Descriptor AXI Master Monitor
SNK_AXIS_MON_AGE NT_ID	int	9	0x09 - Sink- ingress AXIS monitor (rapids TASK-015)
SRC_AXIS_MON_AGE NT_ID	int	10	0x0A - Source- egress AXIS monitor (rapids TASK-015)
ACLK_MHZ	int	100	clk in MHz: the AXIS monitors' microsecond tick
MON_UNIT_ID	int	1	0x1
MON_MAX_TRANSACTION	int	16	-
USE_AXI_MONITORS	int	1	-
USE_ROW_COL_MAJOR ADDRESSING	int	1	-
GEN_MON	bit	1'b1	-
NC	int	NUM_CHANNELS	-
AW	int	ADDR_WIDTH	-
DW	int	DATA_WIDTH	-
IW	int	AXI_ID_WIDTH	-
SD	int	SRAM_DEPTH	-
SCW	int	SEG_COUNT_WIDTH	-
CIW	int	(NC > 1) ? \$clog2(NC) : 1	-

Parameter	Type	Default	Notes
SW	int	DW / 8	-

4.2 Port List (324 ports)

The module is two independent halves. Ports both halves carry are prefixed `src_ / snk_`; the direction-unique ports (the source's AXI read master and AXIS egress, the sink's AXI write master and AXIS ingress) carry no prefix. Sections follow the declaration order in the RTL, so a reader can walk this page and the module side by side. Widths use the parameter aliases.

4.2.1 Clock and Reset (shared - single clock/reset for the whole wrapper)

Table 1.2.2: Clock and Reset (shared - single clock/reset for the whole wrapper)

Signal	Direction	Width	Description
clk	input	1	-
rst_n	input	1	-

4.3 SOURCE HALF (u_src) - shared-infrastructure ports (src_ prefixed)

4.3.1 APB Programming Interface

Table 1.2.3: APB Programming Interface

Signal	Direction	Width	Description
src_apb_valid	input	NC	-
src_apb_ready	output	NC	-
src_apb_addr	input	NC x AW	-

4.3.2 Per-channel configuration

Table 1.2.4: Per-channel configuration

Signal	Direction	Width	Description
src_cfg_chann el_enable	input	NC	-
src_cfg_chann el_reset	input	NC	-

4.3.3 Scheduler Configuration (global)

Table 1.2.5: Scheduler Configuration (global)

Signal	Direction	Width	Description
src_cfg_sched_enable	input	1	-
src_cfg_sched_timeout_cycles	input	32	-
src_cfg_sched_timeout_limit	input	8	-
src_cfg_sched_timeout_enable	input	1	-
src_cfg_sched_err_enable	input	1	-
src_cfg_sched_compl_enable	input	1	-
src_cfg_sched_perf_enable	input	1	-

4.3.4 Descriptor Engine Configuration (global)

Table 1.2.6: Descriptor Engine Configuration (global)

Signal	Direction	Width	Description
src_cfg_desceng_enable	input	1	-
src_cfg_desceng_prefetch	input	1	-
src_cfg_desceng_fifo_thresh	input	4	-
src_cfg_desceng_addr0_base	input	AW	-
src_cfg_desceng_addr0_limit	input	AW	-
src_cfg_desceng_addr1_base	input	AW	-
src_cfg_desceng_addr1_limit	input	AW	-

4.3.5 Control Engine Configuration (Phase 2, global)

Table 1.2.7: Control Engine Configuration (Phase 2, global)

Signal	Direction	Width	Description
src_cfg_ctrlr_d_max_try	input	9	-
src_tick_lus	input	1	-

4.3.6 Descriptor AXI Monitor Configuration

Table 1.2.8: Descriptor AXI Monitor Configuration

Signal	Direction	Width	Description
src_cfg_desc_mon_enable	input	1	-
src_cfg_desc_mon_err_enable	input	1	-
src_cfg_desc_mon_perf_enable	input	1	-
src_cfg_desc_mon_perf_run	input	1	-
src_cfg_desc_mon_timeout_enable	input	1	-
src_cfg_desc_mon_timeout_cycles	input	32	-
src_cfg_desc_mon_latency_thresh	input	32	-
src_cfg_desc_mon_pkt_mask	input	16	-
src_cfg_desc_mon_err_select	input	4	-
src_cfg_desc_mon_err_mask	input	8	-
src_cfg_desc_mon_timeout_mask	input	8	-
src_cfg_desc_mon_compl_mask	input	8	-

Signal	Direction	Width	Description
k			
src_cfg_desc_mon_thresh_mask	input	8	-
src_cfg_desc_mon_perf_mask	input	8	-
src_cfg_desc_mon_addr_mask	input	8	-
src_cfg_desc_mon_debug_mask	input	8	-

4.3.7 AXIS data-path monitor-lite configuration (rapids TASK-015)

Table 1.2.9: AXIS data-path monitor-lite configuration (rapids TASK-015)

Signal	Direction	Width	Description
src_cfg_axis_mon_enable	input	1	-
src_cfg_axis_mon_err_enable	input	1	-
src_cfg_axis_mon_compl_enable	input	1	-
src_cfg_axis_mon_perf_enable	input	1	-
src_cfg_axis_mon_timeout_enable	input	1	-
src_cfg_axis_mon_timeout_cycles	input	32	-
src_cfg_axis_mon_latency_thresh	input	32	-
src_cfg_axis_mon_pkt_mask	input	16	-

4.3.8 Status

Table 1.2.10: Status

Signal	Direction	Width	Description
src_system_idle	output	1	-
src_descriptor_engine_idle	output	NC	-
src_scheduler_idle	output	NC	-
src_scheduler_state	output	NC x 7	-
src_sched_error	output	NC	-

4.3.9 Descriptor AXI Monitor Status

Table 1.2.11: Descriptor AXI Monitor Status

Signal	Direction	Width	Description
src_cfg_sts_desc_mon_busy	output	1	-
src_cfg_sts_desc_mon_active_txns	output	8	-
src_cfg_sts_desc_mon_error_count	output	16	-
src_cfg_sts_desc_mon_txn_count	output	32	-
src_cfg_sts_desc_mon_conflict_error	output	1	-

4.3.10 AXIS data-path monitor-lite status (rapids TASK-015)

Table 1.2.12: AXIS data-path monitor-lite status (rapids TASK-015)

Signal	Direction	Width	Description
src_cfg_sts_axis_mon_busy	output	1	-
src_cfg_sts_axis_mon_packet_count	output	32	-

Signal	Direction	Width	Description
src_cfg_sts_axis_mon_error_count	output	16	-
src_cfg_sts_axis_mon_dropped_count	output	16	-

4.3.11 Descriptor AXI Monitor perf window (feeds SRC_MON.DAXMON_PERF_*).

Table 1.2.13: Descriptor AXI Monitor perf window (feeds SRC_MON.DAXMON_PERF_*).

Signal	Direction	Width	Description
src_sts_desc_mon_win_active	output	1	-
src_sts_desc_mon_win_cycles	output	32	-
src_sts_desc_mon_prod_cycles	output	32	-
src_sts_desc_mon_bp_cycles	output	32	-
src_sts_desc_mon_starv_cycles	output	32	-
src_sts_desc_mon_idle_cycles	output	32	-
src_sts_desc_mon_beat_count	output	32	-
src_sts_desc_mon_byte_count	output	64	-
src_sts_desc_mon_burst_count	output	32	-

4.3.12 Descriptor Fetch AXI Master

Table 1.2.14: Descriptor Fetch AXI Master

Signal	Direction	Width	Description
src_m_axi_des c_arvalid	output	1	-
src_m_axi_des c_arready	input	1	-
src_m_axi_des c_araddr	output	AW	-
src_m_axi_des c_arlen	output	8	-
src_m_axi_des c_arsize	output	3	-
src_m_axi_des c_arburst	output	2	-
src_m_axi_des c_arid	output	IW	-
src_m_axi_des c_arlock	output	1	-
src_m_axi_des c_arcache	output	4	-
src_m_axi_des c_arprot	output	3	-
src_m_axi_des c_arqos	output	4	-
src_m_axi_des c_arregion	output	4	-
src_m_axi_des c_rvalid	input	1	-
src_m_axi_des c_rready	output	1	-
src_m_axi_des c_rdata	input	256	-
src_m_axi_des c_rresp	input	2	-
src_m_axi_des c_rlast	input	1	-
src_m_axi_des c_rid	input	IW	-

4.3.13 Control Read AXI Master (32-bit) [Phase 2]

Table 1.2.15: Control Read AXI Master (32-bit) [Phase 2]

Signal	Direction	Width	Description
src_m_axi_ctr lrd_arvalid	output	1	-
src_m_axi_ctr lrd_arready	input	1	-
src_m_axi_ctr lrd_araddr	output	AW	-
src_m_axi_ctr lrd_arlen	output	8	-
src_m_axi_ctr lrd_arsize	output	3	-
src_m_axi_ctr lrd_arburst	output	2	-
src_m_axi_ctr lrd_arid	output	IW	-
src_m_axi_ctr lrd_arlock	output	1	-
src_m_axi_ctr lrd_arcache	output	4	-
src_m_axi_ctr lrd_arprot	output	3	-
src_m_axi_ctr lrd_arqos	output	4	-
src_m_axi_ctr lrd_arregion	output	4	-
src_m_axi_ctr lrd_rvalid	input	1	-
src_m_axi_ctr lrd_rready	output	1	-
src_m_axi_ctr lrd_rdata	input	32	-
src_m_axi_ctr lrd_rresp	input	2	-
src_m_axi_ctr lrd_rlast	input	1	-
src_m_axi_ctr lrd_rid	input	IW	-

4.3.14 Control Write AXI Master (32-bit) [Phase 2]

Table 1.2.16: Control Write AXI Master (32-bit) [Phase 2]

Signal	Direction	Width	Description
src_m_axi_ctr lwr_awvalid	output	1	-
src_m_axi_ctr lwr_awready	input	1	-
src_m_axi_ctr lwr_awaddr	output	AW	-
src_m_axi_ctr lwr_awlen	output	8	-
src_m_axi_ctr lwr_awsz	output	3	-
src_m_axi_ctr lwr_awburst	output	2	-
src_m_axi_ctr lwr_awid	output	IW	-
src_m_axi_ctr lwr_awlock	output	1	-
src_m_axi_ctr lwr_awcache	output	4	-
src_m_axi_ctr lwr_awprot	output	3	-
src_m_axi_ctr lwr_awqos	output	4	-
src_m_axi_ctr lwr_awregion	output	4	-
src_m_axi_ctr lwr_wvalid	output	1	-
src_m_axi_ctr lwr_wready	input	1	-
src_m_axi_ctr lwr_wdata	output	32	-
src_m_axi_ctr lwr_wstrb	output	4	-
src_m_axi_ctr lwr_wlast	output	1	-
src_m_axi_ctr lwr_bvalid	input	1	-
src_m_axi_ctr lwr_bready	output	1	-

Signal	Direction	Width	Description
src_m_axi_ctr lwr_bid	input	IW	-
src_m_axi_ctr lwr_bresp	input	2	-

4.4 SOURCE HALF (u_src) - direction-unique ports (no prefix)

4.4.1 AXI Transfer Configuration (source-only)

Table 1.2.17: AXI Transfer Configuration (source-only)

Signal	Direction	Width	Description
cfg_axi_rd_xf er_beats	input	8	-
cfg_drain_siz e	input	8	source: beats drained per AXIS packet

4.4.2 Source Path - AXIS Master Interface (SRAM -> Network); tid = channel id

Table 1.2.18: Source Path - AXIS Master Interface (SRAM -> Network); tid = channel id

Signal	Direction	Width	Description
m_axis_tdata	output	DW	-
m_axis_tstrb	output	SW	-
m_axis_tlast	output	1	-
m_axis_tid	output	AXIS_ID_WIDT H	-
m_axis_tdest	output	AXIS_DEST_WI DTH	-
m_axis_tuser	output	AXIS_USER_WI DTH	-
m_axis_tvalid	output	1	-
m_axis_tready	input	1	-

4.4.3 AXI4 Master - Data Read (Memory -> Source SRAM)

Table 1.2.19: AXI4 Master - Data Read (Memory -> Source SRAM)

Signal	Direction	Width	Description
m_axi_rd_arid	output	IW	-
m_axi_rd_aradr	output	AW	-
m_axi_rd_arlen	output	8	-
m_axi_rd_arsize	output	3	-
m_axi_rd_arburst	output	2	-
m_axi_rd_arvalid	output	1	-
m_axi_rd_arready	input	1	-
m_axi_rd_rid	input	IW	-
m_axi_rd_rdata	input	DW	-
m_axi_rd_rresp	input	2	-
m_axi_rd_rlast	input	1	-
m_axi_rd_rvalid	input	1	-
m_axi_rd_rready	output	1	-

4.5 SINK HALF (u_snk) - shared-infrastructure ports (snk_prefixed)

4.5.1 APB Programming Interface

Table 1.2.20: APB Programming Interface

Signal	Direction	Width	Description
snk_apb_valid	input	NC	-
snk_apb_ready	output	NC	-
snk_apb_addr	input	NC x AW	-

4.5.2 Per-channel configuration

Table 1.2.21: Per-channel configuration

Signal	Direction	Width	Description
snk_cfg_chann el_enable	input	NC	-
snk_cfg_chann el_reset	input	NC	-

4.5.3 Scheduler Configuration (global)

Table 1.2.22: Scheduler Configuration (global)

Signal	Direction	Width	Description
snk_cfg_sched _enable	input	1	-
snk_cfg_sched _timeout_cycles	input	32	-
snk_cfg_sched _timeout_limit	input	8	-
snk_cfg_sched _timeout_enable	input	1	-
snk_cfg_sched _err_enable	input	1	-
snk_cfg_sched _compl_enable	input	1	-
snk_cfg_sched _perf_enable	input	1	-

4.5.4 Descriptor Engine Configuration (global)

Table 1.2.23: Descriptor Engine Configuration (global)

Signal	Direction	Width	Description
snk_cfg_desce ng_enable	input	1	-
snk_cfg_desce ng_prefetch	input	1	-
snk_cfg_desce ng_fifo_thresh	input	4	-
snk_cfg_desce	input	AW	-

Signal	Direction	Width	Description
ng_addr0_base			
snk_cfg_desceng_addr0_limit	input	AW	-
snk_cfg_desceng_addr1_base	input	AW	-
snk_cfg_desceng_addr1_limit	input	AW	-

4.5.5 Control Engine Configuration (Phase 2, global)

Table 1.2.24: Control Engine Configuration (Phase 2, global)

Signal	Direction	Width	Description
snk_cfg_ctrlrd_max_try	input	9	-
snk_tick_lus	input	1	-

4.5.6 Descriptor AXI Monitor Configuration

Table 1.2.25: Descriptor AXI Monitor Configuration

Signal	Direction	Width	Description
snk_cfg_desc_mon_enable	input	1	-
snk_cfg_desc_mon_err_enable	input	1	-
snk_cfg_desc_mon_perf_enable	input	1	-
snk_cfg_desc_mon_perf_run	input	1	-
snk_cfg_desc_mon_timeout_enable	input	1	-
snk_cfg_desc_mon_timeout_cycles	input	32	-
snk_cfg_desc_mon_latency_thresh	input	32	-
snk_cfg_desc_mon_latency_thresh	input	16	-

Signal	Direction	Width	Description
mon_pkt_mask			
snk_cfg_desc_mon_err_select	input	4	-
snk_cfg_desc_mon_err_mask	input	8	-
snk_cfg_desc_mon_timeout_mask	input	8	-
snk_cfg_desc_mon_compl_mask	input	8	-
snk_cfg_desc_mon_thresh_mask	input	8	-
snk_cfg_desc_mon_perf_mask	input	8	-
snk_cfg_desc_mon_addr_mask	input	8	-
snk_cfg_desc_mon_debug_mask	input	8	-

4.5.7 AXIS data-path monitor-lite configuration (rapids TASK-015)

Table 1.2.26: AXIS data-path monitor-lite configuration (rapids TASK-015)

Signal	Direction	Width	Description
snk_cfg_axis_mon_enable	input	1	-
snk_cfg_axis_mon_err_enable	input	1	-
snk_cfg_axis_mon_compl_enable	input	1	-
snk_cfg_axis_mon_perf_enable	input	1	-
snk_cfg_axis_mon_timeout_enable	input	1	-
snk_cfg_axis_mon_data	input	32	-

Signal	Direction	Width	Description
mon_timeout_cycles			
snk_cfg_axis_mon_latency_thresh	input	32	-
snk_cfg_axis_mon_pkt_mask	input	16	-

4.5.8 Status

Table 1.2.27: Status

Signal	Direction	Width	Description
snk_system_idle	output	1	-
snk_descriptor_engine_idle	output	NC	-
snk_scheduler_idle	output	NC	-
snk_scheduler_state	output	NC x 7	-
snk_sched_error	output	NC	-

4.5.9 Descriptor AXI Monitor Status

Table 1.2.28: Descriptor AXI Monitor Status

Signal	Direction	Width	Description
snk_cfg_sts_desc_mon_busy	output	1	-
snk_cfg_sts_desc_mon_active_txns	output	8	-
snk_cfg_sts_desc_mon_error_count	output	16	-
snk_cfg_sts_desc_mon_txn_count	output	32	-
snk_cfg_sts_desc_mon_conflict_error	output	1	-

4.5.10 AXIS data-path monitor-lite status (rapids TASK-015)

Table 1.2.29: AXIS data-path monitor-lite status (rapids TASK-015)

Signal	Direction	Width	Description
snk_cfg_sts_axis_mon_busy	output	1	-
snk_cfg_sts_axis_mon_packet_count	output	32	-
snk_cfg_sts_axis_mon_error_count	output	16	-
snk_cfg_sts_axis_mon_dropped_count	output	16	-

4.5.11 Descriptor AXI Monitor perf window (feeds SNK_MON.DAXMON_PERF_*).

Table 1.2.30: Descriptor AXI Monitor perf window (feeds SNK_MON.DAXMON_PERF_*).

Signal	Direction	Width	Description
snk_sts_desc_mon_win_active	output	1	-
snk_sts_desc_mon_win_cycles	output	32	-
snk_sts_desc_mon_prod_cycles	output	32	-
snk_sts_desc_mon_bp_cycles	output	32	-
snk_sts_desc_mon_starv_cycles	output	32	-
snk_sts_desc_mon_idle_cycles	output	32	-
snk_sts_desc_mon_beat_count	output	32	-
snk_sts_desc_mon_total_cycles	output	64	-

Signal	Direction	Width	Description
mon_byte_count			
snk_sts_desc_mon_burst_count	output	32	-

4.5.12 Descriptor Fetch AXI Master

Table 1.2.31: Descriptor Fetch AXI Master

Signal	Direction	Width	Description
snk_m_axi_desc_arvalid	output	1	-
snk_m_axi_desc_arready	input	1	-
snk_m_axi_desc_araddr	output	AW	-
snk_m_axi_desc_arlen	output	8	-
snk_m_axi_desc_arsize	output	3	-
snk_m_axi_desc_arburst	output	2	-
snk_m_axi_desc_arid	output	IW	-
snk_m_axi_desc_arlock	output	1	-
snk_m_axi_desc_arcache	output	4	-
snk_m_axi_desc_arprot	output	3	-
snk_m_axi_desc_arqos	output	4	-
snk_m_axi_desc_arregion	output	4	-
snk_m_axi_desc_rvalid	input	1	-
snk_m_axi_desc_rready	output	1	-
snk_m_axi_desc_rdata	input	256	-

Signal	Direction	Width	Description
snk_m_axi_desc_rresp	input	2	-
snk_m_axi_desc_rlast	input	1	-
snk_m_axi_desc_rid	input	IW	-

4.5.13 Control Read AXI Master (32-bit) [Phase 2]

Table 1.2.32: Control Read AXI Master (32-bit) [Phase 2]

Signal	Direction	Width	Description
snk_m_axi_ctr_lrd_arvalid	output	1	-
snk_m_axi_ctr_lrd_arready	input	1	-
snk_m_axi_ctr_lrd_araddr	output	AW	-
snk_m_axi_ctr_lrd_arlen	output	8	-
snk_m_axi_ctr_lrd_arsize	output	3	-
snk_m_axi_ctr_lrd_arburst	output	2	-
snk_m_axi_ctr_lrd_arid	output	IW	-
snk_m_axi_ctr_lrd_arlock	output	1	-
snk_m_axi_ctr_lrd_arcache	output	4	-
snk_m_axi_ctr_lrd_arprot	output	3	-
snk_m_axi_ctr_lrd_arqos	output	4	-
snk_m_axi_ctr_lrd_arregion	output	4	-
snk_m_axi_ctr_lrd_rvalid	input	1	-
snk_m_axi_ctr_lrd_rready	output	1	-
snk_m_axi_ctr_lrd_rdata	input	32	-

Signal	Direction	Width	Description
snk_m_axi_ctr lrd_rresp	input	2	-
snk_m_axi_ctr lrd_rlast	input	1	-
snk_m_axi_ctr lrd_rid	input	IW	-

4.5.14 Control Write AXI Master (32-bit) [Phase 2]

Table 1.2.33: Control Write AXI Master (32-bit) [Phase 2]

Signal	Direction	Width	Description
snk_m_axi_ctr lwr_awvalid	output	1	-
snk_m_axi_ctr lwr_awready	input	1	-
snk_m_axi_ctr lwr_awaddr	output	AW	-
snk_m_axi_ctr lwr_awlen	output	8	-
snk_m_axi_ctr lwr_awsz	output	3	-
snk_m_axi_ctr lwr_awburst	output	2	-
snk_m_axi_ctr lwr_awid	output	IW	-
snk_m_axi_ctr lwr_awlock	output	1	-
snk_m_axi_ctr lwr_awcache	output	4	-
snk_m_axi_ctr lwr_awprot	output	3	-
snk_m_axi_ctr lwr_awqos	output	4	-
snk_m_axi_ctr lwr_awregion	output	4	-
snk_m_axi_ctr lwr_wvalid	output	1	-
snk_m_axi_ctr lwr_wready	input	1	-
snk_m_axi_ctr lwr_wdata	output	32	-

Signal	Direction	Width	Description
snk_m_axi_ctr lwr_wstrb	output	4	-
snk_m_axi_ctr lwr_wlast	output	1	-
snk_m_axi_ctr lwr_bvalid	input	1	-
snk_m_axi_ctr lwr_bready	output	1	-
snk_m_axi_ctr lwr_bid	input	IW	-
snk_m_axi_ctr lwr_bresp	input	2	-

4.6 Monitor Bus (SINGLE aggregated stream for the whole core)

The two halves' monitor outputs are merged through a top-level monbus_arbiter, so the core exposes exactly one monitor stream.

4.6.1 Monitor Bus

Table 1.2.34: Monitor Bus

Signal	Direction	Width	Description
mon_valid	output	1	-
mon_ready	input	1	-
mon_packet	output	monitor_com mon_pkg::mo nitor_packe t_t	-
mon_timestamp	output	monitor_com mon_pkg::mo nbus_timest amp_t	-

4.7 SINK HALF (u_snk) - direction-unique ports (no prefix)

4.7.1 AXI Transfer Configuration (sink-only)

Table 1.2.35: AXI Transfer Configuration (sink-only)

Signal	Direction	Width	Description
cfg_axi_wr_xf er_beats	input	8	-

Signal	Direction	Width	Description
cfg_alloc_size	input	8	sink: SRAM alloc size per AXIS fill

4.7.2 Sink Path - AXIS Slave Interface (Network -> SRAM); tid = channel id

Table 1.2.36: Sink Path - AXIS Slave Interface (Network -> SRAM); tid = channel id

Signal	Direction	Width	Description
s_axis_tdata	input	DW	-
s_axis_tstrb	input	SW	-
s_axis_tlast	input	1	-
s_axis_tid	input	AXIS_ID_WIDTH	-
s_axis_tdest	input	AXIS_DEST_WIDTH	-
s_axis_tuser	input	AXIS_USER_WIDTH	-
s_axis_tvalid	input	1	-
s_axis_tready	output	1	-

4.7.3 AXI4 Master - Data Write (Sink SRAM -> Memory)

Table 1.2.37: AXI4 Master - Data Write (Sink SRAM -> Memory)

Signal	Direction	Width	Description
m_axi_wr_awid	output	IW	-
m_axi_wr_awaddr	output	AW	-
m_axi_wr_awlen	output	8	-
m_axi_wr_awsiz	output	3	-
m_axi_wr_awburst	output	2	-
m_axi_wr_awlock	output	1	-
m_axi_wr_awcache	output	4	-

Signal	Direction	Width	Description
m_axi_wr_awprot	output	3	-
m_axi_wr_awqs	output	4	-
m_axi_wr_awregion	output	4	-
m_axi_wr_awvalid	output	1	-
m_axi_wr_awready	input	1	-
m_axi_wr_wdata	output	DW	-
m_axi_wr_wstrb	output	(DW/8)	-
m_axi_wr_wlast	output	1	-
m_axi_wr_wvalid	output	1	-
m_axi_wr_wready	input	1	-
m_axi_wr_bid	input	IW	-
m_axi_wr_bresp	input	2	-
m_axi_wr_bvalid	input	1	-
m_axi_wr_bready	output	1	-

4.8 Debug Interface (per half, src_dbg_ / snk_dbg_ prefixed)

4.8.1 Source half debug

Table 1.2.38: Source half debug

Signal	Direction	Width	Description
src_dbg_rd_all_complete	output	NC	-
src_dbg_r_bits_rcvd	output	32	-
src_dbg_sram_writes	output	32	-

Signal	Direction	Width	Description
src_dbg_arb_request	output	NC	-
src_dbg_src_sram_bridge_pending	output	NC	-
src_dbg_src_sram_bridge_output_valid	output	NC	-
src_dbg_axis_beats_sent	output	32	-
src_dbg_axis_packets_sent	output	32	-

4.8.2 Sink half debug

Table 1.2.39: Sink half debug

Signal	Direction	Width	Description
snk_dbg_snk_sram_bridge_pending	output	NC	-
snk_dbg_snk_sram_bridge_output_valid	output	NC	-
snk_dbg_axis_beats_received	output	32	-
snk_dbg_axis_packets_received	output	32	-

4.8.3 Active-channel sideband for per-channel bus instrumentation

(*axi_bus_meter*). The W bus carries no width, so the write engine's channel index must travel out of band to reach the meter at the top.

Table 1.2.40: Active-channel sideband for per-channel bus instrumentation

Signal	Direction	Width	Description
snk_active_channel_id	output	CIW	-
snk_active_channel_valid	output	1	-

Verified: every row above is a declared port of `rapids_core_beats` and every declared port has a row (324 of 324), generated from the module declaration on 2026-09-28. Regenerate rather than hand-edit when the interface changes.

5 Clocks and Reset

Module: `rapids_core_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

5.1 Clock Domain

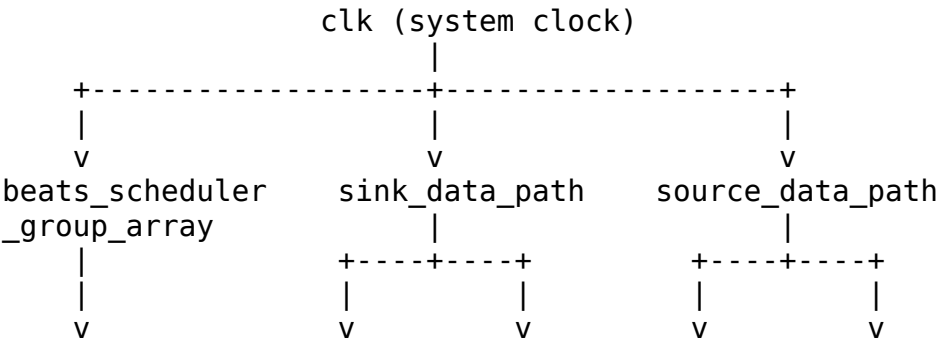
RAPIDS Beats operates in a single clock domain for simplified design and maximum throughput.

5.1.1 Clock Specifications

Table 1.3.1: Clock Specifications

Parameter	Specification
Clock Signal	clk (also <code>axi_aclk</code> in some submodules)
Frequency Range	100 - 500 MHz
Duty Cycle	45% - 55%
Clock Jitter	< 100 ps peak-to-peak

5.1.2 Clock Distribution



descriptor_engine	snk_sram_ctrl	axi_write_engine	axi_read_engine	src_sram_ctrl
-------------------	---------------	------------------	-----------------	---------------

5.2 Reset Specification

RAPIDS uses asynchronous assert, synchronous deassert reset methodology.

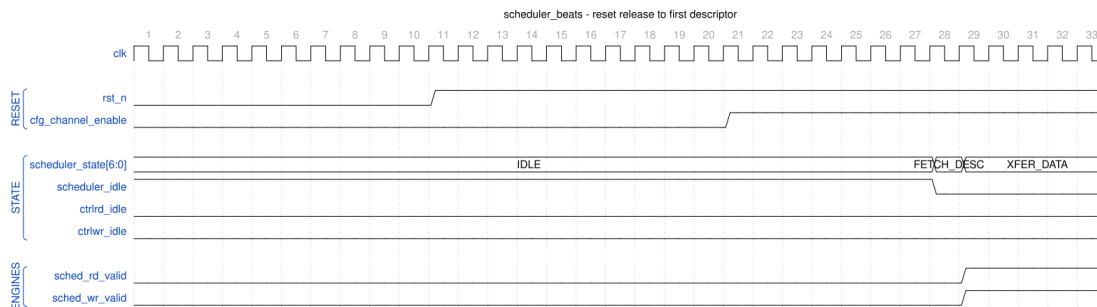
5.2.1 Reset Signal

Table 1.3.2: Reset Specifications

Parameter	Specification
Reset Signal	rst_n (also axi_aresetn in some submodules)
Polarity	Active-low
Assert	Asynchronous (immediate)
Deassert	Synchronous (on rising clock edge)
Minimum Duration	10 clock cycles

5.2.2 Reset Timing Diagram

5.2.3 Figure 1.3.1: Reset Timing



scheduler_beats - reset release to first descriptor

Source: [scheduler_reset.json](#), captured from `dv/tests/fub_beats/test_scheduler_beats.py` (basic transfer, the same run as Figure 2.1.3) with WAVES=1.

Reading it: `rst_n` is held low for ten cycles and released on a clock edge. Every state register is already at its reset value while reset is asserted – the FSM sits in IDLE, `scheduler_idle` is high, both control engines report idle and neither engine request is raised. Nothing changes at the

release edge itself; the scheduler stays in IDLE until the testbench raises `cfg_channel_enable` ten cycles later and a descriptor becomes available, at which point it steps through `FETCH_DESC` to `XFER_DATA` and raises the read and write requests together.

5.3 Reset Behavior

5.3.1 On Reset Assert

All modules initialize to known states:

Table 1.3.3: Reset States

Component	Reset State
Scheduler FSM	IDLE
Descriptor Engine	IDLE, FIFOs empty
AXI Read Engine	Idle, no outstanding transactions
AXI Write Engine	Idle, no outstanding transactions
SRAM Pointers	All zeros
Alloc/Drain Controllers	Empty, full space available
MonBus	No packets pending

5.3.2 Reset Sequence

1. External reset asserted (`rst_n = 0`)
 - All flip-flops async reset
 - All outputs driven to safe states
 - No AXI transactions
 2. System stabilization (minimum 10 cycles)
 - Clock running and stable
 - Configuration registers programmed (optional)
 3. Reset deasserted (`rst_n = 1`, sync to `clk` rising edge)
 - FSMs begin operation
 - Ready to accept descriptors/data
-

5.4 Configuration During Reset

Some configuration signals MUST be stable before reset deassertion:

Table 1.3.4: Configuration Timing Requirements

Signal	Requirement
src_cfg_channel_enable / snk_cfg_channel_enable	Stable before rst_n=1 (one per half)
src_cfg_sched_timeout_cycles / snk_cfg_sched_timeout_cycles	Stable before rst_n=1 (one per half)
cfg_axi_rd_xfer_beats	Stable before rst_n=1
cfg_axi_wr_xfer_beats	Stable before rst_n=1

5.5 Reset Macros

RAPIDS uses standardized reset macros from `reset_defs.svh`:

```
`include "reset_defs.svh"

// Standard reset pattern
`ALWAYS_FF_RST(clk, rst_n,
  if (`RST_ASSERTED(rst_n)) begin
    r_state <= IDLE;
    r_counter <= '0;
  end else begin
    r_state <= w_next_state;
    r_counter <= w_next_counter;
  end
)
```

5.6 Power-On Reset Considerations

- External reset should be held for minimum 10 clock cycles after clock is stable
 - Configuration registers should be programmed before or during reset
 - No AXI transactions until reset is deasserted and channels are enabled
-

6 Scheduler

Module: scheduler.sv **Location:** projects/components/dma-ip/rapids/rtl/fub/ **Status:** Implemented **Last Updated:** 2026-09-30

6.1 Overview

The scheduler coordinates one channel. It accepts a 256-bit descriptor from the descriptor engine, turns the descriptor's byte length and byte addresses into beat counts for the AXI engines, tells the data paths how the packet is laid out, and tracks progress until every source beat is read and every destination beat is committed.

Compared with the RAPIDS Beats scheduler, the state machine, the timeout logic, the control-descriptor path and the MonBus events are the same. Three things change: the beat math, the address advance, and the packet record.

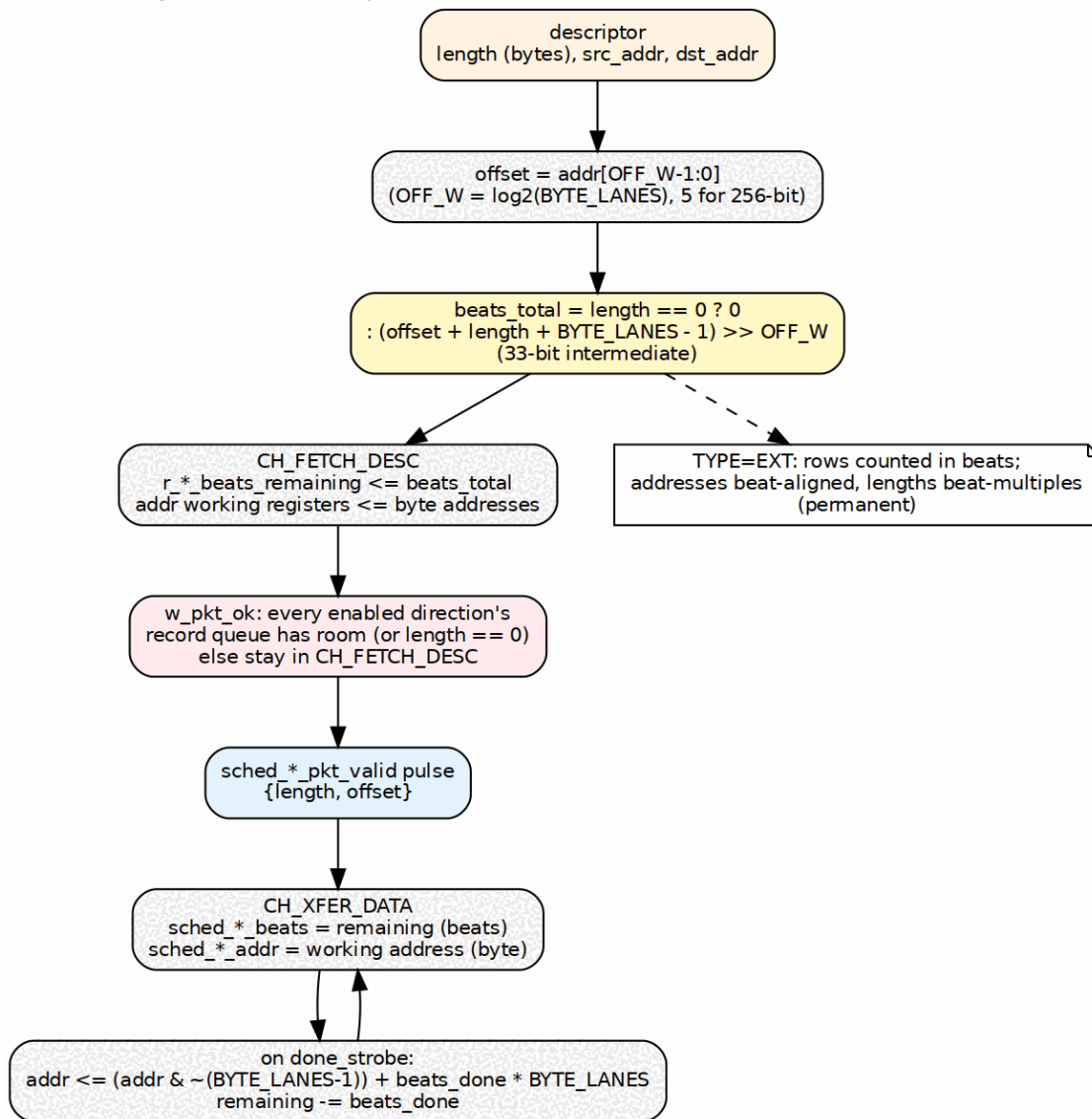
6.1.1 Key Features

- **Byte length:** the descriptor's 32-bit length is a byte count.
- **Byte addresses:** the offset of a transfer inside its first beat is `addr[OFF_W-1:0]`.
- **Beat math:** beats moved are derived from offset and length, in a 33-bit intermediate.
- **Packet record:** one {bytes, offset} pulse per DATA descriptor, per enabled direction, held back while the data path cannot take it.
- **Concurrent read and write:** both directions run inside one XFER_DATA state.
- **Recoverable write-progress timeout:** strikes escalate to a sticky error only when a limit is set.
- **Control descriptors:** CTRL_READ and CTRL_WRITE drive the control engines, unchanged.
- **Directional halves:** EN_READ and EN_WRITE select the source or sink half.

6.1.2 Block Diagram

6.1.3 Figure 2.1.1: Scheduler Byte Math

scheduler: bytes in, beats out (per enabled direction)



Scheduler byte math

Source: assets/graphviz/03_scheduler_byte_math.dot

6.2 Parameters

Table 2.1.1: Scheduler Parameters

Parameter	Type	Default	Description
CHANNEL_ID	int	0	Channel index of this scheduler
NUM_CHANNELS	int	8	Channel count
CHAN_WIDTH	int	$\lceil \log_2(\text{NUM_CHANNELS}) \rceil$	Channel field width
ADDR_WIDTH	int	64	Address width
DATA_WIDTH	int	512	Data width in bits; the board design uses 256
MON_AGENT_ID	logic [15:0]	16'h0040	MonBus agent ID
MON_UNIT_ID	logic [7:0]	8'h01	MonBus unit ID
MON_CHANNEL_ID	logic [8:0]	9'h000	MonBus base channel ID
DESC_WIDTH	int	256	Descriptor width; fixed at 256 (elaboration fails otherwise)
EN_READ	bit	1	Enable the read direction for DATA descriptors
EN_WRITE	bit	1	Enable the write direction for DATA descriptors
USE_ROW_COL_MAJOR_ADDRESSING	int	1	Enable TYPE=EXT run-base generators
BYTE_LANES	int	DATA_WIDTH/8	Bytes per beat
OFF_W	int	$\lceil \log_2(\text{BYTE_LANES}) \rceil$	Offset field width

BYTE_LANES and OFF_W are the two parameters that do not exist in the Beats scheduler. A directional half sets exactly one of EN_READ and EN_WRITE: the source half reads only, the sink half writes only.

6.3 Port List

6.3.1 Clock, Reset and Configuration

Table 2.1.2: Clock, Reset and Configuration Ports

Signal	Dir	Width	Description
clk	in	1	Clock
rst_n	in	1	Active-low reset
cfg_channel_enable	in	1	Channel enable; IDLE leaves only when set
cfg_channel_reset	in	1	Channel reset; forces IDLE and clears sticky state
cfg_sched_timeout_cycles	in	32	Write-progress timeout window in cycles
cfg_sched_timeout_limit	in	8	Consecutive windows before escalation; 0 = never
cfg_sched_timeout_enable	in	1	Enable timeout detection

6.3.2 Status

Table 2.1.3: Status Ports

Signal	Dir	Width	Description
scheduler_idle	out	1	High in CH_IDLE with no channel reset pending
scheduler_state	out	7	One-hot state (see Table 2.1.12)
sched_error	out	1	Sticky, high in CH_ERROR
dbg_descriptor_error	out	1	Latched fatal descriptor-level error
dbg_read_error_sticky	out	1	Latched read-engine error
dbg_write_error_sticky	out	1	Latched write-engine error
dbg_timeout_expired	out	1	Live: the timeout window has elapsed

6.3.3 Descriptor Engine Interface

Table 2.1.4: Descriptor Engine Interface

Signal	Dir	Width	Description
descriptor_valid	in	1	A descriptor is available
descriptor_ready	out	1	High in CH_IDLE and CH_NEXT_DESC
descriptor_packet	in	256	Descriptor (layout in Table 2.1.10)
descriptor_ext_packet	in	256	Extended chunk; used only when TYPE=EXT
descriptor_error	in	1	Descriptor fetch or parse error

6.3.4 Data Read Interface (to AXI Read Engine)

Table 2.1.5: Data Read Interface

Signal	Dir	Width	Description
sched_rd_valid	out	1	Read request
sched_rd_addr	out	ADDR_WIDTH	Source byte address (working register)
sched_rd_beats	out	32	Beats remaining to read
sched_rd_pkt_valid	out	1	Packet record pulse to the source egress
sched_rd_pkt_ready	in	1	Source egress record queue has room
sched_rd_pkt_bytes	out	32	Descriptor length in bytes
sched_rd_pkt_offset	out	OFF_W	Source byte offset inside the first beat

6.3.5 Data Write Interface (to AXI Write Engine)

Table 2.1.6: Data Write Interface

Signal	Dir	Width	Description
sched_wr_valid	out	1	Write request

Signal	Dir	Width	Description
sched_wr_ready	in	1	Write engine can take the channel
sched_wr_addr	out	ADDR_WIDTH	Destination byte address (working register)
sched_wr_beats	out	32	Beats remaining to issue
sched_wr_pkt_valid	out	1	Packet record pulse to the sink ingress
sched_wr_pkt_ready	in	1	Sink ingress record queue has room
sched_wr_pkt_bytes	out	32	Descriptor length in bytes
sched_wr_pkt_offset	out	OFF_W	Destination byte offset inside the first beat

6.3.6 Completion Interface

Table 2.1.7: Completion and Error Interface

Signal	Dir	Width	Description
sched_rd_done_strobe	in	1	A read burst completed
sched_rd_beats_done	in	32	Beats in that burst
sched_wr_done_strobe	in	1	A write burst was issued (AW handshake)
sched_wr_beats_done	in	32	Beats in that burst
sched_wr_commit_strobe	in	1	A write burst was committed (B response)
sched_wr_commit_beats	in	32	Beats in that burst
sched_rd_error	in	1	Read engine error
sched_wr_error	in	1	Write engine error, including the sink packet length error

6.3.7 Control Engine Interface

Table 2.1.8: Control Engine Interface

Signal	Dir	Width	Description
ctrlrd_valid	out	1	CTRL_READ request
ctrlrd_ready	in	1	Engine accepted the request
ctrlrd_addr	out	ADDR_WIDTH	Poll address (descriptor source slot)
ctrlrd_data	out	32	Expected value (dst_addr[31:0])
ctrlrd_mask	out	32	Compare mask (dst_addr[63:32])
ctrlrd_error	in	1	Engine error (retries exhausted or AXI error)
ctrlrd_idle	in	1	Engine idle
ctrlwr_valid	out	1	CTRL_WRITE request
ctrlwr_ready	in	1	Engine accepted the request
ctrlwr_addr	out	ADDR_WIDTH	Doorbell address (descriptor source slot)
ctrlwr_data	out	32	Doorbell data (dst_addr[31:0])
ctrlwr_error	in	1	Engine error
ctrlwr_idle	in	1	Engine idle

6.3.8 Monitor Bus Interface

Table 2.1.9: Monitor Bus Interface

Signal	Dir	Width	Description
i_mon_time	in	timestamp	Time base for packets
mon_valid	out	1	Packet valid
mon_ready	in	1	Consumer ready
mon_packet	out	128	Monitor packet
mon_timestamp	out	64	Side-band

Signal	Dir	Width	Description
			timestamp

The control interface tables do not change from RAPIDS Beats. The packet-record ports in Tables 2.1.5 and 2.1.6 are the new ports.

6.4 Descriptor Format

Table 2.1.10: Descriptor Layout

Bits	Field	Notes
[63:0]	src_addr	Byte address
[127:64]	dst_addr	Byte address
[159:128]	length	Bytes
[191:160]	next_descriptor_ptr	Chain pointer
[192]	valid	Must be 1
[193]	gen_irq	Emit an IRQ event on completion
[194]	last	End of chain
[195]	error	Error flag
[199:196]	channel_id	
[207:200]	priority	
[209:208]	opcode	DATA = 0, CTRL_READ = 1, CTRL_WRITE = 2
[212:210]	type	LEGACY = 0, EXT = 1

For CTRL_READ and CTRL_WRITE the address and data slots are reinterpreted as in Tables 2.1.8; the length field does not describe a byte count and no packet record is issued.

6.5 Byte-to-Beat Math

For each enabled direction the scheduler forms the offset from the low OFF_W bits of that side's address and derives the number of memory beats the transfer touches.

```

w_rd_off      = src_addr[OFF_W-1:0]
w_wr_off      = dst_addr[OFF_W-1:0]
w_rd_span     = 33'(w_rd_off) + 33'(length) + 33'(BYTE_LANES-1)
w_wr_span     = 33'(w_wr_off) + 33'(length) + 33'(BYTE_LANES-1)
w_rd_beats_total = (length == 0) ? 0 : 32'(w_rd_span >> OFF_W)
w_wr_beats_total = (length == 0) ? 0 : 32'(w_wr_span >> OFF_W)

```

The span is 33 bits wide so a length near 4 GiB does not overflow. The read and write beat totals are computed separately because the two addresses can have different offsets; a 60-byte transfer read from offset 5 and written at offset 0 touches three source beats and two destination beats on a 32-byte design.

Table 2.1.11: Beat Totals for a 32-byte Beat

Length (bytes)	Offset	Beats
0	any	0
1	0	1
32	0	1
32	1	2
60	5	3
60	0	2
64	0	2

The values load in CH_FETCH_DESC:

- `r_beats_remaining` loads `EN_READ ? rd_total : wr_total`.
- `r_read_beats_remaining` loads `EN_READ ? rd_total : 0`.
- `r_write_beats_remaining` and `r_write_beats_to_commit` load `EN_WRITE ? wr_total : 0`.

A disabled direction therefore loads zero, so its valid never asserts and its completion term is true from the start. Completion collapses onto the enabled direction.

For TYPE=EXT the run counters load `min(run_size, beats_total)` instead of the whole total, and `stream_run_addr_gen.cfg_total_beats` takes the beats total. TYPE=EXT is beat-aligned by design: addresses are aligned and lengths are beat multiples, so the offset is zero and the math reduces to the Beats result.

6.6 Address Advance

The working address registers start at the descriptor's byte addresses. On each done strobe the address moves forward by the beats reported:

```
r_src_addr <= {r_src_addr[AW-1:OFF_W], {OFF_W{1'b0}}} +  
(AW'(sched_rd_beats_done) << OFF_W) // on sched_rd_done_strobe  
r_dst_addr <= {r_dst_addr[AW-1:OFF_W], {OFF_W{1'b0}}} +  
(AW'(sched_wr_beats_done) << OFF_W) // on sched_wr_done_strobe
```

The low bits are cleared before the add. Only the first burst of a descriptor starts inside a beat; every later burst starts on a beat boundary, so aligning down first is correct for both. The write address advances on the AW handshake (issue), not on the B response. The B response decrements `r_write_beats_to_commit` instead.

The scheduler reports the working address to the engines unmodified. The engines align it (see [AXI Read Engine](#)); the offset reaches the data path through the packet record, not through the address bus.

6.7 Packet Records

A DATA descriptor with a non-zero length causes one record pulse per enabled direction, on the cycle the scheduler leaves CH_FETCH_DESC:

```
w_pkt_ok = !w_is_data || length == 0 ||  
           ((!EN_READ || sched_rd_pkt_ready) &&  
            (!EN_WRITE || sched_wr_pkt_ready))  
  
sched_{rd,wr}_pkt_valid = w_state_fetch_desc && r_descriptor.valid &&  
w_is_data &&  
                           EN_{READ,WRITE} && w_pkt_ok && (length != 0)  
sched_{rd,wr}_pkt_bytes = r_descriptor.length  
sched_{rd,wr}_pkt_offset = w_{rd,wr}_off
```

Both directions share `w_pkt_ok`, so each enabled direction sees exactly one pulse. `w_pkt_ok` also gates the FETCH_DESC to XFER_DATA transition: while a record queue is full, the scheduler holds in CH_FETCH_DESC and the pulse does not fire. Control descriptors and zero-length transfers carry no record. See [Packet Record Interface](#).

6.8 Finite State Machine

The state is one-hot and is `rapids_pkg::channel_state_t`.

Table 2.1.12: State Encoding

State	Encoding	Description
CH_IDLE	7'b0000001	Waiting for a descriptor and <code>cfg_channel_enable</code>
CH_FETCH_DESC	7'b0000010	Descriptor latched; working registers and counters load; record pulses
CH_XFER_DATA	7'b0000100	Read and write engines run concurrently; control ops issue
CH_COMPLETE	7'b0001000	Descriptor done; chain decision
CH_NEXT_DESC	7'b0010000	Waiting for the chained descriptor
CH_ERROR	7'b0100000	Sticky; only channel reset leaves it
CH_RESERVED	7'b1000000	Unused

Transition priority is: registered channel reset to IDLE, then a hard error or timeout escalation to ERROR, then the normal case.

Table 2.1.13: Transitions

From	To	Condition
IDLE	FETCH_DESC	<code>descriptor_valid</code> && <code>cfg_channel_enable</code>
FETCH_DESC	XFER_DATA	<code>r_descriptor.valid</code> && <code>w_pkt_ok</code>
FETCH_DESC	FETCH_DESC	<code>r_descriptor.valid</code> and <code>w_pkt_ok</code> false (record queue full)
FETCH_DESC	ERROR	descriptor valid bit clear
XFER_DATA	COMPLETE	<code>w_exec_complete</code>
COMPLETE	NEXT_DESC	<code>next_descriptor_ptr != 0</code> && !last
COMPLETE	IDLE	otherwise
NEXT_DESC	FETCH_DESC	<code>descriptor_valid</code>
ERROR	ERROR	always; channel reset is the only exit
any other	ERROR	default

`w_exec_complete` is `w_transfer_complete` for a DATA descriptor and `w_ctrl_complete` for a control descriptor. `w_transfer_complete` is the read remaining count reaching zero together

with the commit remaining count reaching zero. `w_ctrl_complete` is the control request having been accepted and the control engine having returned to idle.

`scheduler_idle` is `CH_IDLE && !r_channel_reset_active`. `sched_error` is the ERROR state.

6.9 Operation

6.9.1 Data Request Outputs

`sched_rd_valid` is high in `XFER_DATA` for a DATA descriptor while `!w_read_complete`, `!w_sched_rd_completing_this_cycle` and, for `TYPE=EXT`, `!w_rd_need_base`.

`sched_rd_beats` is the read remaining count, or for `EXT` the current run's remainder.

`sched_wr_valid` is high in `XFER_DATA` for a DATA descriptor while `r_write_beats_remaining != 0`, the commit count has not reached zero, `!w_sched_wr_completing_this_cycle` and `!w_wr_need_base`.

The look-ahead terms drop the valid on the same cycle the last done strobe arrives. Without them the engine would see a still-asserted valid for one more cycle and issue an extra burst.

6.9.2 Write Completion

The write direction has two counters. `r_write_beats_remaining` decrements on the AW done strobe and drives burst sizing and the destination address. `r_write_beats_to_commit` decrements on the B commit strobe and gates `w_write_complete`. The channel therefore completes only after every issued burst has been acknowledged.

6.9.3 Control Descriptors

`CTRL_READ` and `CTRL_WRITE` do not use the data engines. `ctrlrd_valid` and `ctrlwr_valid` assert in `XFER_DATA` until the engine accepts (`r_ctrl_issued`), and completion is the engine returning to idle after acceptance.

6.10 Error Handling

`w_hard_error` is the OR of:

- `descriptor_error`, `sched_rd_error`, `sched_wr_error`
- the sticky read and write error flags
- `ctrlrd_error` while the descriptor is `CTRL_READ`
- `ctrlwr_error` while the descriptor is `CTRL_WRITE`

A hard error, or a timeout escalation, moves the FSM to CH_ERROR. The sink packet length check reports through sched_wr_error (see [Sink Data Path AXIS](#)), so a stream packet whose byte count differs from the descriptor length is a hard error for that channel. Sticky read and write flags and the descriptor error clear in CH_IDLE.

6.11 Recoverable Write-Progress Timeout

The timeout counter increments while sched_wr_valid && !sched_wr_ready. It resets on sched_wr_done_strobe or sched_wr_commit_strobe, and also when the condition is false. When the counter reaches cfg_sched_timeout_cycles with cfg_sched_timeout_enable set, w_timeout_expired pulses and the counter re-arms.

Each expiry adds one strike, in an 8-bit counter that saturates. Strikes clear on write progress, in CH_IDLE, and on channel reset. Escalation is:

```
w_timeout_escalate = (cfg_sched_timeout_limit != 0) && (strikes >=
cfg_sched_timeout_limit)
```

A limit of zero never escalates: the timeout is then purely informational.

6.12 MonBus Events

Packets are PktTypeCompletion from PROTOCOL_CORE unless noted.

Table 2.1.14: MonBus Events

State	Event	Data
CH_FETCH_DESC	RAPIDS_EVENT_DESC_START	descriptor length (bytes)
CH_COMPLETE	RAPIDS_EVENT_IRQ when gen_irq, else RAPIDS_EVENT_DESC_COMPL ETE	
CH_ERROR	RAPIDS_EVENT_ERROR (PktTypeError), once per error episode	{29'h0, wr_sticky, rd_sticky, 33'h0}

There are no events during XFER_DATA. The only difference from RAPIDS Beats is that the DESC_START data field is a byte count.

6.13 Waveforms

6.13.0.1 Waveform 2.1: Byte Descriptor Start

The beat-count arithmetic and the record pulse are combinational functions of the latched descriptor, so the start of a descriptor is short:

cycle	:	0	1	2	3	
state	:	IDLE	FETCH	XFER	XFER	
descriptor_valid	:	1	0	0	0	
sched_wr_pkt_ready	:	1	1	1	1	
sched_wr_pkt_valid	:	0	1	0	0	(one pulse, the last FETCH cycle)
r_write_beats_remaining	:	0	0	N	N	(loads at the end of FETCH)
sched_wr_valid	:	0	0	1	1	

If sched_wr_pkt_ready is low in FETCH, the state holds and the pulse is delayed until it rises.

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

7 Descriptor Engine Specification

Module: descriptor_engine_beats.sv **Location:** projects/components/dma-ip/rapids/rtl/fub_beats/ **Status:** Implemented **Last Updated:** 2025-01-10

7.1 Overview

The Descriptor Engine fetches and manages 256-bit descriptors from memory via AXI4. It maintains a prefetch FIFO to hide memory latency and provides parsed descriptors to the scheduler.

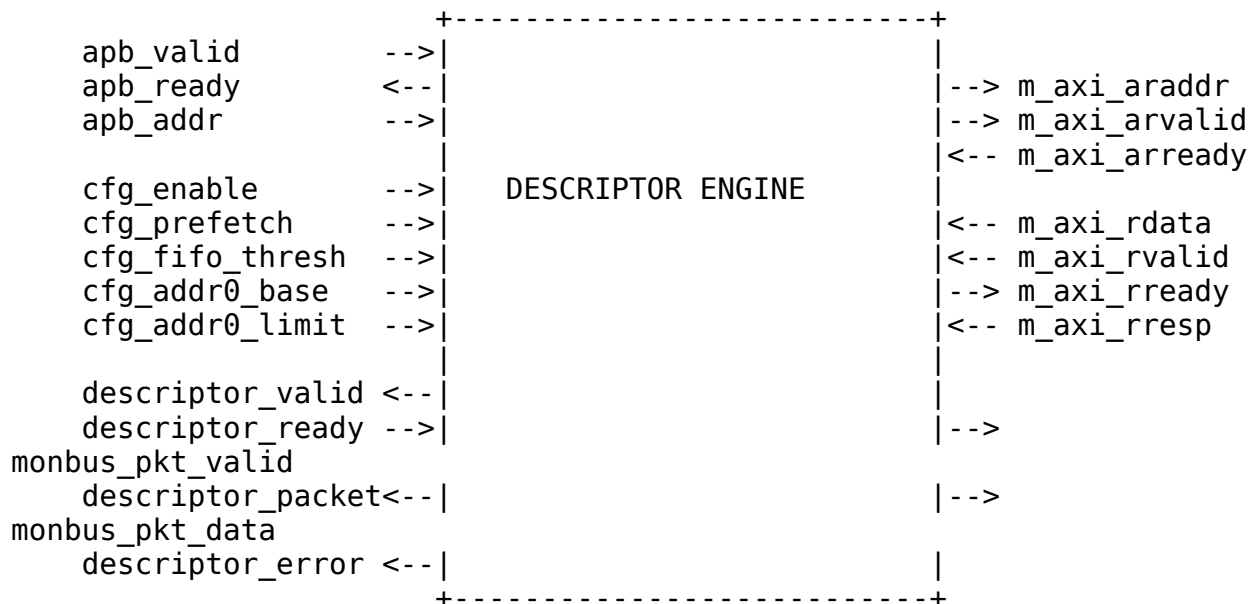
7.1.1 Key Features

- **AXI4 Descriptor Fetch:** Single-beat 256-bit reads
- **Prefetch FIFO:** Configurable depth for latency hiding
- **Address Range Checking:** Validates descriptor addresses
- **Error Detection:** Address violations, AXI errors

- **MonBus Integration:** Fetch events and error reporting

7.1.2 Block Diagram

7.1.3 Figure 2.2.1: Descriptor Engine Block Diagram



7.2 Parameters

```

parameter int CHANNEL_ID = 0;           // Channel identifier
parameter int NUM_CHANNELS = 8;         // Total channels
parameter int ADDR_WIDTH = 64;         // Address bus width
parameter int FIFO_DEPTH = 4;          // Prefetch FIFO
depth

// Monitor Bus Parameters
parameter logic [7:0] MON_AGENT_ID = 8'h10; // Descriptor Engine
Agent ID
parameter logic [3:0] MON_UNIT_ID = 4'h1;   // Unit identifier
  
```

: Table 2.2.1: Descriptor Engine Parameters

7.3 Port List

7.3.1 Clock and Reset

Table 2.2.2: Clock and Reset

Signal	Direction	Width	Description
clk	input	1	System clock
rst_n	input	1	Active-low asynchronous reset

7.3.2 APB Programming Interface

Table 2.2.3: APB Programming Interface

Signal	Direction	Width	Description
apb_valid	input	1	Kick-off valid (start descriptor chain)
apb_ready	output	1	Ready to accept kick-off
apb_addr	input	AW	First descriptor address

7.3.3 Configuration Interface

Table 2.2.4: Configuration Interface

Signal	Direction	Width	Description
cfg_prefetch_enable	input	1	Enable prefetch chaining (0 = on-demand, ~1 ahead)
cfg_fifo_threshold	input	4	Prefetch depth (descriptors buffered ahead when enabled)
cfg_addr0_base	input	AW	Valid address range 0 base
cfg_addr0_limit	input	AW	Valid address range 0 limit

Signal	Direction	Width	Description
cfg_addr1_base	input	AW	Valid address range 1 base
cfg_addr1_limit	input	AW	Valid address range 1 limit

7.3.4 AXI4 Read Master Interface

Table 2.2.5: AXI4 Read Master Interface

Signal	Direction	Width	Description
ar_valid	output	1	Address valid
ar_ready	input	1	Address ready
ar_addr	output	ADDR_WIDTH	Descriptor fetch address
ar_len	output	8	Burst length - 1
ar_size	output	3	Burst size (log2 bytes)
ar_burst	output	2	Burst type (INCR)
ar_id	output	AXI_ID_WIDTH	Transaction ID
ar_lock	output	1	Lock type
ar_cache	output	4	Cache attributes
ar_prot	output	3	Protection attributes
ar_qos	output	4	Quality of service
ar_region	output	4	Region identifier
r_valid	input	1	Data valid
r_ready	output	1	Data ready
r_data	input	256	Descriptor payload (fixed 256-bit)
r_resp	input	2	Read response

Signal	Direction	Width	Description
r_last	input	1	Last beat
r_id	input	AXI_ID_WIDTH	Response ID

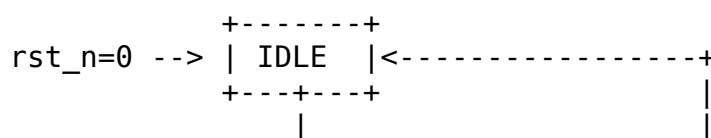
7.3.5 Scheduler Interface

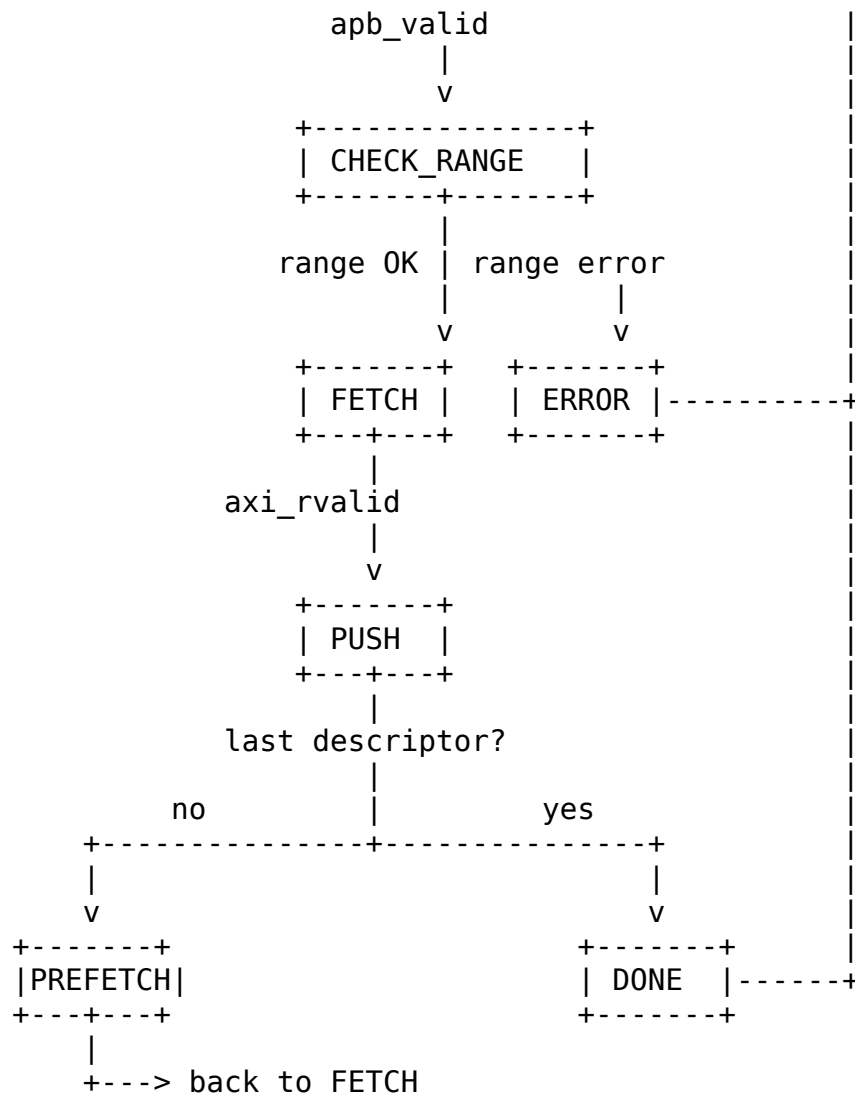
Table 2.2.6: Scheduler Interface

Signal	Direction	Width	Description
descriptor_valid	output	1	Descriptor valid
descriptor_ready	input	1	Scheduler ready
descriptor_packet	output	256	Parsed descriptor (fixed 256-bit)
descriptor_ext_packet	output	256	Extended chunk 1 (0 when unused)
descriptor_error	output	1	Error flag
descriptor_end_of_stream	output	1	End of Stream
descriptor_end_of_line	output	1	End of Line
descriptor_end_of_data	output	1	End of Data
descriptor_type	output	2	Packet type
channel_idle	input	1	Scheduler idle (enables APB kick-off)
descriptor_engine_idle	output	1	Engine idle

7.4 FSM States

7.4.1 Figure 2.2.2: Descriptor Engine FSM

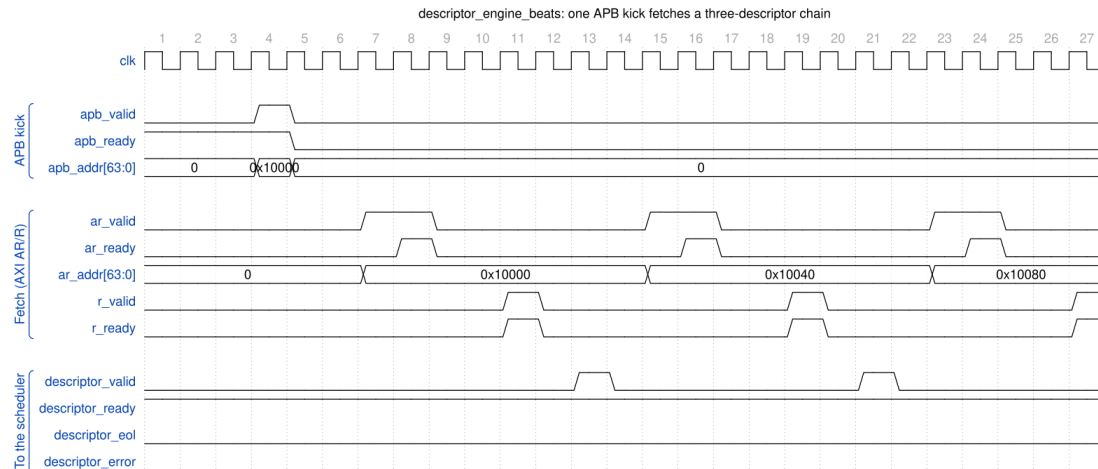




7.5 Operation

7.5.1 Descriptor Chain Processing

7.5.2 Figure 2.2.3: Descriptor Chain Timing



Descriptor Engine - three-descriptor chain

Source: [descriptor_engine_chain.json](#), captured from `dv/tests/fub_beats/test_descriptor_engine_beats.py` (descriptor_chaining, 32 channels, seed 7) with WAVES=1.

Reading it: one apb_valid kick carries the head address (0x10000). The engine issues an AXI read for it (ar_valid, ar_addr), the data returns on r_valid, and the parsed descriptor is presented on descriptor_valid to a ready scheduler. Because the descriptor's next_ptr points at 0x10040 and last is clear, the engine fetches the next one on its own – and the one after that at 0x10080 – with no further APB traffic; descriptor_eol marks the last of the chain. Each fetch costs the same AR -> R -> present latency, so a chain streams at one descriptor per fetch round trip.

7.6 Address Range Checking

Descriptors must fall within configured valid address ranges:

```
Valid if: (addr >= cfg_addr0_base && addr <= cfg_addr0_limit)
          || (addr >= cfg_addr1_base && addr <= cfg_addr1_limit)
```

Invalid addresses generate an error and halt the descriptor chain.

7.7 Prefetch and Autonomous Chaining

`cfg_prefetch_enable` and `cfg_fifo_threshold` are functional prefetch controls (previously declared-but-unused, which left chaining permanently greedy). They throttle how far ahead the engine fetches chained descriptors by comparing the output descriptor FIFO occupancy (`w_desc_fifo_count`) against a computed `w_prefetch_limit`:

```
if      (!cfg_prefetch_enable)      w_prefetch_limit = 1;    // on-
demand
else if (cfg_fifo_threshold == 0)    w_prefetch_limit = 1;    // guard
(0 would stall)
else                                  w_prefetch_limit =
cfg_fifo_threshold;
```

```
w_prefetch_allows = (w_desc_fifo_count < w_prefetch_limit);
```

- **Prefetch OFF (`cfg_prefetch_enable = 0`):** limit forced to 1, so a chained fetch is only allowed when the output FIFO has drained – the engine stays roughly one descriptor ahead of the scheduler (on-demand).
- **Prefetch ON:** the engine buffers up to `cfg_fifo_threshold` descriptors ahead.

7.7.1 Chaining Decision

A chain fetch issues immediately when all of the following hold:

```
w_should_chain = w_chain_eligible    // valid non-last descriptor, in
range, no error
                && w_desc_committed    // current descriptor accepted
into output FIFO
                && w_prefetch_allows    // occupancy below prefetch
limit
                && w_desc_addr_fifo_wr_ready; // address FIFO has a
slot
```

7.7.2 Deferred (Throttle-Blocked) Chain Fetch

If a descriptor commits and is chain-eligible but the fetch cannot issue this cycle (throttled by `w_prefetch_allows`, or the address FIFO is full), the owed fetch is recorded rather than dropped:

- `r_chain_pending` is set and the next-descriptor pointer is held in `r_pending_chain_addr`.
- The deferred fetch is reissued (`w_pending_push_fire`) once the output FIFO drains enough that `w_prefetch_allows` is true again and the address FIFO has a slot, clearing `r_chain_pending`.

Last Updated: 2026-07-02

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

8 AXI Read Engine

Module: `axi_read_engine.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub/`
Status: Implemented **Last Updated:** 2026-09-30

8.1 Overview

The AXI read engine turns per-channel read requests from the schedulers into AXI4 read bursts and routes the returning data to the SRAM controller. It is a streaming pipeline with no state machine. The channel identity travels in the AXI ID.

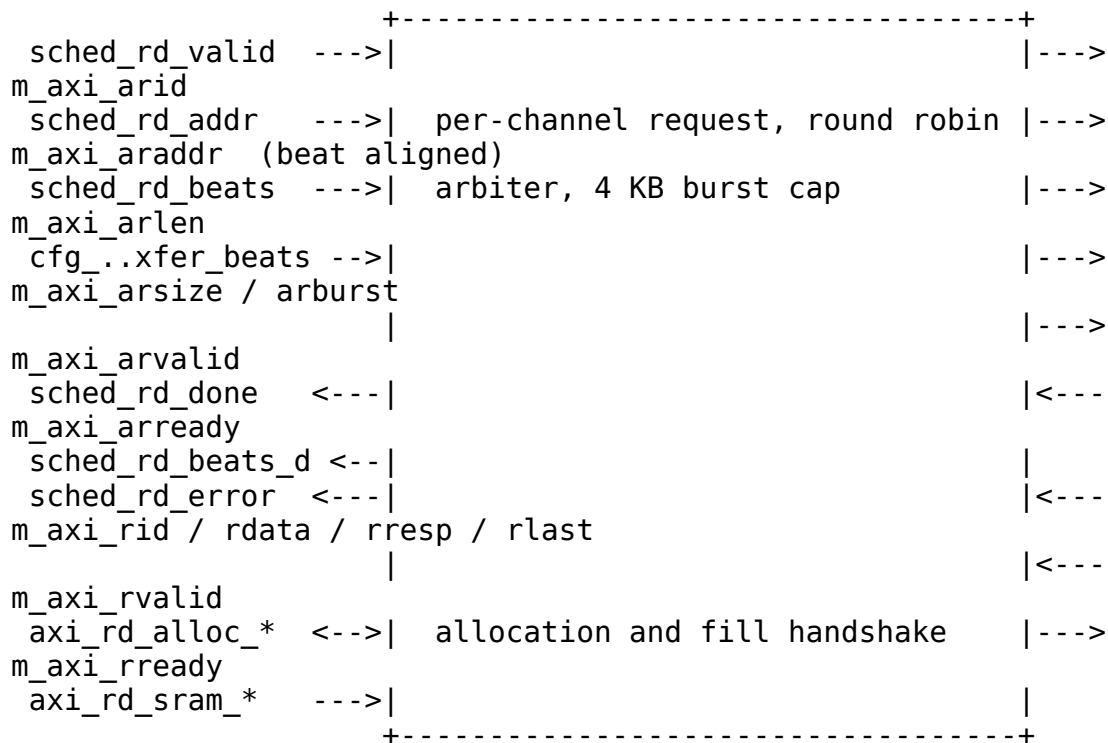
The engine is beat based, as in RAPIDS Beats. The scheduler has already converted the descriptor's byte length into a beat count. Two things change for byte-granular RAPIDS: the engine must cope with a channel address that is not beat aligned, and it must never let a burst cross a 4 KB boundary.

8.1.1 Key Features

- **Streaming pipeline:** no FSM, round-robin arbitration across channels.
- **Aligned AR address:** ARADDR is the channel address rounded down to a beat boundary. The lanes below the offset are read and ignored by the data path.
- **4 KB burst cap:** the burst length is limited so that no burst crosses a 4 KB boundary, whatever the start address.
- **Configurable burst length:** `cfg_axi_rd_xfer_beats`, clamped to what the buffer can hold (rapids BUG-009).
- **Space-checked:** a channel is only granted when the SRAM controller has room for the whole burst.
- **Sticky error:** SLVERR or DECERR on R sets `sched_rd_error` for the channel.

8.1.2 Block Diagram

8.1.3 Figure 2.3.1: AXI Read Engine Block Diagram



8.2 Parameters

Table 2.3.1: AXI Read Engine Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Number of channels. Channel id is carried in the AXI ID.
ADDR_WIDTH	64	AXI address width.
DATA_WIDTH	512	AXI data width in bits. The Genesys 2 build uses 256.
ID_WIDTH	8	AXI ID width.
SEG_COUNT_WIDTH	8	Width of the per-channel free-space count from the SRAM controller. Sets the burst clamp.

Parameter	Default	Description
PIPELINE	1	1 allows the next AR before the previous burst's data has returned. 0 waits for the burst to complete.
AR_MAX_OUTSTANDING	8	Maximum outstanding AR transactions.
STROBE_EVERY_BEAT	0	Declared for compatibility with STREAM. The completion strobe in this engine is generated on the AR handshake regardless of its value.

The module also accepts the short aliases NC, AW, DW, IW, SCW and CIW (the channel id width, $\$clog2(NUM_CHANNELS)$).

8.2.1 Derived Constants

Table 2.3.2: AXI Read Engine Derived Constants

Constant	Definition	Meaning
BYTES_PER_BEAT	DW / 8	Bytes per beat.
AXSIZE	$\$clog2(BYTES_PER_BEAT)$	Value driven on ARSIZE. Full-width beats.
SD_BEATS	$1 \ll (SCW - 1)$	Buffer depth in beats implied by SEG_COUNT_WIDTH.
XFER_MAX	$SD_BEATS < 256 ? SD_BEATS - 1 : 254$	Largest burst length value the engine will use.

8.3 Port List

8.3.1 Clock, Reset and Configuration

Table 2.3.3: Clock, Reset and Configuration

Signal	Direction	Width	Description
clk	input	1	Clock.
rst_n	input	1	Active-low reset.
cfg_axi_rd_xfer_beats	input	8	Configured burst length (ARLEN value, 0 to 255). Clamped to XFER_MAX.

8.3.2 Scheduler Interface

Table 2.3.4: Scheduler Interface

Signal	Direction	Width	Description
sched_rd_val_id	input	NC	Channel has a read request.
sched_rd_addr	input	NC x AW	Working source byte address.
sched_rd_beats	input	NC x 32	Beats still to read for the channel.
sched_rd_done_strobe	output	NC	Registered pulse, one cycle after the AR handshake for the channel.
sched_rd_beats_done	output	NC x 32	Beats issued by that AR (ARLEN + 1).
sched_rd_error	output	NC	Sticky read error for the channel, set by any non-OKAY R response.

The address is a byte address and may carry an offset in its low bits. The scheduler advances it to the next beat boundary after each done strobe, so only the first burst of a descriptor can see a non-zero offset.

8.3.3 SRAM Allocation and Fill Interface

Table 2.3.5: SRAM Allocation and Fill Interface

Signal	Direction	Width	Description
axi_rd_alloc_req	output	1	Registered pulse, one cycle after the AR handshake, reserving space for the burst.
axi_rd_alloc_size	output	8	Beats to reserve.
axi_rd_alloc_id	output	IW	Channel that owns the reservation.
axi_rd_alloc_space_free	input	NC x SCW	Free space per channel. A channel is granted only if it fits the burst.
axi_rd_sram_valid	output	1	Follows m_axi_rvalid.
axi_rd_sram_ready	input	1	SRAM controller accepts the beat. Drives m_axi_rready.
axi_rd_sram_id	output	IW	Follows m_axi_rid.
axi_rd_sram_data	output	DW	Follows m_axi_rdata.

8.3.4 AXI4 Read Master Interface

Table 2.3.6: AXI4 Read Master Interface

Signal	Direction	Width	Description
m_axi_arid	output	IW	Read address ID (the channel).
m_axi_araddr	output	AW	Beat-aligned read address.
m_axi_arlen	output	8	Burst length minus one.
m_axi_arsize	output	3	AXSIZE, full-width beats.

Signal	Direction	Width	Description
m_axi_arburst	output	2	INCR.
m_axi_arvalid	output	1	Address valid.
m_axi_arready	input	1	Address ready.
m_axi_rid	input	IW	Response ID.
m_axi_rdata	input	DW	Read data.
m_axi_rresp	input	2	Read response.
m_axi_rlast	input	1	Last beat of the burst.
m_axi_rvalid	input	1	Data valid.
m_axi_rready	output	1	Data ready.

8.3.5 Debug

Table 2.3.7: Debug Outputs

Signal	Direction	Width	Description
dbg_rd_all_complete	output	NC	Channel has no outstanding reads.
dbg_r_beats_rcvd	output	32	Total R beats received.
dbg_sram_writes	output	32	Total beats written to the SRAM controller.
dbg_arb_request	output	NC	Arbiter request vector.

8.4 Operation

8.4.1 Burst Length Cap (rapids BUG-009)

cfg_axi_rd_xfer_beats is an ARLEN value, but a burst must fit in the buffer that stages it. The engine clamps the configured value to XFER_MAX, giving the working limit w_xfer_cfg. Without the clamp an ARLEN of 255 wraps the 8-bit ARLEN + 1 and a channel is granted with no space reserved. This is the same fix as in RAPIDS Beats.

8.4.2 Byte-Granular Burst Sizing

The engine sizes each burst from the channel's current byte address and remaining beats. Three steps produce ARLEN:

1. Distance to the next 4 KB boundary, in beats: $w_beats_to_4k = (4096 - \{sched_rd_addr[11:AXSIZE], AXSIZE'b0\}) \gg AXSIZE$. The offset bits are dropped first, so the distance is measured from the aligned beat.
2. Burst cap: $w_cap_beats = \min(w_xfer_cfg + 1, w_beats_to_4k)$.
3. Burst length: $w_transfer_size = (sched_rd_beats \leq w_cap_beats) ? sched_rd_beats - 1 : w_cap_beats - 1$.

m_axi_araddr is $\{sched_rd_addr[grant][AW-1:AXSIZE], AXSIZE'b0\}$. The offset never reaches the bus. The first beat of the burst carries the bytes before the offset, and the data path discards them.

Because the cap always rounds toward the boundary, a burst can be shorter than the configured length, and a later burst restarts on the boundary. The scheduler needs no knowledge of this. It only sees `beats_done` per burst.

8.4.3 Streaming Pipeline

An arbiter picks a channel that has `sched_rd_valid` set and enough `axi_rd_alloc_space_free` for its burst. When the AR handshakes, the engine pulses `axi_rd_alloc_req` one cycle later with the burst size so the SRAM controller reserves the space, and pulses `sched_rd_done_strobe` for the channel. R beats pass straight through: `axi_rd_sram_valid`, `axi_rd_sram_id` and `axi_rd_sram_data` follow the R channel, and `m_axi_rready` is `axi_rd_sram_ready`.

With `PIPELINE = 1` the next AR for a channel can issue while earlier data is still in flight. With `PIPELINE = 0` the channel waits for the burst to finish. The top and core default to 1.

8.4.4 Figure 2.3.2: Burst Segmentation with a Mid-Beat Start

The example uses 32-byte beats (`OFF_W 5`) and a configured burst of 16 beats. A descriptor reads 8,200 bytes from address `0xFF8`. The scheduler computes offset `0x18` (24) and `beats_total = (24 + 8200 + 31) >> 5 = 257`.

Table 2.3.8: Burst Segmentation Example

Burst	Channel address	ARADDR	Beats to 4 KB	ARLEN	Beats left after
1	0x0FF8	0x0FE0	1	0	256
2	0x1000	0x1000	128	15	240
3	0x1200	0x1200	112	15	224
...				15	continues

The first burst is one beat because the aligned start is 32 bytes below the 4 KB boundary. From then on each burst is the configured 16 beats until the remainder runs out. The scheduler advances its address to the next beat boundary after each done strobe, which is why burst 2 starts at `0x1000`, not `0x0FF8` plus a beat.

8.4.5 Done Strobe

The engine pulses `sched_rd_done_strobe[ch]` once per AR handshake, with `sched_rd_beats_done[ch]` equal to `ARLEN + 1`. The scheduler subtracts this from its remaining count and advances the address. The strobe therefore reports beats issued, not beats returned.

8.5 Error Handling

Table 2.3.9: Error Handling

Error	Detection	Response
AXI SLVERR	<code>m_axi_rresp == 2'b10</code> on an accepted beat	Set sticky <code>sched_rd_error</code> for the channel
AXI DECERR	<code>m_axi_rresp == 2'b11</code> on an accepted beat	Set sticky <code>sched_rd_error</code> for the channel
Any other non-zero response	<code>m_axi_rresp != 2'b00</code> on an accepted beat	Same: the test is not-OKAY, and the channel is taken from RID

The scheduler turns the sticky error into its ERROR state and a MonBus event. The engine does not abort the burst: the remaining R beats still pass through, so the interconnect is not left mid-burst.

8.6 Differences from RAPIDS Beats

Table 2.3.10: Read Engine Delta

Item	RAPIDS Beats	RAPIDS
ARADDR	Channel address as given (always aligned)	Aligned down to the beat
Burst cap	Configured length and remaining beats	Also capped at the 4 KB boundary
Everything else		Unchanged

The beat-count chapter for the beats engine is [AXI Read Engine \(RAPIDS Beats\)](#). The scheduler that feeds this engine is described in [Scheduler](#).

Last Updated: 2026-09-30

9 AXI Write Engine

Module: `axi_write_engine.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub/`
Status: Implemented **Last Updated:** 2026-09-30

9.1 Overview

The AXI write engine drains data from the SRAM controller and issues AXI4 write bursts to system memory. It is a streaming pipeline with no state machine. It also tracks the write responses, which is how the scheduler learns that data has been committed.

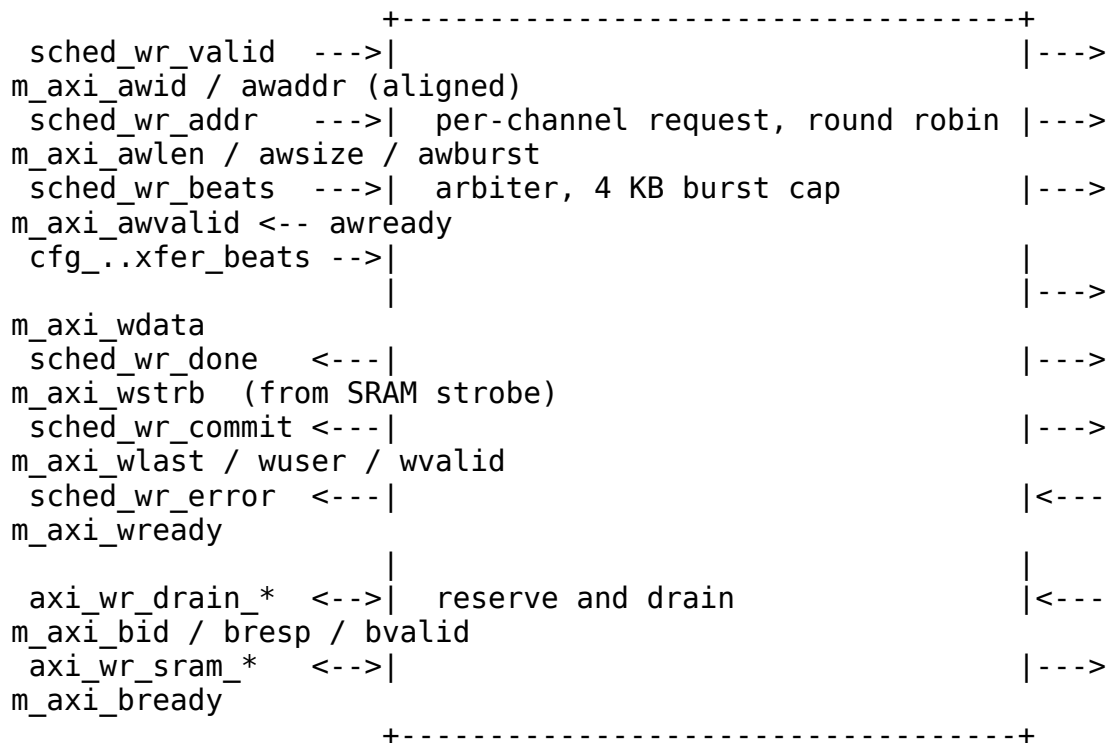
Byte-granular RAPIDS changes the engine in three places. Write strobes now come from the SRAM instead of being all ones, so a partial first or last beat writes only the bytes that belong to the packet. Burst lengths are capped at the 4 KB boundary. AWADDR is beat aligned.

9.1.1 Key Features

- **Streaming pipeline:** no FSM, round-robin arbitration across channels.
- **Real write strobes:** `m_axi_wstrb` is the strobe word stored beside each data beat (`axi_wr_sram_strb`). Lanes outside the packet are not written.
- **Aligned AW address:** AWADDR is the channel address rounded down to a beat boundary.
- **4 KB burst cap:** the same cap as the read engine, applied to both the burst length and the final-burst test.
- **Configurable burst length:** `cfg_axi_wr_xfer_beats`, clamped to what the buffer can hold (rapids BUG-009).
- **Two completion strobes:** done on the AW handshake, commit on the B response.
- **Sticky error:** a non-OKAY B response sets `sched_wr_error`.

9.1.2 Block Diagram

9.1.3 Figure 2.4.1: AXI Write Engine Block Diagram



9.2 Parameters

Table 2.4.1: AXI Write Engine Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Number of channels. Channel id is carried in the AXI ID.
ADDR_WIDTH	64	AXI address width.
DATA_WIDTH	512	AXI data width in bits. The Genesys 2 build uses 256.
ID_WIDTH	8	AXI ID width.
USER_WIDTH	8	Width of m_axi_wuser, which carries the active channel.
SEG_COUNT_WIDTH	8	Width of the per-channel available-data count from the

Parameter	Default	Description
PIPELINE	1	SRAM controller. Sets the burst clamp. 1 allows further AWs before earlier B responses return. 0 waits for the B.
AW_MAX_OUTSTANDING	8	Maximum outstanding AW transactions per channel.
W_PHASE_FIFO_DEPTH	64	Depth of the W-phase transaction FIFO, kept in order with the AW issue.
B_PHASE_FIFO_DEPTH	16	Depth of the B-phase transaction FIFO used to match responses.

The module also accepts the short aliases NC, AW, DW, IW, UW, SCW and CIW (the channel id width).

9.2.1 Derived Constants

Table 2.4.2: AXI Write Engine Derived Constants

Constant	Definition	Meaning
BYTES_PER_BEAT	DW / 8	Bytes per beat.
AXSIZE	$\$clog2(BYTES_PER_BEAT)$	Value driven on AWSIZE.
XFER_MAX	$\min(2^{(SCW-1)} - 1, 254)$	Largest burst length value the engine will use.

9.3 Port List

9.3.1 Clock, Reset and Configuration

Table 2.4.3: Clock, Reset and Configuration

Signal	Direction	Width	Description
clk	input	1	Clock.
rst_n	input	1	Active-low reset.

Signal	Direction	Width	Description
cfg_axi_wr_xfer_beats	input	8	Configured burst length (AWLEN value, 0 to 255). Clamped to XFER_MAX.

9.3.2 Scheduler Interface

Table 2.4.4: Scheduler Interface

Signal	Direction	Width	Description
sched_wr_val_id	input	NC	Channel has a write request.
sched_wr_ready	output	NC	Pulses when the B response of the last burst of a descriptor arrives.
sched_wr_addr	input	NC x AW	Working destination byte address.
sched_wr_beats	input	NC x 32	Beats still to write for the channel.
sched_wr_burst_len	input	NC x 8	Requested burst length. Present on the port; the engine sizes bursts from cfg_axi_wr_xfer_beats and the 4 KB cap.
sched_wr_done_strobe	output	NC	Pulses on the AW handshake for the channel.
sched_wr_beats_done	output	NC x 32	Beats issued by that AW (AWLEN + 1).
sched_wr_commit_strobe	output	NC	Pulses when the B response for the channel arrives.
sched_wr_commit_beats	output	NC x 32	Beats confirmed by that B response, recovered from the B-phase FIFO.
sched_wr_error	output	NC	Sticky write error for the channel.

The done strobe advances the scheduler's destination address, which lets the next AW issue without waiting for a response. The commit strobe is what the scheduler counts when it decides the descriptor is complete. Both strobes are per channel and report beat counts, not bytes.

9.3.3 SRAM Reservation and Drain Interface

Table 2.4.5: SRAM Drain Interface

Signal	Direction	Width	Description
axi_wr_drain_req	output	NC	Pulses for the granted channel on the AW handshake.
axi_wr_drain_size	output	NC x 8	Beats to reserve (AWLEN + 1).
axi_wr_drain_data_avail	input	NC x SCW	Data available per channel after reservations.
axi_wr_sram_valid	input	NC	Per-channel valid, registered. Used for arbitration.
axi_wr_sram_valid_comb	input	NC	Per-channel valid, combinational. Gates m_axi_wvalid.
axi_wr_sram_drain	output	1	Drain request. Equal to wvalid && wready.
axi_wr_sram_id	output	CIW	Channel selected for the drain.
axi_wr_sram_data	input	DW	Data from the selected channel.
axi_wr_sram_strb	input	DW/8	Byte strobes stored with that data beat. New in RAPIDS.

9.3.4 AXI4 Write Master Interface

Table 2.4.6: AXI4 Write Master Interface

Signal	Direction	Width	Description
m_axi_awid	output	IW	Write address ID (the

Signal	Direction	Width	Description
m_axi_awaddr	output	AW	channel). Beat-aligned write address.
m_axi_awlen	output	8	Burst length minus one.
m_axi_awsiz	output	3	AXSIZE, full-width beats.
m_axi_awburst	output	2	INCR.
m_axi_awvalid	output	1	Address valid.
m_axi_awready	input	1	Address ready.
m_axi_wdata	output	DW	Write data.
m_axi_wstrb	output	DW/8	Write strobes, equal to axi_wr_sram_strb.
m_axi_wlast	output	1	Last beat of the burst.
m_axi_wuser	output	UW	Channel of the beat being written.
m_axi_wvalid	output	1	Data valid.
m_axi_wready	input	1	Data ready.
m_axi_bid	input	IW	Response ID.
m_axi_bresp	input	2	Write response.
m_axi_bvalid	input	1	Response valid.
m_axi_bready	output	1	Response ready.

9.3.5 Debug and Status

Table 2.4.7: Debug and Status Outputs

Signal	Direction	Width	Description
dbg_wr_all_complete	output	NC	Channel has no outstanding writes.
dbg_aw_transactions	output	32	Total AW handshakes.
dbg_w_beats	output	32	Total W beats.
o_active_channel_id	output	CIW	Channel whose burst is driving, or about to

Signal	Direction	Width	Description
<code>o_active_channel_valid</code>	output	1	drive, the W bus. <code>o_active_channel_id</code> is meaningful.

9.4 Operation

9.4.1 Burst Length Cap (rapids BUG-009)

`cfg_axi_wr_xfer_beats` is an AWLEN value, but a burst cannot be larger than the SRAM that stages it: the engine waits for the whole burst to be present before issuing the AW, and a burst the buffer cannot hold would wait forever. The engine clamps the configured value to `XFER_MAX`. Without the clamp an AWLEN of 255 wraps the 8-bit `AWLEN + 1` and the engine issues an AW on an empty buffer. This is the same fix as in RAPIDS Beats.

9.4.2 Byte-Granular Burst Sizing

The sizing follows the read engine:

1. `w_beats_to_4k` is the distance from the aligned channel address to the next 4 KB boundary, in beats.
2. `w_cap_beats` = $\min(w_xfer_cfg + 1, w_beats_to_4k)$.
3. `w_transfer_size` (the AWLEN) is `sched_wr_beats - 1` if `sched_wr_beats` $\leq$ `w_cap_beats`, otherwise `w_cap_beats - 1`.

A channel may issue when the buffer holds at least `w_transfer_size + 1` beats (`w_has_data`), or when this is the final burst. The final-burst test is `sched_wr_beats > 0`, `sched_wr_beats` $\leq$ `w_cap_beats`, and the available data covers all remaining beats. The 4 KB cap therefore applies to both tests, so a descriptor that ends just past a boundary still finishes on a short final burst.

`m_axi_awaddr` is `{sched_wr_addr[ch][AW-1:AXSIZE], AXSIZE'b0}`. The offset of a descriptor that starts mid-beat never reaches the bus. The first W beat of that descriptor carries `WSTRB` bits only for the bytes at and above the offset, and the last carries only the bytes up to the end of the packet.

9.4.3 Write Strobes

Each SRAM word holds a data beat and its strobe word side by side (see [Sink Data Path](#)). The write engine takes both from the drain port and presents them together, so the strobe always matches the data. In RAPIDS Beats `WSTRB` was tied high.

9.4.4 Streaming Pipeline

Each channel with `sched_wr_valid` and a burst that passes the data test requests arbitration. On grant the AW is issued, and on the AW handshake the engine pulses `axi_wr_drain_req` with the burst size so the drain controller reserves the beats. W beats stream from the drain port that `axi_wr_sram_id` selects. `axi_wr_sram_drain` equals `m_axi_wvalid && m_axi_wready`. `m_axi_wlast` closes each burst, and `m_axi_wuser` carries the channel.

With `PIPELINE = 1` a channel may issue further AWs before earlier B responses return, up to `AW_MAX_OUTSTANDING`. With `PIPELINE = 0` the channel waits for each B. That costs throughput, so the top and core default to 1.

9.4.5 Two Completion Strobes

Table 2.4.8: Completion Strobes

Strobe	Raised when	Scheduler use
<code>sched_wr_done_strobe</code>	AW handshake for the channel	Advance the destination address and the issued-beat count.
<code>sched_wr_committ_strobe</code>	B response for the channel	Count committed beats. The descriptor is complete when all beats are committed.

`sched_wr_ready` is a third, coarser pulse: it fires when the B response belongs to the last burst of a descriptor.

9.5 Error Handling

Table 2.4.9: Error Handling

Error	Detection	Response
AXI SLVERR	<code>m_axi_bresp == 2'b10</code>	Set sticky <code>sched_wr_error</code> for the channel
AXI DECERR	<code>m_axi_bresp == 2'b11</code>	Set sticky <code>sched_wr_error</code> for the channel
Any other non-zero response	<code>m_axi_bresp != 2'b00</code>	Same: the test is not-OKAY, and the channel is taken from BID

The sink data path ORs `sched_wr_error` with its own packet-length error before passing it to the scheduler.

9.6 Differences from RAPIDS Beats

Table 2.4.10: Write Engine Delta

Item	RAPIDS Beats	RAPIDS
WSTRB	All ones	axi_wr_sram_strb
SRAM drain port	Data only	Data plus strobe word
AWADDR	Channel address as given (always aligned)	Aligned down to the beat
Burst cap and final-burst test	Configured length and remaining beats	Also capped at the 4 KB boundary
Everything else		Unchanged

The beats chapter is [AXI Write Engine \(RAPIDS Beats\)](#). The read-side counterpart is [AXI Read Engine](#).

Last Updated: 2026-09-30

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

10 Beats Alloc Control Specification

Module: alloc_ctrl_beats.sv **Location:**

projects/components/dma-ip/rapids/rtl/fub_beats/ **Status:** Implemented **Last Updated:** 2025-01-10

No longer in the SRAM datapath (2026-09-26). RAPIDS' SRAM path now uses STREAM's stream_alloc_ctrl / stream_drain_ctrl / stream_latency_bridge, reached through the snk/src_sram_controller_beats naming wrappers (bdf4e0dff). This FUB is still built and still verified – it keeps its own test, its own filelist and its rapids_all.f entry – it simply is not instantiated by the SRAM

controllers any more. Treat this page as the FUB’s own specification, not as a description of the current SRAM path.

10.1 Overview

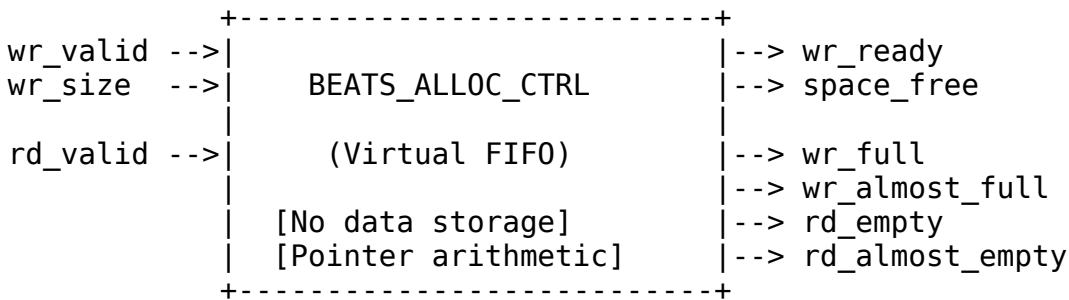
The Beats Alloc Control is a “virtual FIFO” that tracks space allocation without storing actual data. It uses FIFO pointer logic to manage pre-allocation for AXI engines, preventing race conditions between allocation and actual data arrival.

10.1.1 Key Features

- **Virtual FIFO:** Pointer tracking without data storage
- **Variable-Size Allocation:** Supports multi-beat allocation requests
- **Single-Beat Release:** Data writes release one beat at a time
- **Status Flags:** Full, almost full, empty, almost empty
- **Space Tracking:** Reports available space for flow control

10.1.2 Block Diagram

10.1.3 Figure 2.5.1: Beats Alloc Control Block Diagram



10.2 Concept: Virtual FIFO

Unlike a traditional FIFO that stores data, **alloc_ctrl_beats** only tracks **space reservations**:

Traditional FIFO: Virtual FIFO (alloc_ctrl):

 wr_data --> [storage] -> rd_data wr_size --> [pointers] -->
space_free

 Data moves through FIFO Only pointers move, no data

10.2.1 Use Case: Pre-Allocation for AXI Reads

1. AXI engine requests N beats of space
 - alloc_ctrl.wr_valid = 1
 - alloc_ctrl.wr_size = N
 - wr_ptr advances by N
2. As AXI read data arrives (1 beat at a time)
 - alloc_ctrl.rd_valid = 1 (each beat)
 - rd_ptr advances by 1
3. Space becomes available again
 - space_free reflects available capacity

10.3 Parameters

```
parameter int DEPTH = 512;           // Virtual FIFO depth
parameter int ALMOST_WR_MARGIN = 1;  // Almost full margin
parameter int ALMOST_RD_MARGIN = 1;  // Almost empty margin
parameter int REGISTERED = 1;        // Registered outputs

// Derived
parameter int AW = $clog2(DEPTH);    // Address width
```

: Table 2.5.1: Beats Alloc Control Parameters

10.4 Port List

10.4.1 Clock and Reset

Table 2.5.2: Clock and Reset

Signal	Direction	Width	Description
axi_aclk	input	1	System clock
axi_aresetn	input	1	Active-low reset

10.4.2 Write Interface (Allocation Requests)

Table 2.5.3: Write Interface

Signal	Direction	Width	Description
wr_valid	input	1	Allocate space

Signal	Direction	Width	Description
wr_size	input	8	Number of beats to allocate
wr_ready	output	1	Space available for allocation

10.4.3 Read Interface (Data Written)

Table 2.5.4: Read Interface

Signal	Direction	Width	Description
rd_valid	input	1	One beat of data written
rd_ready	output	1	Not full (can accept)

10.4.4 Status Outputs

Table 2.5.5: Status Outputs

Signal	Direction	Width	Description
space_free	output	AW+1	Available space in beats
wr_full	output	1	No space available
wr_almost_full	output	1	Near full
rd_empty	output	1	No allocations pending
rd_almost_empty	output	1	Near empty

10.5 Operation

10.5.1 Pointer Arithmetic

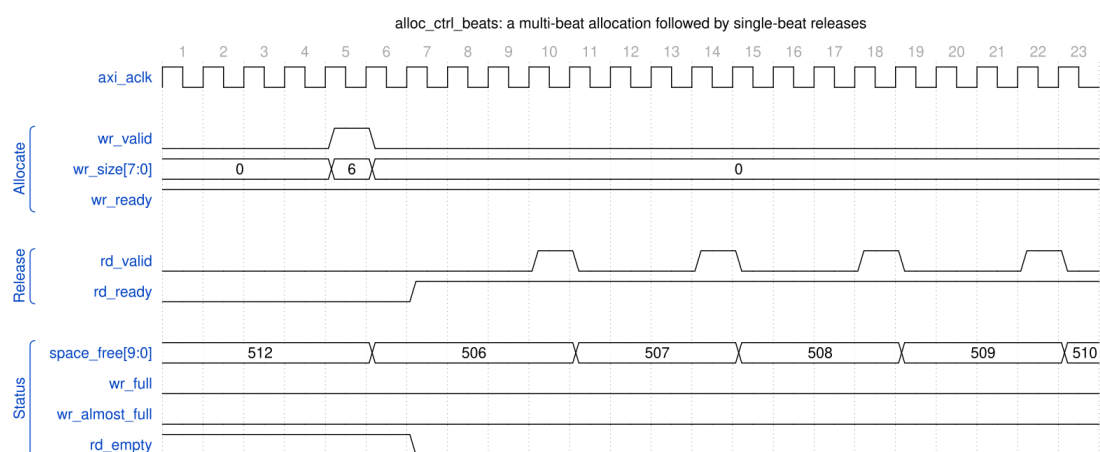
$\text{space_free} = \text{DEPTH} - (\text{wr_ptr} - \text{rd_ptr})$

Full condition: $\text{wr_ptr} - \text{rd_ptr} \geq \text{DEPTH}$

Empty condition: $\text{wr_ptr} == \text{rd_ptr}$

10.5.2 Timing Diagram

10.5.3 Figure 2.5.2: Allocation and Release Timing



Allocation control - allocate then release

Source: [alloc_ctrl_beats_alloc_release.json](#), captured from `dv/tests/fub_beats/test_alloc_ctrl_beats.py` (basic_alloc_drain, depth 512, seed 7) with WAVES=1.

Reading it: one `wr_valid` handshake with `wr_size = 6` reserves six beats and `space_free` drops 512 to 506 the next cycle; `rd_empty` falls because reserved space is now outstanding. Each later `rd_valid` handshake releases one beat and `space_free` climbs back one at a time (507, 508, ...). `wr_full` and `wr_almost_full` stay low: the virtual FIFO is nowhere near its depth.

10.6 Integration Example

```
// Source path: Pre-allocate SRAM space before AXI read
alloc_ctrl_beats #(
    .DEPTH(SRAM_DEPTH),
    .ALMOST_WR_MARGIN(8)
) u_src_alloc (
    .axi_clk      (clk),
```

```

.axi_aresetn    (rst_n),

// Scheduler allocates space before issuing AXI read
.wr_valid      (sched_rd_valid),
.wr_size       (sched_rd_beats[7:0]),
.wr_ready      (alloc_ready),

// AXI read data arrival releases allocation
.rd_valid      (m_axi_rvalid && m_axi_rready),
.rd_ready      (), // Not used for single-beat

// Status
.space_free     (sram_space_available),
.wr_full       (sram_full),
.wr_almost_full (sram_almost_full)
);

```

Last Updated: 2025-01-10

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

11 Beats Drain Control Specification

Module: `drain_ctrl_beats.sv` **Location:**

`projects/components/dma-ip/rapids/rtl/fub_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

No longer in the SRAM datapath (2026-09-26). RAPIDS' SRAM path now uses STREAM's `stream_alloc_ctrl / stream_drain_ctrl / stream_latency_bridge`, reached through the `snk/src_sram_controller_beats` naming wrappers (`bdf4e0dff`). This FUB is still built and still verified – it keeps its own test, its own filelist and its `rapids_all.f` entry – it simply is not instantiated by the SRAM controllers any more. Treat this page as the FUB's own specification, not as a description of the current SRAM path.

11.1 Overview

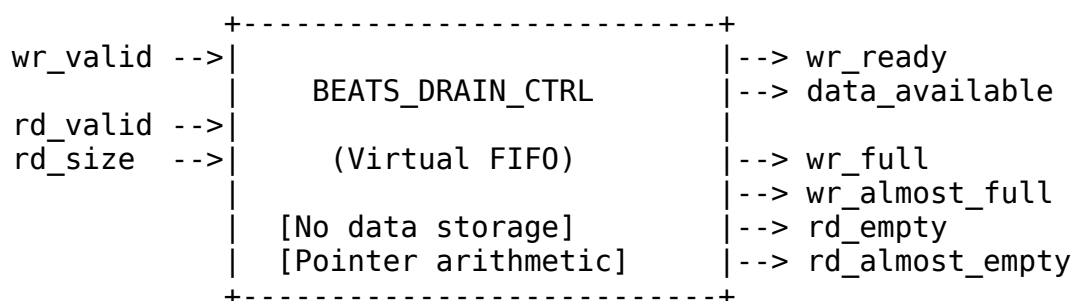
The Beats Drain Control is a “virtual FIFO” that tracks data availability without storing actual data. It complements `alloc_ctrl_beats` by tracking when data has been written to SRAM and is available for draining.

11.1.1 Key Features

- **Virtual FIFO:** Pointer tracking without data storage
- **Single-Beat Write:** Data arrivals tracked one beat at a time
- **Variable-Size Drain:** Supports multi-beat drain requests
- **Status Flags:** Full, almost full, empty, almost empty
- **Data Availability:** Reports beats available for drain

11.1.2 Block Diagram

11.1.3 Figure 2.6.1: Beats Drain Control Block Diagram



11.2 Concept: Data Availability Tracking

The `drain_ctrl` tracks when data has been written to SRAM:

`alloc_ctrl`: Tracks SPACE reservations (allocation -> actual write)
`drain_ctrl`: Tracks DATA availability (data write -> drain request)

11.2.1 Complementary Roles

Source Path:

1. Scheduler allocates space (`alloc_ctrl.wr`)
2. AXI reads data into SRAM
3. Data written to SRAM (`drain_ctrl.wr`)
4. Drain consumer reads (`drain_ctrl.rd`)

Sink Path:

1. Fill allocates space (`alloc_ctrl.wr`)

2. Fill writes to SRAM (alloc_ctrl.rd)
3. Data available in SRAM (drain_ctrl.wr)
4. AXI write drains SRAM (drain_ctrl.rd)

11.3 Parameters

```

parameter int DEPTH = 512;           // Virtual FIFO depth
parameter int ALMOST_WR_MARGIN = 1;  // Almost full margin
parameter int ALMOST_RD_MARGIN = 1;  // Almost empty margin
parameter int REGISTERED = 1;        // Registered outputs

// Derived
parameter int AW = $clog2(DEPTH);    // Address width

```

: Table 2.6.1: Beats Drain Control Parameters

11.4 Port List

11.4.1 Clock and Reset

Table 2.6.2: Clock and Reset

Signal	Direction	Width	Description
axi_aclk	input	1	System clock
axi_aresetn	input	1	Active-low reset

11.4.2 Write Interface (Data Written to SRAM)

Table 2.6.3: Write Interface

Signal	Direction	Width	Description
wr_valid	input	1	One beat written to SRAM
wr_ready	output	1	Can track more data

11.4.3 Read Interface (Drain Requests)

Table 2.6.4: Read Interface

Signal	Direction	Width	Description
rd_valid	input	1	Drain request
rd_size	input	8	Number of beats to drain
rd_ready	output	1	Data available for drain

11.4.4 Status Outputs

Table 2.6.5: Status Outputs

Signal	Direction	Width	Description
data_available	output	AW+1	Beats available to drain
wr_full	output	1	Tracking full
wr_almost_full	output	1	Near full
rd_empty	output	1	No data to drain
rd_almost_empty	output	1	Near empty

11.5 Operation

11.5.1 Pointer Arithmetic

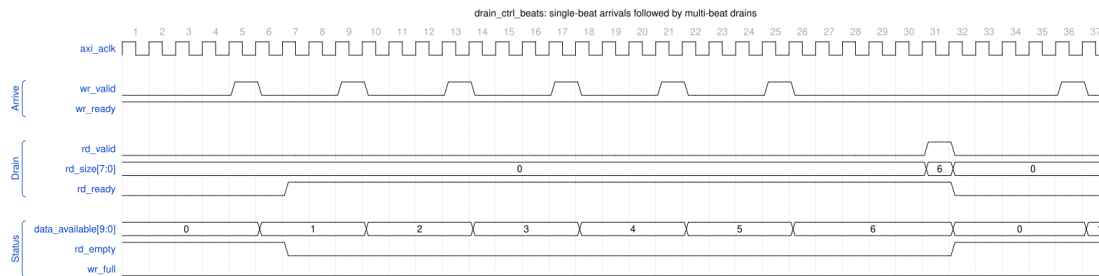
$\text{data_avail} = \text{wr_ptr} - \text{rd_ptr}$

Full condition: $\text{wr_ptr} - \text{rd_ptr} \geq \text{DEPTH}$

Empty condition: $\text{wr_ptr} == \text{rd_ptr}$

11.5.2 Timing Diagram

11.5.3 Figure 2.6.2: Data Arrival and Drain Timing



Drain control - single-beat arrivals then a multi-beat drain

Source: [drain_ctrl_beats_arrive_drain.json](#), captured from `dv/tests/fub_beats/test_drain_ctrl_beats.py` (basic_write_drain, depth 512, seed 7) with WAVES=1.

Reading it: six `wr_valid` handshakes, one beat each, raise `data_available` 1, 2, ... 6 and drop `rd_empty` on the first arrival. A single `rd_valid` handshake with `rd_size` = 6 then reserves all six at once: `data_available` returns to 0 and `rd_empty` rises the next cycle. The counter tracks beats, not bytes, and never touches the data itself.

11.6 Integration Example

// Sink path: Track data availability for AXI write engine

```
drain_ctrl_beats #(
    .DEPTH(SRAM_DEPTH),
    .ALMOST_RD_MARGIN(8)
) u_snk_drain (
    .axi_aclk      (clk),
    .axi_aresetn   (rst_n),
```

// Fill writes data to SRAM

```
.wr_valid      (fill_valid && fill_ready),
.wr_ready      (), // Single-beat, always ready
```

// AXI write engine drains data

```
.rd_valid      (axi_wr_drain_req),
.rd_size       (axi_wr_burst_len),
.rd_ready      (data_ready_for_drain),
```

// Status

```
.data_available (beats_to_write),
```

```
); rd_empty (no_data_to_write)
```

11.7 Relationship: alloc_ctrl vs drain_ctrl

Table 2.6.6: alloc_ctrl vs drain_ctrl Comparison

Aspect	alloc_ctrl_beats	drain_ctrl_beats
Tracks	Space reservations	Data availability
Write port	Multi-beat allocation	Single-beat arrival
Read port	Single-beat release	Multi-beat drain
Output	space_free	data_available
Use case	Pre-allocate before fill	Know when to drain

Last Updated: 2025-01-10

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

12 Latency Bridge (beats) Specification

Module: latency_bridge_beats.sv **Location:** projects/components/dma-ip/rapids/rtl/fub_beats/ **Status:** Implemented, tested, not in the SRAM path **Last Updated:** 2026-09-27

Not in the SRAM datapath since 2026-09-26. RAPIDS' SRAM path is STREAM's sram_controller, which carries its own stream_latency_bridge (the same design), reached through the snk/src_sram_controller_beats naming wrappers (bdf4e0dff). This FUB is still built and verified – it keeps its test, its filelist and its rapids_all.f entry – but nothing in rapids_core_beats instantiates it. This page is the FUB's own specification.

12.1 Overview

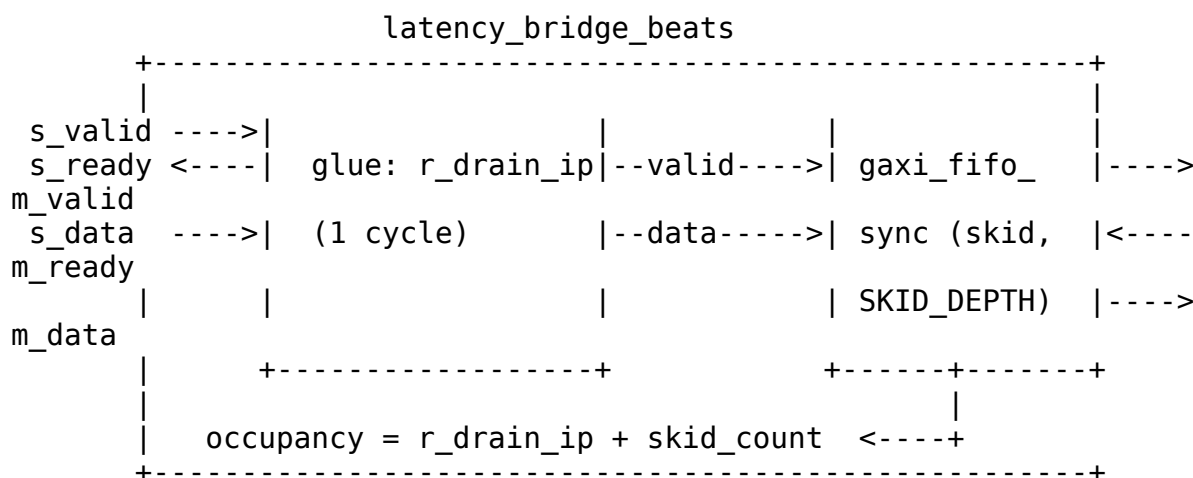
latency_bridge_beats turns a **registered-read FIFO** – one whose data appears the cycle *after* the read handshake – into a plain **valid/ready stream** the consumer can stall at will. It is a data-width bridge: it moves DATA_WIDTH bits per beat and carries nothing else. There is no beat count, no channel ID and no programmable delay in it; an earlier version of this page described such an interface, and it never existed in this module.

12.1.1 Key Features

- **One-cycle glue, one skid buffer:** a single flop (r_drain_ip) remembers that a FIFO read was accepted, so the data that lands the next cycle is pushed into a gaxi_fifo_sync skid of depth SKID_DEPTH
- **Full throughput when the consumer is ready:** back-to-back reads are accepted as long as the skid keeps draining
- **Backpressure absorbed in the skid:** s_ready drops only when a skid write is actually stalled, so the FIFO is never read into a slot that does not exist
- **Occupancy reported:** occupancy counts the beats the bridge holds (the one in flight plus the skid contents), which is what an occupancy-based data-available counter needs
- **Debug taps:** dbg_r_pending and dbg_r_out_valid expose the in-flight flop and the skid output valid, for catching stuck data from a test

12.1.2 Block Diagram

12.1.3 Figure 2.7.1: Latency Bridge Block Diagram



12.2 Concept: bridging a registered read

A FIFO with REGISTERED=1 read (the SRAM style this repo uses so BRAM infers cleanly) answers a read handshake with data **one cycle later**. A consumer speaking valid/ready expects data *with* valid, and may deassert ready at any time. Without a bridge the FIFO has already popped a beat the consumer did not take.

```
cycle 0    s_valid && s_ready          FIFO read accepted
r_drain_ip <= 1
cycle 1    data arrives on s_data      skid_wr_valid = r_drain_ip, data
pushed
cycle N    m_valid && m_ready          consumer drains the skid at its
own pace
```

The bridge never guesses about downstream readiness: s_ready is derived from the skid's room, counting only a write that is *stalled* this cycle as pending (a write that completes this cycle frees its slot at the same edge). That is why full throughput holds while writes complete immediately, and why the FIFO is never over-read when they do not.

12.3 Parameters

```
parameter int DATA_WIDTH = 64;    // Beat width
parameter int SKID_DEPTH = 4;      // Skid buffer depth (2-4
recommended)
parameter int DW = DATA_WIDTH;    // Short alias
```

: Table 2.7.1: Latency Bridge Parameters

occupancy is 3 bits wide and counts 0..5 at the default depth (one beat in flight plus four in the skid); a larger SKID_DEPTH needs the port widened.

12.4 Port List

12.4.1 Clock and Reset

Table 2.7.2: Clock and Reset

Signal	Direction	Width	Description
clk	input	1	System clock
rst_n	input	1	Active-low reset

12.4.2 Upstream Interface (from the registered FIFO)

Table 2.7.3: Upstream Interface

Signal	Direction	Width	Description
s_valid	input	1	FIFO has a beat to read (not empty)
s_ready	output	1	Bridge accepts a read this cycle; the data lands NEXT cycle
s_data	input	DW	Read data, valid one cycle after the s_valid && s_ready handshake

12.4.3 Downstream Interface (to the consumer)

Table 2.7.4: Downstream Interface

Signal	Direction	Width	Description
m_valid	output	1	Beat available
m_ready	input	1	Consumer takes the beat
m_data	output	DW	Beat data, valid with m_valid

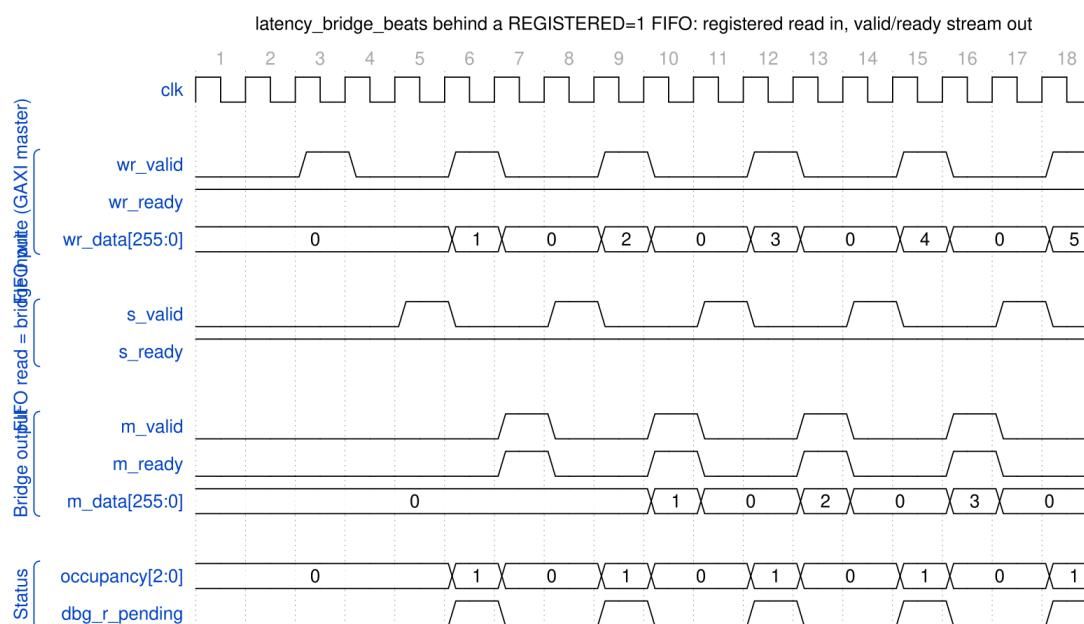
12.4.4 Status and Debug

Table 2.7.5: Status and Debug

Signal	Direction	Width	Description
occupancy	output	3	Beats held by the bridge: the in-flight read plus the skid contents (0..5)
dbg_r_pending	output	1	r_drain_ip: a read was accepted and its data is due next cycle
dbg_r_output_valid	output	1	Skid output valid (mirrors m_valid)

12.5 Operation

12.5.1 Figure 2.7.2: Latency Bridge Timing (consumer always ready)



Latency bridge behind a registered-read FIFO

Source: [latency_bridge_beats_streaming.json](#), captured from `dv/tests/fub_beats/test_latency_bridge_beats.py` (streaming, 256-bit, seed 7) with `WAVES=1`. The test drives a `REGISTERED = 1 gaxi_fifo_sync` in front of the bridge (`dv/tb/latency_bridge_beats_tb_top.sv`), because that is the only way to produce the bridge's upstream contract: a valid/ready master presents data *with* valid, a registered FIFO presents it one cycle *after* the read handshake.

Reading it: each `wr_valid` beat lands in the FIFO and `s_valid` (FIFO not empty) rises a cycle later; the bridge reads it at once (`s_ready` stays high) and `dbg_r_pending` marks the cycle the data is in flight. Two cycles after the write the beat is on `m_valid/m_data` with its payload intact (1, 2, 3, ...), and occupancy shows exactly one beat inside the bridge per transfer. With a consumer that is always ready the bridge adds the FIFO's read latency and no more.

An earlier revision of this test drove `s_*` directly from the BFM and never compared `m_data`; every beat came out 0 and the test passed (rapids TASK-003). (`dv/tests/fub_beats/test_latency_bridge_beats.py`).

12.6 Integration Context

The bridge sits between a `gaxi_fifo_sync` with `REGISTERED=1` and whatever consumes the stream. The same arrangement, with `STREAM`'s twin `stream_latency_bridge`, is what `sram_controller_unit` uses inside the shared SRAM controller today; occupancy there is what the drain accounting must allow for (stream BUG-011).

```
latency_bridge_beats #(
    .DATA_WIDTH (512),
    .SKID_DEPTH (4)
) u_bridge (
    .clk          (clk),
    .rst_n        (rst_n),
    // registered-read FIFO side
    .s_valid      (fifo_rd_valid),
    .s_ready      (fifo_rd_ready),
    .s_data       (fifo_rd_data),
    // consumer side
    .m_valid      (drain_valid),
    .m_ready      (drain_read),
    .m_data       (drain_data),
    // status / debug
    .occupancy    (bridge_occupancy),
    .dbg_r_pending (dbg_bridge_pending),
    .dbg_r_out_valid (dbg_bridge_out_valid)
);
```

12.7 Design Considerations

Table 2.7.6: Skid Depth Selection

SKID_DEPTH	Behaviour
2	Minimum: one stall of the consumer costs a bubble on the next read
4	Default: absorbs a short consumer stall with no upstream bubble
8	Deeper stall absorption; occupancy must be widened past 3 bits

The skid does not need to cover the FIFO's read latency (that is the one `r_drain_ip` flop's job); it needs to cover how long the *consumer* stalls without the bridge stopping upstream reads.

Last Updated: 2026-09-27

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

13 Control-Read Engine Specification

Module: `ctrlrd_engine.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub/`
Status: Implemented **Last Updated:** 2026-07-12

13.1 Overview

The Control-Read Engine implements the CTRL_READ descriptor opcode (`DESC_OP_CTRL_READ = 2'b01`) as a **consumer gate**. When the scheduler decodes a CTRL_READ descriptor, it hands the engine a poll address, an expected value, and a compare mask. The engine issues an AXI4 read to the poll address, applies the mask to the returned data, and compares (`read_data & mask`) against the expected value. If they match, the gate is satisfied and the descriptor chain proceeds; if not, the engine retries. Retries are bounded so a gate that never matches cannot hang the channel.

One Control-Read Engine is instantiated per channel in `scheduler_group_beats.sv` (`u_ctrlrd_engine`), configured for 32-bit AXI reads.

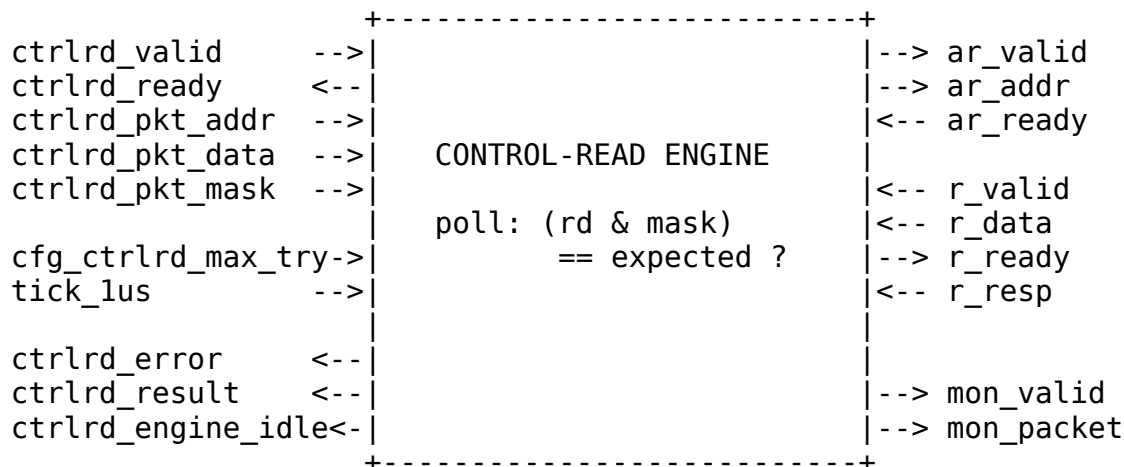
13.1.1 Key Features

- **Consumer Gate Semantics:** Polls memory until `(read_data & mask) == expected`
- **Masked Comparison:** Only the masked bits of the polled word participate in the match
- **Bounded Retry:** `cfg_ctrlrd_max_try` (0-511) caps the number of poll attempts
- **1 microsecond Retry Pacing:** Retries wait for `tick_1us` from the scheduler group
- **Error on Exhaustion:** Asserts `ctrlrd_error` if the gate never matches within budget

- **Null Address Fast-Path:** A poll address of zero completes immediately as a match
- **Channel Reset Support:** `cfg_channel_reset` returns the FSM to idle and drains any read still on the fabric
- **MonBus Integration:** Completion, retry, and error events

13.1.2 Block Diagram

13.1.3 Figure 2.8.1: Control-Read Engine Block Diagram



13.2 Parameters

```

parameter int CHANNEL_ID = 0; // Channel identifier
parameter int NUM_CHANNELS = 32; // Total channels
parameter int CHAN_WIDTH = $clog2(NUM_CHANNELS); // Channel index
width (derived)
parameter int ADDR_WIDTH = 64; // Address bus width
parameter int AXI_DATA_WIDTH = 64; // AXI read data
width (>= 32)
parameter int AXI_ID_WIDTH = 8; // AXI ID width (>=
CHAN_WIDTH)

// Monitor Bus Parameters
parameter logic [15:0] MON_AGENT_ID = 16'h0030; // Control-Read
Engine Agent ID
parameter logic [7:0] MON_UNIT_ID = 8'h02; // Unit identifier
parameter logic [8:0] MON_CHANNEL_ID = 9'h000; // Base channel ID

```

: Table 2.8.1: Control-Read Engine Parameters

Parameter validation enforces `AXI_ID_WIDTH >= CHAN_WIDTH` and `AXI_DATA_WIDTH >= 32` (the engine consumes only the lower 32 bits of the read data). The per-channel instance in `scheduler_group_beats.sv` overrides `AXI_DATA_WIDTH` to 32.

13.3 Port List

13.3.1 Clock and Reset

Table 2.8.2: Clock and Reset

Signal	Direction	Width	Description
clk	input	1	System clock
rst_n	input	1	Active-low asynchronous reset

13.3.2 Scheduler Interface

Table 2.8.3: Scheduler Interface

Signal	Direction	Width	Description
ctrlrd_valid	input	1	Gate request valid from scheduler
ctrlrd_ready	output	1	Engine ready to accept a request
ctrlrd_pkt_a_ddr	input	AW	Poll address (descriptor <code>poll_addr[63:0]</code>)
ctrlrd_pkt_d_ata	input	32	Expected value (descriptor <code>expected[95:64]</code>)
ctrlrd_pkt_m_ask	input	32	Compare mask (descriptor <code>mask[127:96]</code>)
ctrlrd_error	output	1	Gate failed (retries exhausted or AXI error)
ctrlrd_result	output	32	Last read value (lower 32 bits)
ctrlrd_engin	output	1	Engine idle (post-issue

Signal	Direction	Width	Description
e_idle			completion indicator)

13.3.3 Configuration Interface

Table 2.8.4: Configuration Interface

Signal	Direction	Width	Description
cfg_ctrlrd_max_try	input	9	Poll retry budget, 0-511 (from CTRL_CONFIG.CTRLRD_MAX_TRY)
cfg_channel_reset	input	1	Per-channel reset request
tick_lus	input	1	1 microsecond tick that paces retries

13.3.4 AXI4 Read Master Interface

Table 2.8.5: AXI4 Read Master Interface

Signal	Direction	Width	Description
ar_valid	output	1	Read address valid
ar_ready	input	1	Read address ready
ar_addr	output	AW	Read address (poll address)
ar_len	output	8	Burst length (8'h00, single beat)
ar_size	output	3	Burst size (3'b010, 4 bytes)
ar_burst	output	2	Burst type (2'b01, INCR)
ar_id	output	ID	Transaction ID (channel-derived)
ar_lock	output	1	Lock (1'b0, normal access)
ar_cache	output	4	Cache (4'b0010, non-cacheable bufferable)
ar_prot	output	3	Protection (3'b000)

Signal	Direction	Width	Description
ar_qos	output	4	Quality of service (4'h0)
ar_region	output	4	Region (4'h0)
r_valid	input	1	Read data valid (shared channel)
r_ready	output	1	Read data ready
r_data	input	ADW	Read data (lower 32 bits used)
r_id	input	ID	Read response ID (matched to our request)
r_resp	input	2	Read response
r_last	input	1	Read last beat

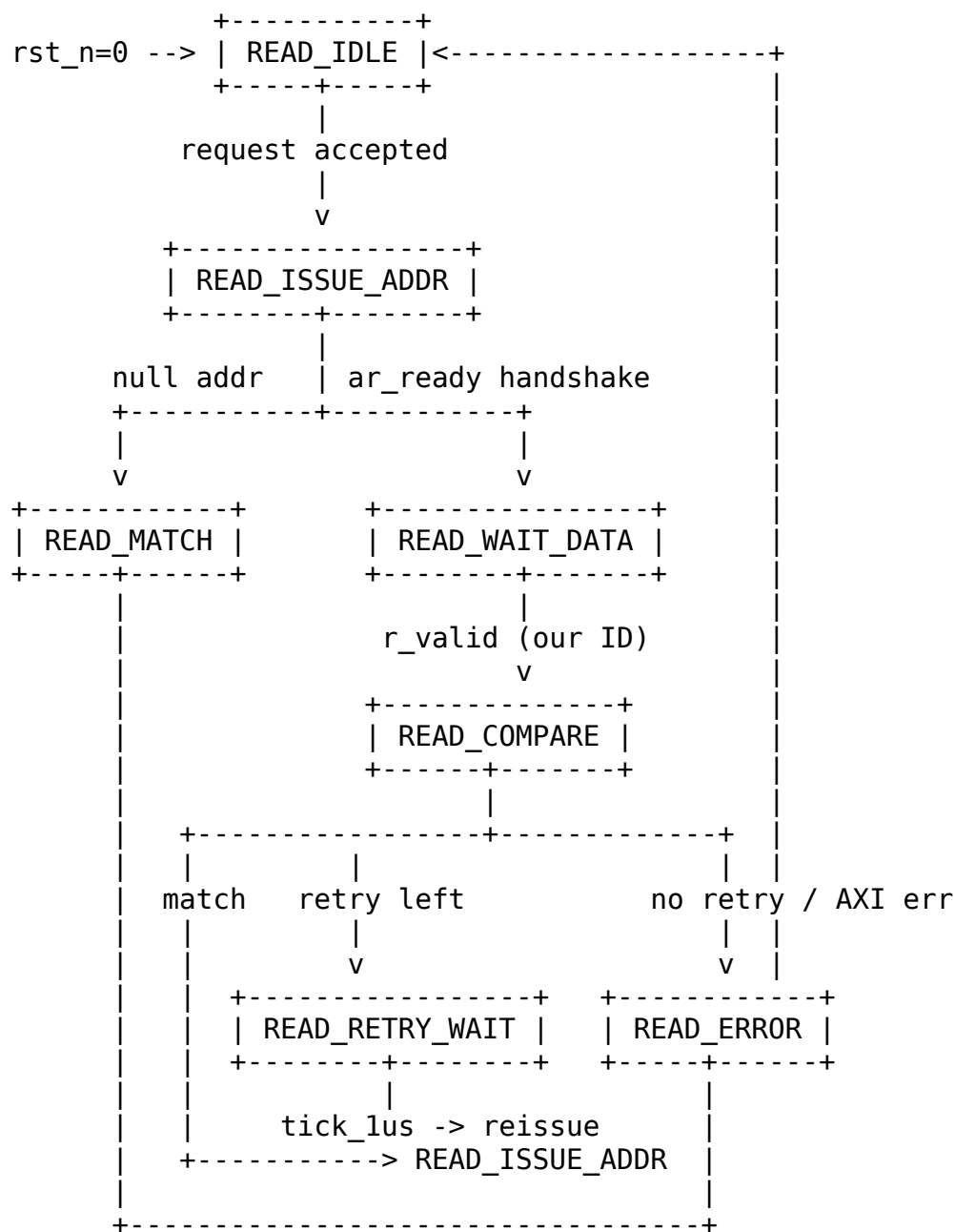
13.3.5 MonBus Interface

Table 2.8.6: MonBus Interface

Signal	Direction	Width	Description
i_mon_time	input	64	Side-band monitor timestamp
mon_valid	output	1	Monitor packet valid
mon_ready	input	1	Monitor consumer ready
mon_packet	output	128	Monitor packet (monitor_packet_t)
mon_timestamp	output	64	Monitor packet timestamp

13.4 FSM States

13.4.1 Figure 2.8.2: Control-Read Engine FSM



State summary:

- **READ_IDLE:** Waits for a request from the scheduler (via a two-deep skid buffer). Latches poll address, expected value, mask, and the AXI ID; initializes the retry counter from `cfg_ctrlrd_max_try`.

- **READ_ISSUE_ADDR:** Drives `ar_valid`. A null address (64'h0) shortcuts to `READ_MATCH`; otherwise, once `ar_ready` handshakes, advances to `READ_WAIT_DATA`.
 - **READ_WAIT_DATA:** Accepts the read response whose `r_id` matches the channel ID, capturing the lower 32 bits and the response code.
 - **READ_COMPARE:** On an AXI error response, goes to `READ_ERROR`. On a masked match, goes to `READ_MATCH`. Otherwise, if retries remain, decrements the counter and goes to `READ_RETRY_WAIT`; if the budget is exhausted, goes to `READ_ERROR`.
 - **READ_RETRY_WAIT:** Waits for `tick_1us`, then reissues the read.
 - **READ_MATCH:** Gate satisfied. Clears the error flag and returns to `READ_IDLE`.
 - **READ_ERROR:** Sets `ctrlrd_error` and returns to `READ_IDLE`.
-

13.5 Operation

13.5.1 Masked Compare

The engine gates on a masked equality check:

```
w_masked_expected = ctrlrd_pkt_data & ctrlrd_pkt_mask
w_masked_actual   = r_data[31:0]    & ctrlrd_pkt_mask
w_data_match      = (w_masked_expected == w_masked_actual)
```

Only the bits set in mask participate. A mask of 32'hFFFFFFF compares the full word; a mask of a single bit turns the gate into a flag-poll.

13.5.2 Retry Budget and Pacing

The retry counter is loaded from `cfg_ctrlrd_max_try` when the request is accepted. On each non-matching, non-error compare with retries remaining, the counter decrements and the engine parks in `READ_RETRY_WAIT` until the next `tick_1us`, which paces successive polls at roughly 1 microsecond intervals. When the counter reaches zero without a match, the engine transitions to `READ_ERROR` and asserts `ctrlrd_error`, so a never-matching gate cannot stall the channel indefinitely.

`cfg_ctrlrd_max_try` is driven by the `CTRL_CONFIG` register at offset 0x240, field `CTRLRD_MAX_TRY[8:0]` (0-511, reset 16).

13.5.3 Null Address Fast-Path

If the latched poll address is 64'h0, the engine treats the gate as already satisfied: READ_ISSUE_ADDR transitions directly to READ_MATCH without issuing an AXI read. This lets software emit an unconditional (no-op) gate.

13.5.4 AXI Read Transaction

Polls are single-beat, 4-byte INCR reads (`ar_len = 0`, `ar_size = 3'b010`, `ar_burst = 2'b01`). The `ar_id` is derived from `CHANNEL_ID`, and only responses whose `r_id` matches are consumed on the shared read data channel. `ctrlrd_result` exposes the last captured 32-bit value; in the per-channel instantiation the scheduler gates on match/error rather than the raw value, so `ctrlrd_result` is left unconnected there.

13.5.5 Channel Reset

`cfg_channel_reset` is registered into `r_channel_reset_active`. While asserted, the engine refuses new requests (`ctrlrd_ready` is forced low), the retry and error state is cleared, and the FSM is driven back to `READ_IDLE`.

What the reset cannot do is cancel a read already on the fabric, because AXI has no cancel. The engine therefore drains it (rapids TASK-014, 2026-09-27):

- an AR that was raised and not yet accepted stays raised until the slave takes it (`r_drain_ar`), then
- the R beat that the accepted AR is owed is taken and discarded (`r_drain_pending`, which alone lifts `r_ready` outside `READ_WAIT_DATA`).

Until the beat is gone `ctrlrd_engine_idle` stays low and the request path stays closed, so the next read on this ID cannot be answered by the stale beat. Before this the reset dropped the FSM to idle with `r_ready` low, the beat sat on the bus, and a re-kicked channel took it as its own answer.

`ctrlrd_engine_idle` asserts only in `READ_IDLE` with no pending request, no active channel reset and nothing left to drain.

13.6 MonBus Reporting

The engine emits monitor packets on notable events:

Table 2.8.7: MonBus Events

Event	Packet Type	Code	Payload
Gate satisfied (non-null)	PktTypeCompletion	CORE_COMPL_CTRLRD_CO	Poll address

Event	Packet Type	Code	Payload
Retry attempt	PktTypePerf	MPLETD CORE_PERF_CTRLRD_RETRY	Remaining retry count
Retries exhausted	PktTypeError	CORE_ERR_CTRLRD_MAX_RETRIES	Poll address
AXI error response	PktTypeError	AXI_ERR_RESP_SLVERR / AXI_ERR_RESP_DECERR	Response code

All packets carry MON_AGENT_ID, MON_UNIT_ID, and MON_CHANNEL_ID, and are timestamped from i_mon_time.

13.7 Descriptor Field Mapping

The CTRL_READ opcode reinterprets the shared 256-bit descriptor (the descriptor engine's existing field extraction is reused):

Table 2.8.8: CTRL_READ Descriptor Field Mapping

Descriptor Field	Bits	Engine Port
poll_addr	[63:0]	ctrlrd_pkt_addr
expected	[95:64]	ctrlrd_pkt_data
mask	[127:96]	ctrlrd_pkt_mask
max_try	[143:128]	Per-descriptor budget (0 = use cfg_ctrlrd_max_try default)

Last Updated: 2026-07-12

RTL Design Sherpa · Learning Hardware Design Through Practice GitHub · Documentation Index · MIT License

14 Control-Write Engine Specification

Module: ctrlwr_engine.sv **Location:** projects/components/dma-ip/rapids/rtl/fub/
Status: Implemented **Last Updated:** 2025-01-10

14.1 Overview

The Control-Write Engine executes a CTRL_WRITE control descriptor (DESC_OP_CTRL_WRITE = 2'b10). A CTRL_WRITE is a producer doorbell: the engine performs a single-beat AXI4 write of the descriptor's 32-bit wr_data value to its 64-bit wr_addr target, then completes so the descriptor chain continues. One instance is dedicated per channel.

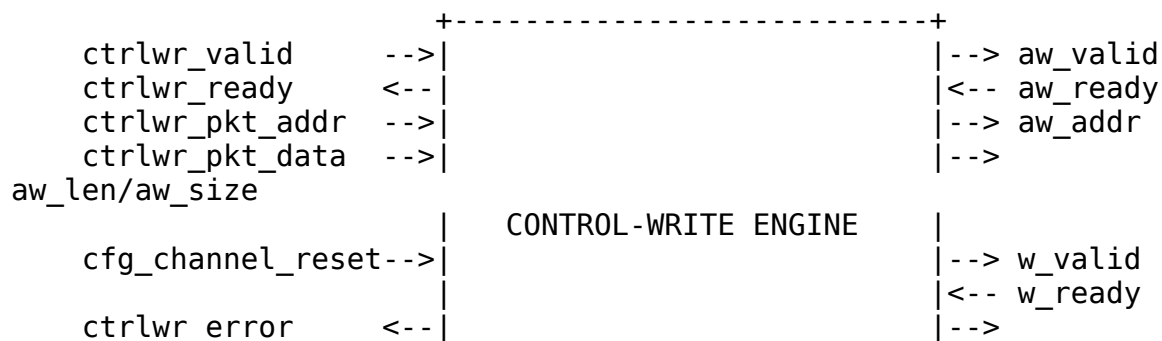
The write target address must be 4-byte aligned because the transfer is a 32-bit (aw_size = 3'b010) single-beat write. A null address (wr_addr == 0) is treated as a no-op and the request completes without issuing an AXI transaction.

14.1.1 Key Features

- **Single-Beat AXI4 Write:** One 32-bit doorbell write per request (aw_len = 0, w_last = 1)
- **Producer Doorbell Semantics:** Write value to target, then release the chain (no poll, no retry)
- **Address Alignment Check:** Requires 4-byte-aligned wr_addr; misalignment raises an error
- **Null-Address Skip:** wr_addr == 0 completes as a no-op
- **Channel Reset Support:** cfg_channel_reset returns the FSM to idle and drains any write still on the fabric
- **MonBus Integration:** Completion and error event reporting

14.1.2 Block Diagram

14.1.3 Figure 2.9.1: Control-Write Engine Block Diagram



```

w_data/w_strb/w_last
  ctrlwr_engine_idle<-|
                        |<-- b_valid
                        |--> b_ready
                        |<-- b_id/b_resp
                        |--> mon_valid
                        |--> mon_packet
                        +-----+

```

14.2 Parameters

```

parameter int CHANNEL_ID = 0;           // Channel identifier
parameter int NUM_CHANNELS = 32;        // Total channels
parameter int CHAN_WIDTH = $clog2(NUM_CHANNELS); // Derived channel
index width
parameter int ADDR_WIDTH = 64;          // Address bus width
parameter int AXI_ID_WIDTH = 8;         // AXI transaction ID
width

// Monitor Bus Parameters
parameter logic [15:0] MON_AGENT_ID = 16'h0020; // Ctrlwr Engine
Agent ID
parameter logic [7:0] MON_UNIT_ID = 8'h01;     // Unit identifier
parameter logic [8:0] MON_CHANNEL_ID = 9'h000; // Base channel ID

```

: Table 2.9.1: Control-Write Engine Parameters

Note: AXI_ID_WIDTH must be $\geq$ CHAN_WIDTH (checked by an elaboration-time \$fatal). When instantiated in scheduler_group_beats.sv as u_ctrlwr_engine, MON_AGENT_ID is overridden to 16'h0021 (CTRLWR_MON_AGENT_ID) to distinguish it from the Control-Read Engine's 16'h0020.

14.3 Port List

14.3.1 Clock and Reset

Table 2.9.2: Clock and Reset

Signal	Direction	Width	Description
clk	input	1	System clock
rst_n	input	1	Active-low asynchronous

Signal	Direction	Width	Description
			reset

14.3.2 Configuration Interface

Table 2.9.3: Configuration Interface

Signal	Direction	Width	Description
cfg_channel_reset	input	1	Per-channel reset: blocks new requests, drains transaction, forces idle

14.3.3 Scheduler Interface

Table 2.9.4: Scheduler Interface

Signal	Direction	Width	Description
ctrlwr_valid	input	1	Request valid (doorbell write requested)
ctrlwr_ready	output	1	Ready to accept request
ctrlwr_pkt_addr	input	AW	Doorbell target address (wr_addr, 4-byte aligned)
ctrlwr_pkt_data	input	32	Doorbell write value (wr_data)
ctrlwr_error	output	1	Error flag (alignment or AXI response error)
ctrlwr_engine_idle	output	1	Engine idle (in WRITE_IDLE, no active transaction, not in channel reset)

14.3.4 AXI4 Write Master Interface

Table 2.9.5: AXI4 Write Master Interface

Signal	Direction	Width	Description
aw_valid	output	1	Write address valid
aw_ready	input	1	Write address ready
aw_addr	output	AW	Write address (= wr_addr)

Signal	Direction	Width	Description
aw_len	output	8	Burst length (0 = single beat)
aw_size	output	3	Transfer size (3'b010 = 4 bytes / 32-bit)
aw_burst	output	2	Burst type (2'b01 = INCR)
aw_id	output	AXI_ID_WID TH	Transaction ID (derived from CHANNEL_ID)
aw_lock	output	1	Lock type (0 = normal)
aw_cache	output	4	Cache type (4'b0010 = non-cacheable bufferable)
aw_prot	output	3	Protection type (3'b000)
aw_qos	output	4	Quality of service (0)
aw_region	output	4	Region identifier (0)
w_valid	output	1	Write data valid
w_ready	input	1	Write data ready
w_data	output	32	Write data (= wr_data)
w_strb	output	4	Write strobes (4'b1111 = all bytes)
w_last	output	1	Last beat (1 = single beat)
b_valid	input	1	Write response valid
b_ready	output	1	Write response ready
b_id	input	AXI_ID_WID TH	Response ID (matched against expected AXI ID)
b_resp	input	2	Write response code

14.3.5 MonBus Interface

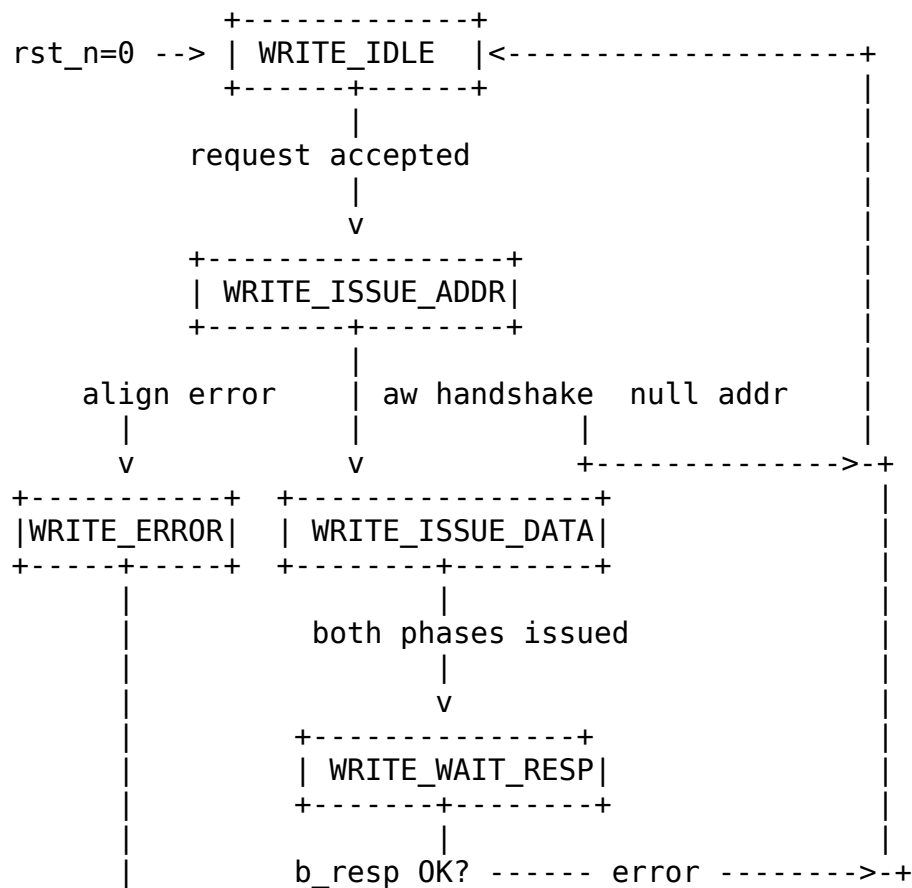
Table 2.9.6: MonBus Interface

Signal	Direction	Width	Description
i_mon_time	input	64	Monitor timestamp side-band

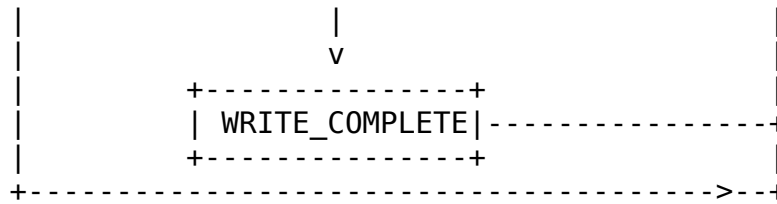
Signal	Direction	Width	Description
mon_valid	output	1	input Monitor packet valid
mon_ready	input	1	Monitor packet ready
mon_packet	output	128	Monitor packet (monitor_packet_t)
mon_timestamp	output	64	Monitor packet timestamp

14.4 FSM States

14.4.1 Figure 2.9.2: Control-Write Engine FSM



(WRITE_ERROR)



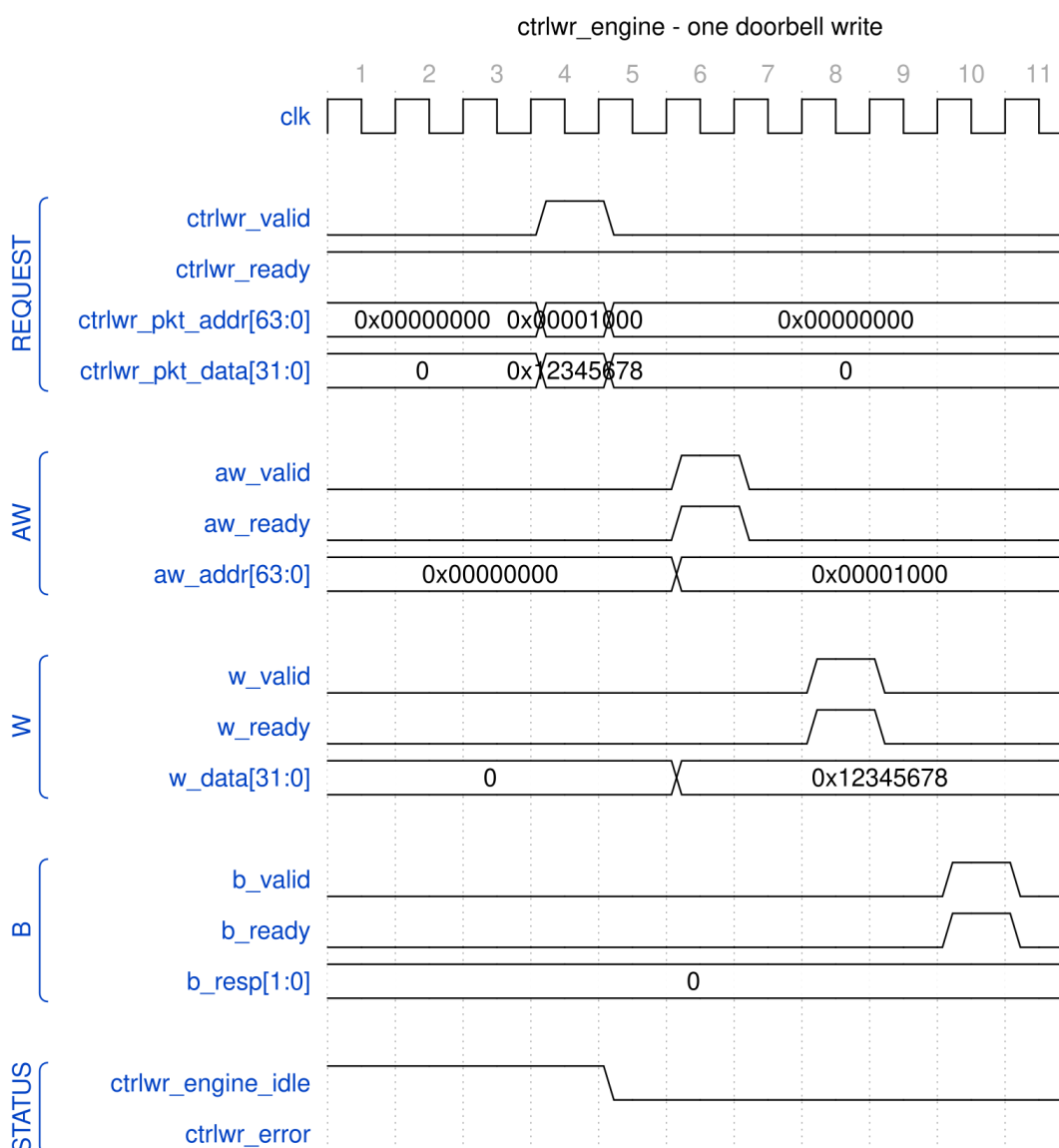
State summary:

- **WRITE_IDLE:** Wait for a request from the request skid buffer; latch `wr_addr/wr_data` and the expected AXI ID.
 - **WRITE_ISSUE_ADDR:** Drive `aw_valid` until `aw_ready`. A misaligned address routes to `WRITE_ERROR`; a null address returns to `WRITE_IDLE` (no-op skip).
 - **WRITE_ISSUE_DATA:** Drive `w_valid` until `w_ready`; advance once both address and data phases have been issued.
 - **WRITE_WAIT_RESP:** Accept the matching b-channel response; a non-OKAY `b_resp` routes to `WRITE_ERROR`, otherwise to `WRITE_COMPLETE`.
 - **WRITE_COMPLETE:** Emit a completion MonBus packet (unless null address) and return to `WRITE_IDLE`.
 - **WRITE_ERROR:** Latch `ctrlwr_error`, emit an error MonBus packet, and return to `WRITE_IDLE`. The error flag persists until reset or `cfg_channel_reset`.
-

14.5 Operation

14.5.1 Doorbell Write Sequence

14.5.2 Figure 2.9.3: Control-Write Doorbell Timing



ctrlwr_engine - one doorbell write

Source: [ctrlwr_engine_doorbell.json](#), captured from `dv/tests/fub/test_ctrlwr_engine.py` (basic_write, channel 0, 64-bit address, REG_LEVEL=GATE) with WAVES=1; the AXI side is the framework write slave.

Reading it: the request handshakes in one cycle (`ctrlwr_ready` is held high by the skid buffer) with the doorbell address `0x1000` and data `0x12345678`, and the engine leaves idle the next cycle. AW goes out two cycles after the request, W two cycles after AW, and the slave answers B (OKAY) two cycles after W; `ctrlwr_engine_idle` returns the cycle after the B handshake, eight cycles after the request. Nothing is pipelined across doorbells: the engine takes one request at a time.

14.5.3 Request Handshake and Skid Buffer

Incoming requests are captured through a depth-2 `gaxi_skid_buffer`. The engine accepts a request (`ctrlwr_ready`) only when not in an active channel reset. The FSM pops the buffered request in `WRITE_IDLE`, latching `{wr_addr, wr_data}` and computing the expected AXI ID from `CHANNEL_ID`:

```
aw_id = {(AXI_ID_WIDTH-CHAN_WIDTH){1'b0}}, CHANNEL_ID[CHAN_WIDTH-1:0];;
```

The b-channel response is matched against this expected ID (`b_id == r_expected_axi_id`) so a shared write master can be demultiplexed per channel.

14.5.4 Address Validation

```
Null (skip)   : wr_addr == 64'h0
Align error  : wr_addr[1:0] != 2'b00 (and not null)
```

A null address completes immediately as a no-op with no AXI activity and no completion packet. A misaligned address never issues on AXI; it transitions to `WRITE_ERROR`, asserts `ctrlwr_error`, and emits an error MonBus packet.

14.5.5 Error Reporting

`ctrlwr_error` is a registered, sticky flag. It is set on an alignment error or on a non-OKAY AXI write response (`b_resp != 2'b00`), and it persists until `rst_n` or `cfg_channel_reset`. The MonBus error packet carries either the offending address (alignment error) or the captured `b_resp` code (AXI response error).

14.5.6 Channel Reset

Asserting `cfg_channel_reset` registers `r_channel_reset_active`, which blocks new requests (`ctrlwr_ready` low), clears the error flag and the phase-issued state, and drives the FSM back to `WRITE_IDLE`.

A write already on the fabric cannot be cancelled, so the engine drains it (rapids TASK-014, 2026-09-27): a raised AW stays up until accepted (`r_drain_aw`), an accepted AW is followed by its W beat with the latched data (`r_drain_w`), and the B response is taken and discarded (`r_drain_b`, which alone lifts `b_ready` outside `WRITE_WAIT_RESP`). The write that was already issued therefore lands. Until B is taken `ctrlwr_engine_idle` stays low and the request path stays closed. Before this the reset abandoned whatever phase was in flight and left a dangling response on the channel's ID.

`ctrlwr_engine_idle` asserts only in `WRITE_IDLE` with no pending request, no active channel reset and nothing left to drain.

14.6 Contrast with the Control-Read Engine

The Control-Write Engine is the producer counterpart to the Control-Read Engine (Section 2.8). Both are per-channel control-descriptor executors, but they implement opposite half of the producer/consumer handshake:

Aspect	Control-Read Engine (2.8)	Control-Write Engine (2.9)
Opcode	DESC_OP_CTRL_READ (2'b01)	DESC_OP_CTRL_WRITE (2'b10)
Role	Consumer gate	Producer doorbell
AXI direction	Read (poll)	Single-beat write
Semantics	Poll until (rd & mask) == expected	Write wr_data to wr_addr, then continue
Retry budget	max_try retry budget (config default)	None (single unconditional write)
Chain effect	Holds off the chain until the gate passes	Releases the chain after the write completes
Descriptor fields	poll_addr / expected / mask / max_try	wr_addr / wr_data

A `CTRL_WRITE` therefore has no maximum-try configuration: it is one unconditional 32-bit write, whereas `CTRL_READ` polls with a bounded retry budget before either passing the gate or timing out.

Last Updated: 2026-07-12

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

15 Scheduler Group

Module: scheduler_group.sv **Location:**

projects/components/dma-ip/rapids/rtl/macro/ **Status:** Implemented **Last Updated:** 2026-09-30

15.1 Overview

The scheduler group wraps one channel: a descriptor engine, the scheduler, the two control engines (control read and control write) and a MonBus arbiter that merges their monitor packets. It is the unit the scheduler group array instantiates once per channel.

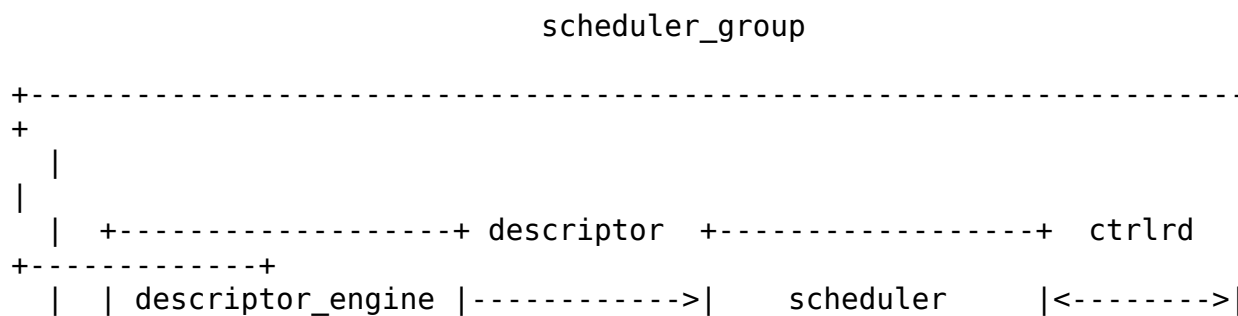
The structure is identical to the RAPIDS Beats group. The one change is the packet-record port set: the scheduler now publishes a per-direction record of the packet's byte length and start offset, and the group passes it through unchanged.

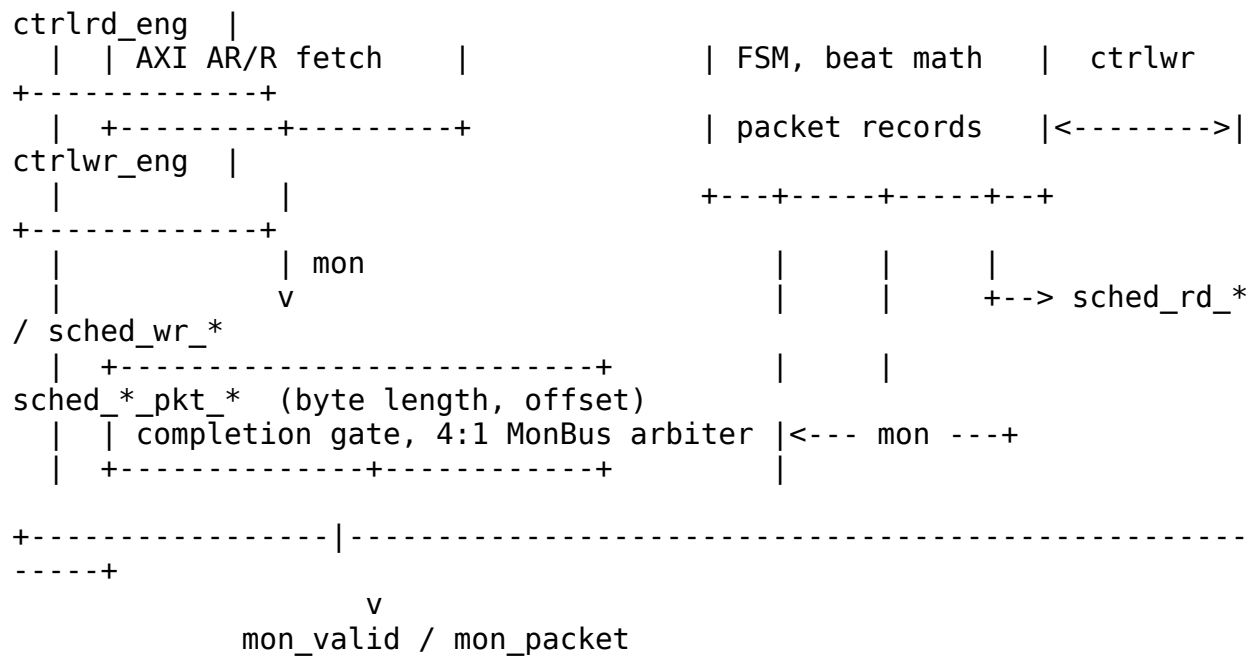
15.1.1 Key Features

- **One channel, four engines:** descriptor engine, scheduler, control read engine, control write engine.
- **Packet-record pass-through:** sched_rd_pkt_* and sched_wr_pkt_* connect the scheduler straight to the group boundary.
- **MonBus aggregation:** a four-client arbiter merges the four engines' packets.
- **Completion gating:** cfg_sched_compl_enable drops CORE Completion packets from the descriptor engine and the scheduler at the arbiter input and acknowledges the emitters, so nothing stalls (rapids ISSUE-005). Errors are never gated.
- **Status aggregation:** idle, FSM state and sticky error are exported.

15.1.2 Block Diagram

15.1.3 Figure 3.1.1: Scheduler Group Block Diagram





15.2 Parameters

Table 3.1.1: Scheduler Group Parameters

Parameter	Default	Description
CHANNEL_ID	0	Channel identifier.
NUM_CHANNELS	8	Total channels.
CHAN_WIDTH	$\$clog2(NUM_CHANNE$ LS)	Channel id width.
ADDR_WIDTH	64	Address width.
DATA_WIDTH	512	Data width in bits. Sets the beat size seen by the scheduler.
AXI_ID_WIDTH	8	AXI ID width for the descriptor and control masters.
DESC_MON_AGENT_ID	16	MonBus agent id of the descriptor engine.
SCHED_MON_AGENT_ID	48	MonBus agent id of the scheduler.

Parameter	Default	Description
CTRLRD_MON_AGENT_ID	32	MonBus agent id of the control read engine.
CTRLWR_MON_AGENT_ID	33	MonBus agent id of the control write engine.
MON_UNIT_ID	1	MonBus unit id.
MON_CHANNEL_ID	0	Base MonBus channel id.
EN_READ	1	Enable the read direction of the scheduler.
EN_WRITE	1	Enable the write direction of the scheduler.
USE_ROW_COL_MAJOR_ADD RESSING	1	Descriptor engine addressing mode.
GEN_MON	1	Enable monitor packet generation.
OFF_W	$\lceil \log_2(\text{DATA_WIDTH}/8) \rceil$	Width of the packet-record byte offset. New in RAPIDS.

15.3 Port List

15.3.1 Clock, Reset and Kick-Off

Table 3.1.2: Clock, Reset and Kick-Off

Signal	Direction	Width	Description
clk	input	1	System clock.
rst_n	input	1	Active-low reset.
apb_valid	input	1	Channel kick-off.
apb_ready	output	1	Descriptor engine ready for kick-off.
apb_addr	input	AW	First descriptor address.

15.3.2 Configuration Interface

Table 3.1.3: Configuration Interface

Signal	Direction	Width	Description
cfg_channel_enable	input	1	Enable this channel.
cfg_channel_reset	input	1	Per-channel soft reset.
cfg_sched_timeout_cycles	input	32	Write-progress timeout window in cycles.
cfg_sched_timeout_limit	input	8	Consecutive timeout windows before a fatal error. 0 means never.
cfg_sched_timeout_enable	input	1	Enable timeout detection.
cfg_sched_err_enable	input	1	Enable error reporting.
cfg_sched_completion_enable	input	1	1 passes CORE Completion packets to the monitor bus; 0 drops them at the arbiter inputs.
cfg_sched_performance_enable	input	1	Enable performance monitoring.
cfg_ctrlrd_max_retry	input	9	Control-read poll retry budget, 0 to 511.
tick_1us	input	1	1 us tick for control-read retry spacing.
cfg_desceng_prefetch	input	1	Enable descriptor prefetch.
cfg_desceng_fifo_thresh	input	4	Prefetch threshold.
cfg_desceng_addr0_base	input	AW	Valid descriptor address range 0 base.
cfg_desceng_addr0_limit	input	AW	Valid descriptor address range 0 limit.
cfg_desceng_	input	AW	Valid descriptor address

Signal	Direction	Width	Description
addr1_base			range 1 base.
cfg_desceng_addr1_limit	input	AW	Valid descriptor address range 1 limit.

15.3.3 Status

Table 3.1.4: Status Interface

Signal	Direction	Width	Description
descriptor_engine_idle	output	1	Descriptor engine idle.
scheduler_idle	output	1	Scheduler idle.
scheduler_state	output	7	Scheduler FSM state, one-hot.
sched_error	output	1	Scheduler error, sticky.

15.3.4 Descriptor AXI Master Interface

Table 3.1.5: Descriptor AXI Master Interface

Signal	Direction	Width	Description
desc_ar_valid / desc_ar_ready	output / input	1	AR handshake.
desc_ar_addr	output	AW	Descriptor fetch address.
desc_ar_len	output	8	Burst length minus one.
desc_ar_size	output	3	Burst size.
desc_ar_burst	output	2	Burst type.
desc_ar_id	output	IW	Transaction ID.
desc_ar_lock, _cache, _prot, _qos, _region	output	1, 4, 3, 4, 4	Attribute fields.
desc_r_valid / desc_r_ready	input / output	1	R handshake.

Signal	Direction	Width	Description
desc_r_data	input	256	Descriptor payload, fixed 256 bits.
desc_r_resp	input	2	Read response.
desc_r_last	input	1	Last beat.
desc_r_id	input	IW	Response ID.

15.3.5 Control Read and Control Write AXI Interfaces

The control read engine fetches a 32-bit semaphore word: `ctrlrd_ar_*` carries the AR channel (address, length, size, burst, id and the same attribute fields as above) and `ctrlrd_r_valid`, `ctrlrd_r_ready`, `ctrlrd_r_data[31:0]`, `ctrlrd_r_id`, `ctrlrd_r_resp` and `ctrlrd_r_last` the R channel. The control write engine writes a 32-bit word: `ctrlwr_aw_*` carries the AW channel, `ctrlwr_w_valid`, `ctrlwr_w_ready`, `ctrlwr_w_data[31:0]`, `ctrlwr_w_strb[3:0]` and `ctrlwr_w_last` the W channel, and `ctrlwr_b_valid`, `ctrlwr_b_ready`, `ctrlwr_b_id` and `ctrlwr_b_resp` the B channel. These are unchanged from RAPIDS Beats. See [Control Read Engine](#) and [Control Write Engine](#).

15.3.6 Scheduler Data Interface

Table 3.1.6: Scheduler Data Interface

Signal	Direction	Width	Description
sched_rd_val id	output	1	Read request.
sched_rd_add r	output	AW	Working source byte address.
sched_rd_beat s	output	32	Beats still to read.
sched_wr_val id	output	1	Write request.
sched_wr_rea dy	input	1	Write engine reports the last burst of the descriptor committed.
sched_wr_add r	output	AW	Working destination byte address.
sched_wr_beat s	output	32	Beats still to write.
sched_rd_don e_strobe	input	1	Read burst issued.
sched_rd_beat	input	32	Beats in that burst.

Signal	Direction	Width	Description
ts_done			
sched_wr_done_strobe	input	1	Write AW issued.
sched_wr_beats_done	input	32	Beats in that AW.
sched_wr_commit_strobe	input	1	Write B response received.
sched_wr_commit_beats	input	32	Beats committed by that B response.
sched_rd_error	input	1	Sticky read error from the read engine.
sched_wr_error	input	1	Sticky write error, including a packet-length error from the sink path.

15.3.7 Packet Record Interface

Table 3.1.7: Packet Record Interface

Signal	Direction	Width	Description
sched_rd_pkt_valid	output	1	One-cycle pulse: a read-side packet record is available.
sched_rd_pkt_ready	input	1	Consumer has room in its record queue.
sched_rd_pkt_bytes	output	32	Packet length in bytes.
sched_rd_pkt_offset	output	OFF_W	Byte offset of the packet start within its first memory beat.
sched_wr_pkt_valid	output	1	One-cycle pulse: a write-side packet record is available.
sched_wr_pkt_ready	input	1	Consumer has room in its record queue.
sched_wr_pkt_bytes	output	32	Packet length in bytes.

Signal	Direction	Width	Description
sched_wr_pkt_offset	output	OFF_W	Byte offset of the packet start within its first memory beat.

The group wires these signals directly to the scheduler. The read-side record goes to the source data path and the write-side record to the sink data path. The scheduler holds a descriptor in its fetch state while a required record queue is full, using the ready inputs. See [Packet Record Interface](#).

15.3.8 MonBus Interface

Table 3.1.8: MonBus Interface

Signal	Direction	Width	Description
i_mon_time	input	monbus_time_stamp_t	Shared monitor timebase.
mon_valid	output	1	Monitor packet valid.
mon_ready	input	1	Consumer ready.
mon_packet	output	monitor_packet_t	Monitor packet.
mon_timestamp	output	monbus_time_stamp_t	Packet timestamp.

15.4 Internal Architecture

15.4.1 Descriptor Path

The descriptor engine fetches 256-bit descriptors from the address range the configuration allows and hands each one to the scheduler with its type, end-of-stream, end-of-list and error flags. The scheduler reports `scheduler_idle` back to the descriptor engine as `channel_idle`. No byte or beat arithmetic happens in the descriptor engine.

15.4.2 Control Engines

When a descriptor asks the scheduler to wait on or update a semaphore, the scheduler talks to the control engines through valid and ready handshakes with an address, data and mask. Both engines share the group's `AXI_ID_WIDTH` and each raises its own MonBus packets.

15.4.3 MonBus Aggregation

A four-client arbiter merges the descriptor engine, scheduler, control read and control write packets. `cfg_sched_compl_enable` gates only the CORE Completion packets of the descriptor

engine and the scheduler: with it low, those packets are consumed at the arbiter input and never reach mon_packet. Error packets are never gated. GEN_MON masks every arbiter input when clear.

15.5 Differences from RAPIDS Beats

Table 3.1.9: Scheduler Group Delta

Item	RAPIDS Beats	RAPIDS
Parameter OFF_W	absent	present, $\$clog2(DATA_WIDTH/8)$
sched_rd_pkt_*, sched_wr_pkt_* ports	absent	present, wired to the scheduler
Everything else		Unchanged

The beats chapter is [Beats Scheduler Group](#). The scheduler is described in [Scheduler](#).

Last Updated: 2026-09-30

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

16 Scheduler Group Array

Module: scheduler_group_array.sv **Location:** projects/components/dma-ip/rapids/rtl/macro/ **Status:** Implemented **Last Updated:** 2026-09-30

16.1 Overview

The scheduler group array instantiates one scheduler group per channel and shares three AXI masters among them: the descriptor fetch master, the control-read master and the control-write

master. It also owns the descriptor-AXI monitor and merges every channel's MonBus output with the monitor's into one stream.

Compared with the RAPIDS Beats array, the sharing, arbitration and monitoring are unchanged. The array adds one parameter, `OFF_W`, and the per-channel packet-record ports, which it connects to each group by channel index.

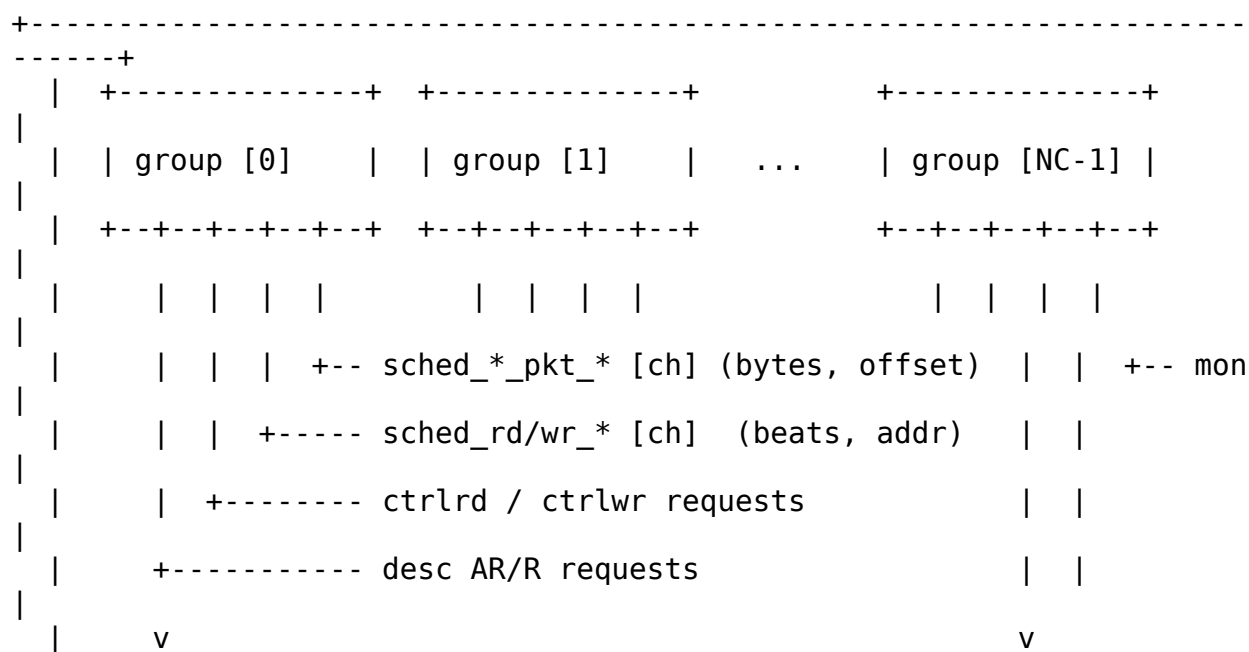
16.1.1 Key Features

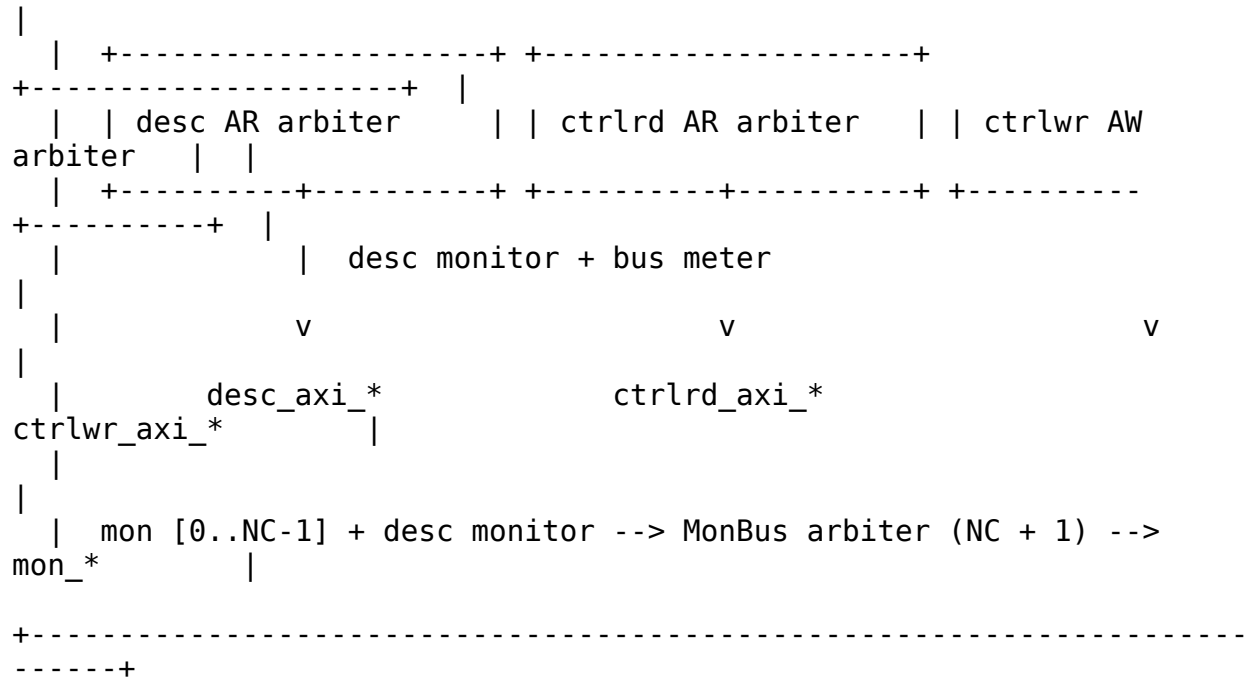
- **One group per channel:** `NUM_CHANNELS` scheduler groups.
- **Shared masters:** three round-robin arbiters with grant acknowledgment (`u_desc_ar_arbiter`, `u_ctrlrd_ar_arbiter`, `u_ctrlwr_aw_arbiter`).
- **Descriptor-AXI monitor:** a lite AXI read monitor on the shared descriptor master, plus an always-on bus meter for the perf window.
- **Unified MonBus:** `NUM_CHANNELS + 1` sources through one round-robin arbiter.
- **Direction gating:** `EN_READ` and `EN_WRITE` let the source array be read-only and the sink array write-only.
- **Per-channel packet records:** vector ports for the byte length and offset of each channel's packet.

16.1.2 Block Diagram

16.1.3 Figure 3.2.1: Scheduler Group Array Block Diagram

scheduler_group_array





16.2 Parameters

Table 3.2.1: Scheduler Group Array Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Number of channels.
CHAN_WIDTH	$\lceil \log_2(\text{NUM_CHANNELS}) \rceil$	Channel id width.
ADDR_WIDTH	64	Address width.
DATA_WIDTH	512	Data width in bits.
AXI_ID_WIDTH	8	AXI ID width.
DESC_MON_BASE_AGENT_ID	16	Base MonBus agent id of the descriptor engines.
SCHED_MON_BASE_AGENT_ID	48	Base MonBus agent id of the schedulers.
DESC_AXI_MON_AGENT_ID	8	MonBus agent id of the descriptor AXI monitor.
MON_UNIT_ID	1	MonBus unit id.
MON_MAX_TRANSACTIONS	16	Transaction table depth of the

Parameter	Default	Description
EN_READ	1	descriptor monitor. Enable the read direction in every scheduler. Clear for a sink array.
EN_WRITE	1	Enable the write direction in every scheduler. Clear for a source array.
USE_AXI_MONITORS	1	0 omits the descriptor-AXI monitor hardware and ties its MonBus source to zero.
USE_ROW_COL_MAJOR_ADDRESSING	1	0 compiles out the extended addressing path.
GEN_MON	1	0 omits the per-channel completion and error emitters.
OFF_W	$\lceil \log_2(\text{DATA_WIDTH}/8) \rceil$	Width of the packet-record byte offset. New in RAPIDS.

16.3 Port List

Vector ports are NUM_CHANNELS wide (NC), indexed by channel.

16.3.1 Clock, Reset and Kick-Off

Table 3.2.2: Clock, Reset and Kick-Off

Signal	Direction	Width	Description
clk	input	1	System clock.
rst_n	input	1	Active-low reset.
apb_valid	input	NC	Per-channel kick-off.
apb_ready	output	NC	Per-channel ready.
apb_addr	input	NC x AW	First descriptor

Signal	Direction	Width	Description
			address per channel.
cfg_channel_enable	input	NC	Per-channel enable.
cfg_channel_reset	input	NC	Per-channel soft reset.

16.3.2 Global Scheduler and Descriptor Engine Configuration

These inputs are applied to every channel.

Table 3.2.3: Global Configuration

Signal	Direction	Width	Description
cfg_sched_enable	input	1	Master scheduler enable.
cfg_sched_timeout_cycles	input	32	Timeout window in cycles.
cfg_sched_timeout_limit	input	8	Consecutive timeouts before a fatal error. 0 means never.
cfg_sched_timeout_enable	input	1	Enable timeout detection.
cfg_sched_error_enable	input	1	Enable error reporting.
cfg_sched_completion_enable	input	1	Enable CORE Completion packets.
cfg_sched_performance_enable	input	1	Enable performance monitoring.
cfg_desceng_enable	input	1	Master descriptor engine enable.
cfg_desceng_prefetch	input	1	Enable descriptor prefetch.
cfg_desceng_fifo_thresh	input	4	Prefetch threshold.
cfg_desceng_addr0_base,	input	AW	Valid descriptor address range 0.

Signal	Direction	Width	Description
_limit			
cfg_desceng_addr1_base, _limit	input	AW	Valid descriptor address range 1.
cfg_ctrlrd_max_try	input	9	Control-read poll retry budget.
tick_1us	input	1	1 us tick for control-read retry spacing.

16.3.3 Descriptor Monitor Configuration and Status

Table 3.2.4: Descriptor Monitor Configuration and Status

Signal	Direction	Width	Description
cfg_desc_mon_enable	input	1	Enable the descriptor AXI monitor.
cfg_desc_mon_err_enable	input	1	Enable error packets.
cfg_desc_mon_perf_enable	input	1	Enable performance packets.
cfg_desc_mon_perf_run	input	1	Perf window control.
cfg_desc_mon_timeout_enable	input	1	Enable timeout packets.
cfg_desc_mon_timeout_cycles	input	32	Timeout threshold.
cfg_desc_mon_latency_threshold	input	32	Latency threshold.
cfg_desc_mon_pkt_mask	input	16	Packet type mask.
cfg_desc_mon_err_select	input	4	Error select.
cfg_desc_mon_err_mask, _timeout_mask, _compl_mask,	input	8 each	Per-class event masks.

Signal	Direction	Width	Description
_thresh_mask, _perf_mask, _addr_mask, _debug_mask			
cfg_sts_desc_mon_busy	output	1	Monitor busy.
cfg_sts_desc_mon_active_txns	output	8	Active transactions.
cfg_sts_desc_mon_error_count	output	16	Error count.
cfg_sts_desc_mon_txn_count	output	32	Transaction count.
cfg_sts_desc_mon_conflict_error	output	1	Conflict error.

16.3.4 Descriptor-AXI Perf Window

The perf counters come from an always-on bus meter on the descriptor R channel, not from the monitor. `cfg_desc_mon_perf_run` opens the window: counters clear on its rising edge, count while it is high and hold while it is low. `USE_AXI_MONITORS` does not gate the meter.

Table 3.2.5: Descriptor-AXI Perf Window

Signal	Direction	Width	Description
sts_desc_mon_win_active	output	1	Window open.
sts_desc_mon_win_cycles	output	32	Cycles in the window.
sts_desc_mon_prod_cycles	output	32	R valid and ready.
sts_desc_mon_bp_cycles	output	32	R valid, not ready.
sts_desc_mon_starv_cycles	output	32	R ready, not valid.
sts_desc_mon_idle_cycles	output	32	Neither valid nor ready.
sts_desc_mon_beat_count	output	32	R handshakes.

Signal	Direction	Width	Description
sts_desc_mon_byte_count	output	64	Bytes moved on R (32 per descriptor beat).
sts_desc_mon_burst_count	output	32	AR handshakes.

16.3.5 Status

Table 3.2.6: Status

Signal	Direction	Width	Description
descriptor_engine_idle	output	NC	Per-channel descriptor engine idle.
scheduler_idle	output	NC	Per-channel scheduler idle.
scheduler_state	output	NC x 7	Per-channel FSM state, one-hot.
sched_error	output	NC	Per-channel scheduler error, sticky.

16.3.6 Shared AXI Masters

The three shared masters carry the same signals as the per-group ports in [Scheduler Group](#), prefixed desc_axi_, ctrlrd_axi_ and ctrlwr_axi_. The descriptor master returns 256-bit data. The control read master returns 32-bit data. The control write master is 32 bits wide with a 4-bit strobe. Each is a single interface: the arbiter selects one channel's request onto it and steers the response back by ID.

16.3.7 Per-Channel Scheduler Data Interface

Table 3.2.7: Per-Channel Scheduler Data Interface

Signal	Direction	Width	Description
sched_rd_val_id	output	NC	Read request per channel.
sched_rd_addr	output	NC x AW	Working source byte address.
sched_rd_beats	output	NC x 32	Beats still to read.
sched_wr_val_id	output	NC	Write request per channel.

Signal	Direction	Width	Description
sched_wr_ready	input	NC	Last burst of the descriptor committed.
sched_wr_addr	output	NC x AW	Working destination byte address.
sched_wr_beats	output	NC x 32	Beats still to write.
sched_rd_done_strobe	input	NC	Read burst issued.
sched_rd_beats_done	input	NC x 32	Beats in that burst.
sched_wr_done_strobe	input	NC	Write AW issued.
sched_wr_beats_done	input	NC x 32	Beats in that AW.
sched_wr_commit_strobe	input	NC	Write B response received.
sched_wr_commit_beats	input	NC x 32	Beats committed.
sched_rd_error	input	NC	Sticky read error.
sched_wr_error	input	NC	Sticky write error.

16.3.8 Per-Channel Packet Record Interface

Table 3.2.8: Per-Channel Packet Record Interface

Signal	Direction	Width	Description
sched_rd_pkt_valid	output	NC	Read-side record pulse per channel.
sched_rd_pkt_ready	input	NC	Read-side consumer has room.
sched_rd_pkt_bytes	output	NC x 32	Packet length in bytes.
sched_rd_pkt_offset	output	NC x OFF_W	Start offset in the first memory beat.
sched_wr_pkt_valid	output	NC	Write-side record pulse per channel.
sched_wr_pkt	input	NC	Write-side consumer has

Signal	Direction	Width	Description
_ready			room.
sched_wr_pkt_bytes	output	NC x 32	Packet length in bytes.
sched_wr_pkt_offset	output	NC x OFF_W	Start offset in the first memory beat.

Element [ch] of each vector connects to the matching port of scheduler_group[ch]. The array adds no logic between them. A source array uses only the read set and a sink array only the write set.

16.3.9 MonBus Interface

Table 3.2.9: MonBus Interface

Signal	Direction	Width	Description
i_mon_time	input	monbus_time stamp_t	Shared monitor timebase.
mon_valid	output	1	Packet valid.
mon_ready	input	1	Consumer ready.
mon_packet	output	monitor_packet_t	Packet.
mon_timestamp	output	monbus_time stamp_t	Timestamp.

16.4 Shared Master Arbitration

Each shared master has a round-robin arbiter over NUM_CHANNELS requesters in acknowledge mode. The grant is held until the granted channel's request is accepted (grant_ack), so a granted channel always sees its own valid and ready. The AR or AW ready seen by a channel is the downstream ready gated by that channel's grant.

16.4.1 Figure 3.2.2: Shared Master Arbitration

```

group[0] --valid--+
group[1] --valid--+--> round-robin arbiter (ack mode) --grant--+-->
mux --> shared AXI master
group[n] --valid--+
                                |
                                grant_ack = grant && valid && ready <----+
R / B <-- routed back to the granted channel by ID

```

16.5 MonBus Aggregation

Table 3.2.10: MonBus Source Assignment

Source	Origin	Content
0 to NC-1	scheduler_group[ch]	Descriptor engine, scheduler, control read and control write packets of the channel
NC	Descriptor AXI monitor	Descriptor fetch monitor packets

One round-robin arbiter with acknowledge mode merges the `NUM_CHANNELS + 1` sources into `mon_*`. Each source carries its own timestamp alongside the packet.

16.6 Differences from RAPIDS Beats

Table 3.2.11: Array Delta

Item	RAPIDS Beats	RAPIDS
Parameter <code>OFF_W</code>	absent	present
<code>sched_rd_pkt_*</code> , <code>sched_wr_pkt_*</code> vectors	absent	present, wired by channel
Everything else		Unchanged

The beats chapter is [Beats Scheduler Group Array](#).

Last Updated: 2026-09-30

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

17 Sink Data Path

Module: `snk_data_path.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro/`
Status: Implemented **Last Updated:** 2026-09-30

17.1 Overview

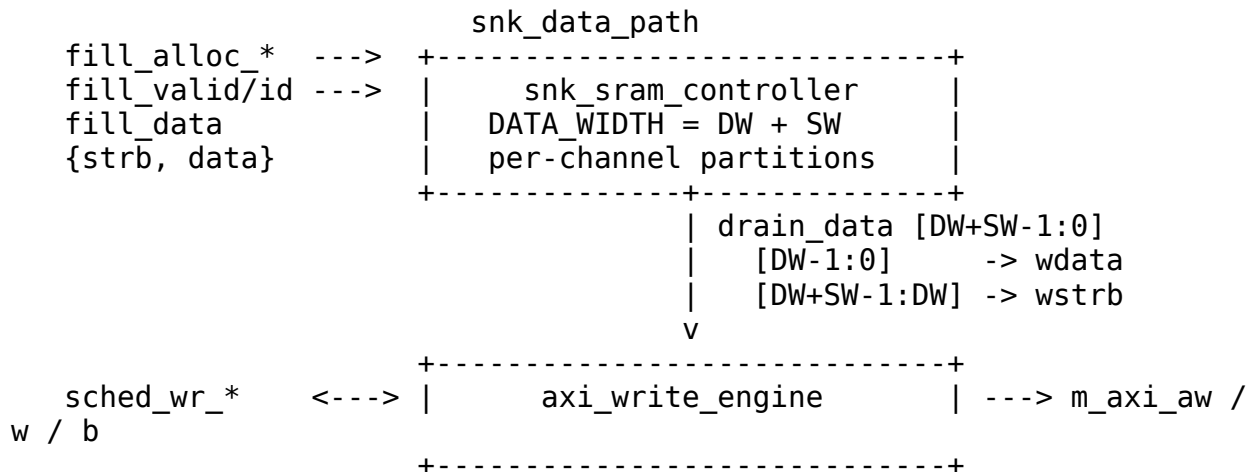
The sink data path joins the per-channel SRAM controller to the AXI write engine. Data enters through the fill interface, waits in the channel's SRAM partition and leaves as AXI4 write bursts.

In RAPIDS every SRAM word carries the byte enables of its data. The fill interface widens from DW to $DW + SW$ bits, where $SW = DW/8$, and the write engine drives `m_axi_wstrb` from the stored enables. Everything else, including the allocation protocol, the scheduler interface and the burst rules, is as in RAPIDS Beats.

17.1.1 Key Features

- **Per-channel SRAM partitions:** flow control by allocation and space-free counts.
- **Byte enables stored with the data:** the SRAM is $DW + SW$ bits wide.
- **AXI write engine:** 4 KB-safe, beat-aligned bursts. See [AXI Write Engine](#).
- **Scheduler integration:** beat counts in, issue and commit strobes out.

17.1.2 Figure 3.3.1: Sink Data Path Block Diagram



17.2 Parameters

Table 3.3.1: Sink Data Path Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Number of channels (alias NC).
ADDR_WIDTH	64	Address width (alias Aw).
DATA_WIDTH	512	Data width in bits (alias Dw). The board uses 256.
AXI_ID_WIDTH	8	AXI ID width (alias Iw).
SRAM_DEPTH	512	Words per channel partition (alias SD).
SEG_COUNT_WIDTH	$\lceil \log_2(\text{SRAM_DEPTH}) \rceil + 1$	Space counter width (alias SCW).
PIPELINE	1	SRAM read pipeline setting.
AW_MAX_OUTSTANDING	8	AW commands the write engine may have in flight.
W_PHASE_FIFO_DEPTH	64	Depth of the write engine W-phase queue.
B_PHASE_FIFO_DEPTH	16	Depth of the write engine B-phase queue.
CIW	$\lceil \log_2(\text{NC}) \rceil$	Channel id width.
SW	DW / 8	Byte enables per beat. New in RAPIDS.

17.3 Interfaces

17.3.1 Fill Interface

Table 3.3.2: Fill Interface

Signal	Direction	Width	Description
fill_alloc_req	input	1	Space allocation request.
fill_alloc_size	input	8	Beats to allocate.

Signal	Direction	Width	Description
fill_alloc_id	input	CIW	Channel to allocate in.
fill_space_free	output	NC x SCW	Free space per channel.
fill_valid	input	1	Fill beat valid.
fill_ready	output	1	Ready for a fill beat.
fill_id	input	CIW	Channel of the fill beat.
fill_data	input	DW + SW	{byte enables, data}. Bits [DW-1:0] are data. Bits [DW+SW-1:DW] are enables.

17.3.2 Configuration

Table 3.3.3: Configuration

Signal	Direction	Width	Description
cfg_axi_wr_xfer_beats	input	8	Write burst size cap in beats, minus one. Renamed from cfg_axi_wr_xfer.

17.3.3 Scheduler Interface

Table 3.3.4: Scheduler Interface

Signal	Direction	Width	Description
sched_wr_val_id	input	NC	Channel requests writes.
sched_wr_ready	output	NC	Last burst of a descriptor committed.
sched_wr_addr	input	NC x AW	Working destination byte address.
sched_wr_beats	input	NC x 32	Beats still to write.
sched_wr_burst_len	input	NC x 8	Requested burst length. Not used for sizing.
sched_wr_done_strobe	output	NC	AW issued.

Signal	Direction	Width	Description
sched_wr_beats_done	output	NC x 32	Beats in that AW.
sched_wr_commit_strobe	output	NC	B response received.
sched_wr_commit_beats	output	NC x 32	Beats committed by that B.
sched_wr_error	output	NC	Sticky write error (bad B response).

17.3.4 AXI4 Write Master

The write master is `m_axi_aw*`, `m_axi_w*` and `m_axi_b*` with IW-bit IDs and DW-bit data. See [AXI4 Interface Specification](#). `m_axi_wstrb` is DW/8 bits and carries the stored byte enables.

17.4 Byte Enable Path

1. The upstream stage presents {strb, data} on `fill_data` with `fill_valid`.
2. The SRAM controller stores the whole DW + SW word. It is instantiated with `DATA_WIDTH = DW + SW`.
3. The drain side returns the same word as `drain_data`.
4. `drain_data[DW-1:0]` goes to `axi_wr_sram_data` and `drain_data[DW+SW-1:DW]` goes to `axi_wr_sram_strb` on the write engine.
5. The write engine drives `m_axi_wdata` and `m_axi_wstrb` from those two.

The SRAM depth counts words, so capacity in beats is unchanged. Only the width grows, by one bit per data byte.

17.5 Differences from RAPIDS Beats

Table 3.3.5: Sink Data Path Delta

Item	RAPIDS Beats	RAPIDS
<code>fill_data</code> width	DW	DW + SW
SRAM word width	DW	DW + SW

Item	RAPIDS Beats	RAPIDS
m_axi_wstrb	all ones	stored per-beat enables
Config port name	cfg_axi_wr_xfer	cfg_axi_wr_xfer_beats
Scheduler port names	sched_wr, sched_wr_done, sched_wr_commit	sched_wr_beats, sched_wr_beats_done, sched_wr_commit_beats
Fill allocation, drain, error export		Unchanged

The beats chapter is [Sink Data Path](#). Its timing diagrams apply to the AW, W and B sequencing unchanged.

Last Updated: 2026-09-30

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

18 Sink Data Path with AXIS (Ingress)

Module: snk_data_path_axis.sv **Location:** projects/components/dma-ip/rapids/rtl/macro/ **Status:** Implemented **Last Updated:** 2026-09-30

18.1 Overview

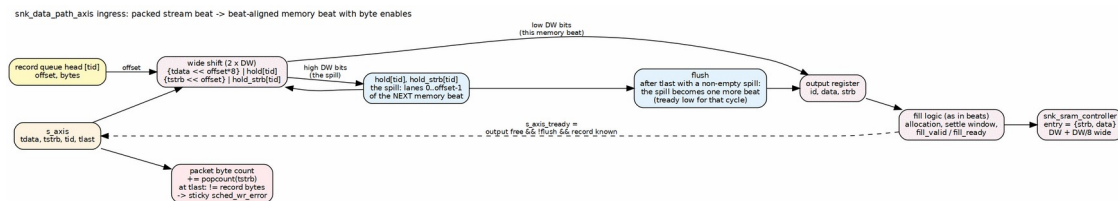
This module puts an AXI-Stream slave in front of the [Sink Data Path](#) and converts a packed byte stream into beat-aligned memory beats with byte enables. It is the block that makes the sink side byte-granular.

A packet arrives with its bytes packed from lane 0. Its destination in memory starts at a byte offset inside the first memory beat. The scheduler supplies that offset in a packet record. The module shifts each stream beat up by the offset, holds the bytes that spill past the beat boundary, and merges them into the next memory beat. After the last stream beat, a non-empty spill becomes one more, partial, memory beat.

18.1.1 Key Features

- **Barrel shift by the destination offset:** stream lane l lands in memory lane $\text{offset} + l$.
- **Byte enables follow the data:** the shifted strobe is stored beside the data and becomes WSTRB.
- **Per-channel state:** spill hold, received-byte count and length-error flag per channel, so packets of different channels interleave by beat.
- **Packet record queue:** four records per channel, fed by the scheduler.
- **Descriptor gating:** the stream is accepted only when the channel has a packet record.
- **Length check:** a packet whose byte count differs from the descriptor sets a sticky channel error.
- **Allocation logic unchanged from RAPIDS Beats,** including the settle wait.

18.1.2 Figure 3.4.1: Ingress Shifter



Ingress shifter

Source: [01_ingress_shifter.dot](#)

18.2 Parameters

Parameters of [Sink Data Path](#) apply unchanged. The additions are:

Table 3.4.1: Additional Parameters

Parameter	Default	Description
AXIS_ID_WIDTH	8	tid width. The low CIW bits

Parameter	Default	Description
AXIS_DEST_WIDTH	4	select the channel. tdest width. Not used to select a channel.
AXIS_USER_WIDTH	1	tuser width. Not used.
SW	DW / 8	Byte lanes per beat.
OFF_W	$\lceil \log_2(SW) \rceil$	Width of the packet-record offset.

18.3 Interfaces

18.3.1 AXI-Stream Slave

Table 3.4.2: AXI-Stream Slave

Signal	Direction	Width	Description
s_axis_tdata	input	DW	Packed data.
s_axis_tstrb	input	SW	Byte strobes. Contiguous from lane 0. Partial only on the tlast beat.
s_axis_tlast	input	1	Last beat of the packet.
s_axis_tid	input	AXIS_ID_WIDTH	Channel select.
s_axis_tdest	input	AXIS_DEST_WIDTH	Unused.
s_axis_tuser	input	AXIS_USER_WIDTH	Unused.
s_axis_tvalid	input	1	Beat valid.
s_axis_tready	output	1	Beat accepted.

s_axis_tready is high when the output register is free, no flush beat is pending and the addressed channel has a packet record queued. A channel's stream therefore stalls until its scheduler has fetched the descriptor. s_axis_tready is a single signal qualified by the TID of the beat on the bus, so a beat for a channel that has no record blocks the stream, and any channel's beats behind it, until that record arrives. Channels on one stream are not isolated from each

other; a system that needs isolation gives each channel its own stream or guarantees the descriptor is queued before the packet is sent.

18.3.2 Scheduler Interface

The write request and completion ports are those of [Sink Data Path](#). The packet record ports are new:

Table 3.4.3: Packet Record Ports

Signal	Direction	Width	Description
sched_wr_pkt_valid	input	NC	Record pulse from the scheduler.
sched_wr_pkt_ready	output	NC	Record queue not full.
sched_wr_pkt_bytes	input	NC x 32	Packet length in bytes.
sched_wr_pkt_offset	input	NC x OFF_W	Destination offset in the first memory beat.

sched_wr_error is the OR of the write engine's B-response error and the packet length error of the channel.

18.3.3 Other Ports

The AXI4 write master is that of [Sink Data Path](#). `cfg_alloc_size` (8 bits) is the allocation segment per request, with 0 treated as 1. The debug outputs are `dbg_sram_bridge_pending`, `dbg_sram_bridge_out_valid`, `dbg_axis_beats_received` and `dbg_axis_packets_received`. `o_active_channel_id` and `o_active_channel_valid` carry the write engine's active channel to the per-channel bus meter.

18.4 Ingress Operation

18.4.1 Packet Records

Each channel has a four-entry queue of {offset, bytes}. A record is pushed on `sched_wr_pkt_valid` when the queue is not full and popped when the packet completes. The head record supplies the offset for every beat of the packet and the expected length at `tlast`.

18.4.2 Placement

For an accepted stream beat, with `off` the head record's offset:

- The wide word is $(tdata \ll off*8) \mid hold_data$, of width $2*DW$.
- The wide strobe is $(tstrb \ll off) \mid hold_strb$, of width $2*SW$.
- The low half goes to the output register as the memory beat and its enables.
- The high half becomes the new hold for the channel.

The first memory beat has its low `off` strobe bits clear, so the bytes below the start address are not written.

18.4.3 Flush and Completion

On `tlast` the received-byte total is compared with the head record. If the wide strobe's high half is non-zero, a flush is scheduled and `s_axis_tready` drops until the hold has been emitted as the last memory beat. The packet then completes and its record pops. If the high half is empty the packet completes on the `tlast` beat itself.

18.4.4 Length Check

`r_pkt_rx_bytes` counts the popcount of `tstrb` over the packet. At `tlast`, a total that differs from the record's byte count sets the channel's sticky error. Nothing is padded or dropped. The error is visible on `sched_wr_error` and the scheduler treats it as fatal.

18.4.5 Allocation

The output register holds one memory beat for the fill interface. Allocation follows the beats design:

- A channel requests `cfg_alloc_size` beats, or the remainder when less is free.
- The next allocation of a channel waits three cycles after its previous one so `fill_space_free` is current.
- A beat leaves the output register only when the fill interface accepts it.

18.5 Differences from RAPIDS Beats

Table 3.4.4: Ingress Delta

Item	RAPIDS Beats	RAPIDS
Stream beat to memory beat	one to one, full lanes	shifted by the offset, spill held
<code>s_axis_tstrb</code>	full only	partial on the <code>tlast</code> beat
Byte enables	none	<code>{strb, data}</code> word

Item	RAPIDS Beats	RAPIDS
into SRAM		
Packet record queue	absent	four per channel
Stream gating	buffers before the descriptor	waits for the packet record
Length mismatch	not checked	sticky channel error

The beats chapter is [Sink Data Path with AXIS](#).

Last Updated: 2026-09-30

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

19 Sink SRAM Controller Specification

Module: `snk_sram_controller_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

19.1 Overview

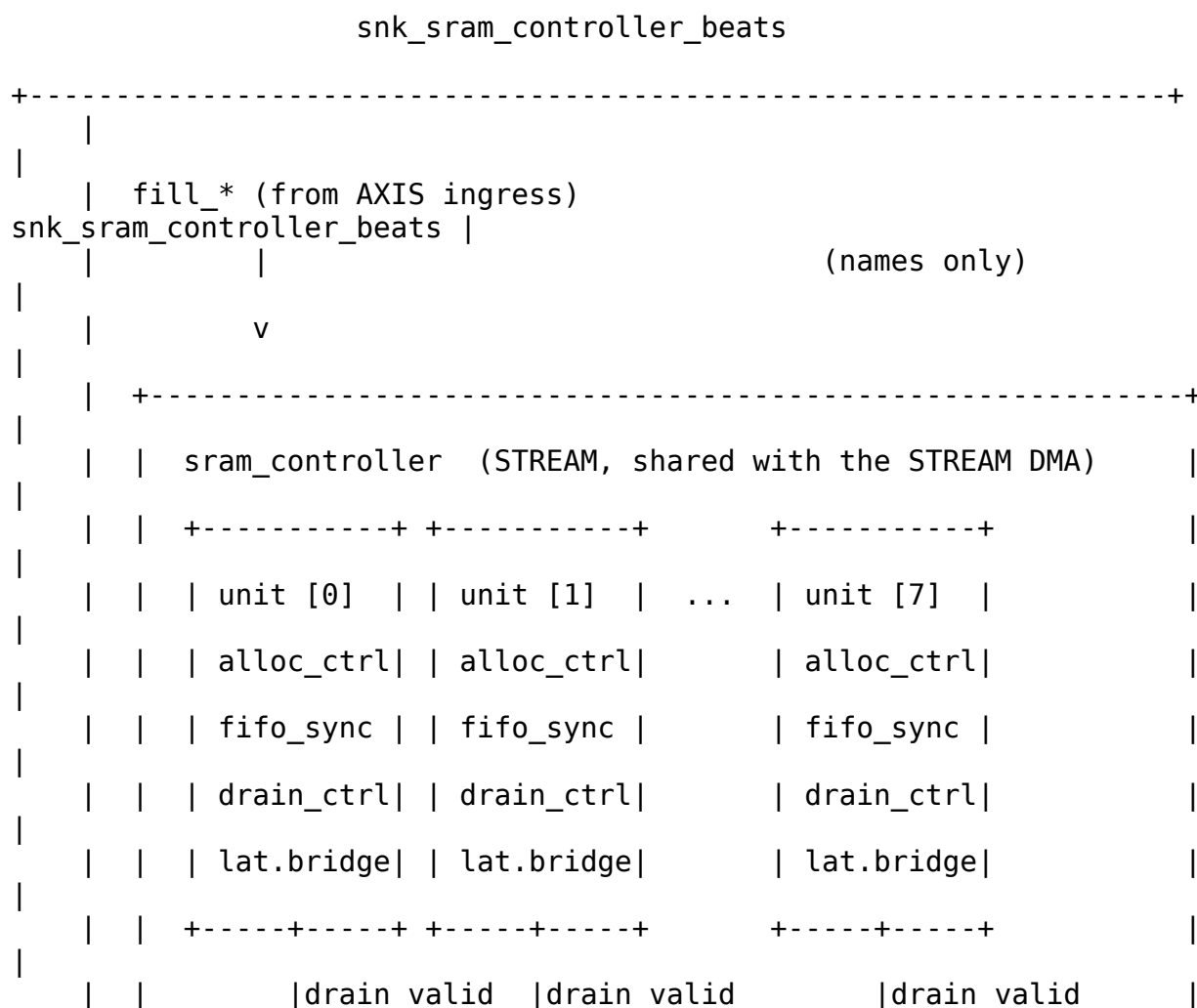
The Sink SRAM Controller is a naming wrapper, with no logic of its own, around STREAM's `sram_controller` – the one per-channel SRAM implementation both DMAs share since `bdf4e0dff`. It maps the RAPIDS `fill_*/drain_*` names onto STREAM's ports; the eight per-channel units (allocation counter, FIFO, drain counter, latency bridge) live inside STREAM's block. It manages 8 channel units, providing per-channel buffering with channel arbitration for the shared AXI write engine.

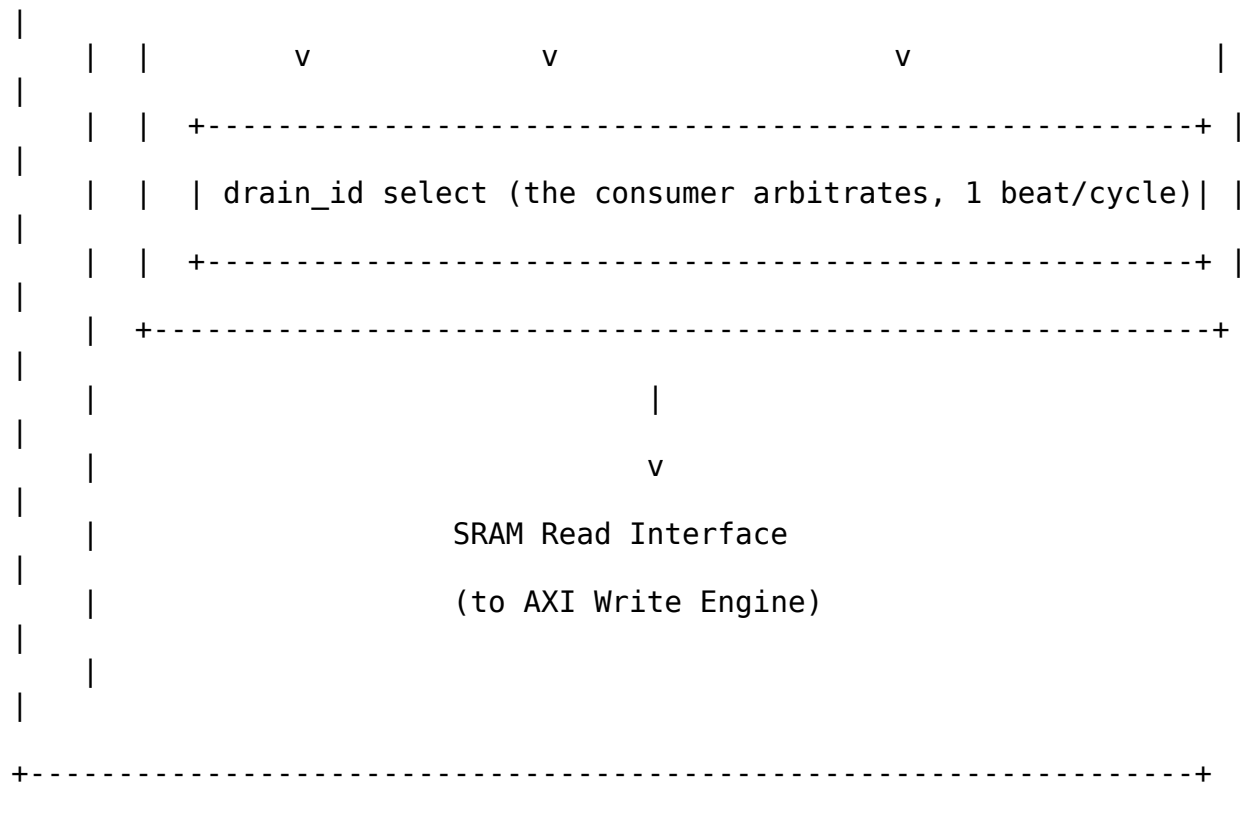
19.1.1 Key Features

- **One shared implementation:** instantiates STREAM's sram_controller once; it holds 8 sram_controller_units, each a gaxi_fifo_sync FIFO with stream_alloc_ctrl, stream_drain_ctrl and stream_latency_bridge
- **Channel Arbitration:** none inside the controller – the AXI write engine picks a channel and presents it on drain_id
- **Flow Control Integration:** per-channel stream_alloc_ctrl / stream_drain_ctrl virtual FIFOs; the drain counter's virtual depth is 2 x SRAM_DEPTH (stream BUG-011)
- **Space Tracking:** Reports available space per channel

19.1.2 Block Diagram

19.1.3 Figure 3.5.1: Sink SRAM Controller Block Diagram





19.2 Parameters

```

parameter int NUM_CHANNELS = 8;
parameter int DATA_WIDTH = 512;
parameter int SRAM_DEPTH = 512; // Per-channel depth
parameter int ADDR_WIDTH = $clog2(SRAM_DEPTH);

```

: Table 3.5.1: Sink SRAM Controller Parameters

19.3 Port List

19.3.1 Clock and Reset

Table 3.5.2: Clock and Reset

Signal	Direction	Width	Description
clk	input	1	System clock
rst_n	input	1	Active-low reset

19.3.2 Fill Interface (ID-Selected)

Table 3.5.3: Fill Interface (ID-Selected)

Signal	Direction	Width	Description
fill_alloc_req	input	1	Allocation request (single, ID-selected)
fill_alloc_size	input	8	Beats to allocate
fill_alloc_id	input	CIW	Transaction ID selects the channel
fill_space_free	output	NC x SCW	Available space per channel
fill_valid	input	1	Fill data valid
fill_ready	output	1	Ready for fill data
fill_id	input	CIW	Transaction ID selects the channel
fill_data	input	DW	Fill data

19.3.3 Arbitrated Drain Interface (to AXI Write Engine)

Table 3.5.4: Arbitrated Drain Interface

Signal	Direction	Width	Description
drain_data_avail	output	NC x SCW	Data available per channel
drain_req	input	NC	Drain reservation request per channel
drain_size	input	NC x 8	Beats to reserve
drain_valid	output	NC	Drain valid, registered (for arbitration)
drain_valid_comb	output	NC	Drain valid, combinational (gates m_axi_wvalid)
drain_read	input	1	Consumer read strobe
drain_id	input	CIW	Channel ID select for drain
drain_data	output	DW	Drain data (muxed from

Signal	Direction	Width	Description
			selected channel)

19.3.4 Debug Interface

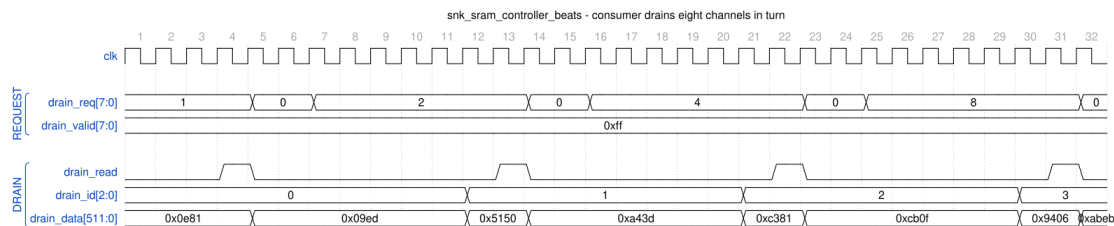
Table 3.5.5: Debug Interface

Signal	Direction	Width	Description
dbg_bridge_p ending	output	NC	Bridge has a pending beat per channel
dbg_bridge_o ut_valid	output	NC	Bridge output valid per channel

19.4 Drain Selection

There is no arbiter inside the controller. Every channel unit raises its own `drain_valid[ch]` / `drain_size[ch]` when it has data; the AXI write engine picks a channel, drives it on `drain_id`, and the controller steers that channel's beats to the single `drain_*` port. Switching channels costs nothing beyond the consumer's own decision, and a channel keeps presenting `drain_valid` until it is empty.

19.4.1 Figure 3.5.2: Drain Selection by the Consumer



snk_sram_controller_beats - consumer drains eight channels in turn

Source: [snk_sram_controller_drain_select.json](#), captured from `dv/tests/macro_beats/test_snk_sram_controller_beats.py` (multi_channel, 8 channels, 512-bit data, 512-entry SRAM, REG_LEVEL=GATE) with WAVES=1.

Reading it: all eight channels have data (`drain_valid = 0xff`) and the consumer walks them in turn. It raises one bit of `drain_req` at a time; the controller answers with `drain_id` naming that channel and `drain_data` presenting its head word on the same cycle, and the consumer's `drain_read` pulse pops it. Each channel's turn lasts nine cycles here because the test releases the request and re-arms the next one; the controller itself needs no gap between channels.

19.5 Integration with AXI Write Engine

```
snk_sram_controller_beats #(
    .NUM_CHANNELS(8),
    .DATA_WIDTH(512),
    .SRAM_DEPTH(512)
) u_snk_sram_ctrl (
    .clk                (clk),
    .rst_n              (rst_n),

    // Per-channel fill interface
    .fill_alloc_req     (fill_alloc_req),
    .fill_alloc_size    (fill_alloc_size),
    .fill_space_free    (fill_space_free),
    .fill_valid         (fill_valid),
    .fill_ready         (fill_ready),
    .fill_data          (fill_data),

    // Arbitrated drain interface
    .drain_req          (sram_drain_req),
    .drain_id           (sram_drain_id),
    .drain_size         (sram_drain_beats),

    // Status
    .drain_data_avail   (channel_data_avail),
    .dbg_bridge_pending (channel_empty),
    .dbg_bridge_out_valid (channel_full)
);
```

Last Updated: 2025-01-10

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

20 Source Data Path Specification

Module: src_data_path_beats.sv **Location:**

projects/components/dma-ip/rapids/rtl/macro_beats/ **Status:** Implemented **Last**

Updated: 2025-01-10

20.1 Overview

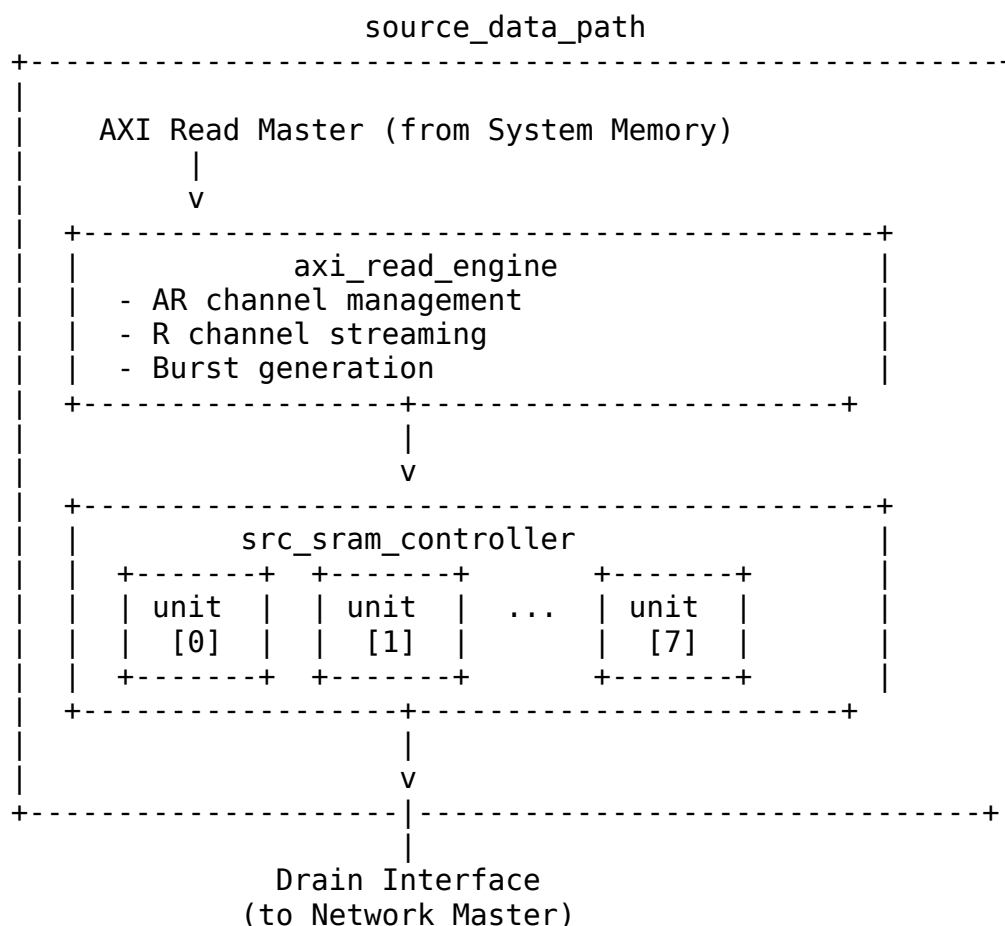
The Source Data Path integrates the AXI read engine and SRAM controller for memory-to-network transfers. Data is read from system memory via AXI4, buffered in SRAM, and delivered through the drain interface to the network.

20.1.1 Key Features

- **AXI Read Engine:** Streaming burst reads from system memory
- **8-Channel SRAM Controller:** Per-channel buffering with flow control
- **Beat-Level Tracking:** Precise flow control at beat granularity
- **Scheduler Integration:** Receives transfer requests from scheduler

20.1.2 Block Diagram

20.1.3 Figure 3.6.1: Source Data Path Block Diagram



20.2 Parameters

```
parameter int NUM_CHANNELS = 8;  
parameter int ADDR_WIDTH = 64;  
parameter int DATA_WIDTH = 512;  
parameter int AXI_ID_WIDTH = 8;  
parameter int SRAM_DEPTH = 512;  
parameter int AR_MAX_OUTSTANDING = 8;  
parameter int R_PHASE_FIFO_DEPTH = 64;
```

: Table 3.6.1: Source Data Path Parameters

20.3 Data Flow

20.3.1 Figure 3.6.2: Source Path Data Flow Sequence

1. Scheduler Read Request
 - sched_rd_valid = 1
 - sched_rd_addr = source address
 - sched_rd_beats = N
 - sched_rd_id = channel

|
v
2. Space Check (stream_alloc_ctrl inside STREAM's sram_controller)
 - Check src_drain_space_free[channel] >= N
 - Allocate N beats of space

|
v
3. AXI Read Transaction
 - axi_read_engine issues AR
 - m_axi_araddr = sched_rd_addr
 - m_axi_arlen = calculated from beats

|
v
4. R Channel Data
 - Data arrives on m_axi_rdata
 - Written to channel's SRAM partition

|
v
5. SRAM Write Complete
 - stream_drain_ctrl notified via the unit's latency bridge
 - Data available for drain

|
v
6. Drain Interface
 - src_drain_valid = 1
 - Data delivered to network master

- |
v
7. Completion
- sched_rd_done_strobe = 1
 - Scheduler notified of completion
-

20.4 Port List

20.4.1 Clock and Reset

Table 3.6.2: Clock and Reset

Signal	Direction	Width	Description
clk	input	1	System clock
rst_n	input	1	Active-low reset

20.4.2 Configuration

Table 3.6.3: Configuration

Signal	Direction	Width	Description
cfg_axi_rd_xfer_beats	input	8	AXI read transfer size in beats (all channels)

20.4.3 Scheduler Interface

Table 3.6.4: Scheduler Interface

Signal	Direction	Width	Description
sched_rd_val_id	input	NC	Channel requests read
sched_rd_addr	input	NC x AW	Source addresses
sched_rd_beats	input	NC x 32	Beats remaining to read
sched_rd_done_strobe	output	NC	Burst completed (pulsed 1 cycle)
sched_rd_beats_done	output	NC x 32	Beats completed in burst
sched_rd_error	output	NC	Sticky error flag per channel

20.4.4 AXI Read Master Interface

Table 3.6.5: AXI Read Master Interface

Signal	Direction	Width	Description
m_axi_arvalid	output	1	AR channel valid
m_axi_arready	input	1	AR channel ready
m_axi_araddr	output	AW	Read address
m_axi_arlen	output	8	Burst length
m_axi_arsize	output	3	Burst size
m_axi_arburst	output	2	Burst type
m_axi_arid	output	ID_W	Transaction ID
m_axi_rvalid	input	1	R channel valid
m_axi_rready	output	1	R channel ready
m_axi_rdata	input	DW	Read data
m_axi_rresp	input	2	Read response
m_axi_rid	input	ID_W	Response ID
m_axi_rlast	input	1	Last beat

20.4.5 Drain Interface (Output to Network)

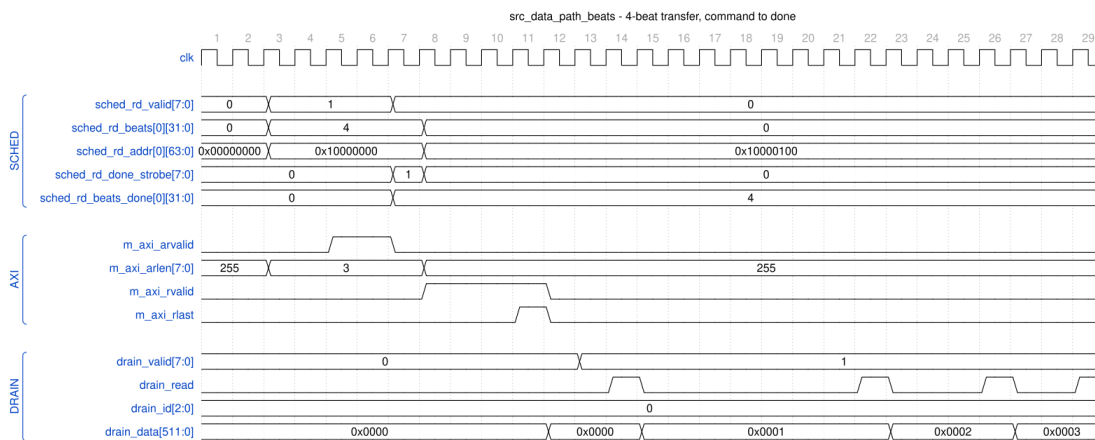
Table 3.6.6: Drain Interface

Signal	Direction	Width	Description
drain_data_avail	output	NC x SCW	Data available per channel
drain_req	input	NC	Drain reservation request per channel
drain_size	input	NC x 8	Beats to reserve
drain_valid	output	NC	Drain data valid per channel
drain_valid_comb	output	NC	Combinational per-channel valid (beat gate)

Signal	Direction	Width	Description
drain_read	input	1	Consumer read strobe
drain_id	input	CIW	Channel ID select for drain
drain_data	output	DW	Drain data (muxed from selected channel)

20.5 Timing Diagram

20.5.1 Figure 3.6.3: Source Path Transfer Timing



src_data_path_beats - 4-beat transfer, command to done

Source: [src_data_path_transfer.json](#), captured inside `rapids_core_beats.u_src.u_source_data_path`, captured from `dv/tests/top_beats/test_rapids_core_beats.py` (source path, channel 0, 4 beats, 512-bit data, `REG_LEVEL=GATE`) with `WAVES=1`.

Reading it: the scheduler holds `sched_rd_valid[0]` with 4 beats at 0x10000000. The read engine issues the AR (`arlen = 3`) two cycles later; the four R beats land from cycle 8, `rlast` on the fourth, and the cycle after it the engine pulses `sched_rd_done_strobe[0]` with `beats_done = 4` – the scheduler advances its address to 0x10000100 on that strobe. The data reaches the source SRAM two cycles after the last R beat (`drain_valid[0]` rises) and the AXIS side drains it one word per `drain_read` pulse at the pace the egress consumer allows.

20.6 Integration Example

```
src_data_path_beats #(
    .NUM_CHANNELS(8),
    .ADDR_WIDTH(64),
    .DATA_WIDTH(512),
    .SRAM_DEPTH(512)
) u_source_path (
    .clk                (clk),
    .rst_n              (rst_n),

    // Scheduler interface
    .sched_rd_valid     (sched_rd_valid),
    .sched_rd_addr      (sched_rd_addr),
    .sched_rd_beats     (sched_rd_beats),
    .sched_rd_done_strobe (sched_rd_done_strobe),
    .sched_rd_beats_done (sched_rd_beats_done),
    .sched_rd_error     (sched_rd_error),

    // AXI read master
    .m_axi_arvalid      (src_axi_arvalid),
    .m_axi_arready      (src_axi_arready),
    .m_axi_araddr       (src_axi_araddr),
    .m_axi_arlen        (src_axi_arlen),
    .m_axi_rvalid       (src_axi_rvalid),
    .m_axi_rready       (src_axi_rready),
    .m_axi_rdata        (src_axi_rdata),
    .m_axi_rlast        (src_axi_rlast),

    // Drain interface
    .drain_valid        (src_drain_valid),
    .drain_data         (src_drain_data),
    .drain_id           (src_drain_id),
    .drain_data_avail   (src_data_avail)
);
```

Last Updated: 2025-01-10

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

21 Source Data Path with AXIS (Egress)

Module: `src_data_path_axis.sv` **Location:**

`projects/components/dma-ip/rapids/rtl/macro/` **Status:** Implemented **Last Updated:** 2026-09-30

21.1 Overview

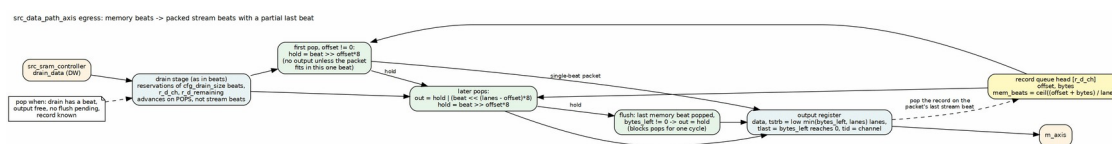
This module puts an AXI-Stream master behind the source data path. It pops beat-aligned memory beats from the SRAM and re-packs them into stream beats that start at lane 0, so a packet leaves as exactly the bytes its descriptor named.

A descriptor's source address may start in the middle of a beat. The scheduler supplies that offset and the byte length in a packet record. The module shifts each memory beat down by the offset, holds the bytes that belong to the next stream beat, and marks the last stream beat with a contiguous strobe and `tlast`. One descriptor produces one AXI-Stream packet.

21.1.1 Key Features

- **Barrel shift by the source offset:** memory lane offset + 1 goes out on stream lane 1.
- **Partial last beat:** `tstrb` is the low $\min(\text{bytes_left}, \text{SW})$ lanes and `tlast` marks the final beat.
- **Packets follow descriptors**, not drain reservations.
- **Two-stage reservation and drain**, unchanged from RAPIDS Beats. The drain stage advances on memory-beat pops, not on stream handshakes.
- **Per-channel state:** hold, bytes-left and beats-left per channel, so reservations of different channels interleave.
- **Packet record queue:** four records per channel, fed by the scheduler.

21.1.2 Figure 3.7.1: Egress Shifter



Egress shifter

Source: [02_egress_shifter.dot](#)

21.2 Parameters

Table 3.7.1: Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Channels (alias NC).
ADDR_WIDTH	64	Address width (alias Aw).
DATA_WIDTH	512	Data width in bits (alias DW). The board uses 256.
AXI_ID_WIDTH	8	AXI ID width (alias IW).
SRAM_DEPTH	512	Words per channel partition.
SEG_COUNT_WIDTH	$\lceil \log_2(\text{SRAM_DEPTH}) \rceil + 1$	Space counter width.
PIPELINE	1	SRAM read pipeline setting.
AR_MAX_OUTSTANDING	8	ARs the read engine may have in flight.
AXIS_ID_WIDTH	8	tid width.
AXIS_DEST_WIDTH	4	tdest width.
AXIS_USER_WIDTH	1	tuser width.
SW	DW / 8	Byte lanes per beat.
OFF_W	$\lceil \log_2(\text{SW}) \rceil$	Width of the packet-record offset.

21.3 Interfaces

21.3.1 Configuration

Table 3.7.2: Configuration

Signal	Direction	Width	Description
cfg_axi_rd_xfer_beats	input	8	Read burst size cap in beats, minus one.
cfg_drain_size	input	8	Beats per drain reservation. 0 is treated as 1.

21.3.2 Scheduler Interface

Table 3.7.3: Scheduler Interface

Signal	Direction	Width	Description
sched_rd_val id	input	NC	Channel requests reads.
sched_rd_add r	input	NC x AW	Working source byte address.
sched_rd_bea ts	input	NC x 32	Beats still to read.
sched_rd_pkt _valid	input	NC	Record pulse from the scheduler.
sched_rd_pkt _ready	output	NC	Record queue not full.
sched_rd_pkt _bytes	input	NC x 32	Packet length in bytes.
sched_rd_pkt _offset	input	NC x OFF_W	Source offset in the first memory beat.
sched_rd_don e_strobe	output	NC	AR issued.
sched_rd_bea ts_done	output	NC x 32	Beats in that AR.
sched_rd_err or	output	NC	Sticky read error.

21.3.3 AXI-Stream Master

Table 3.7.4: AXI-Stream Master

Signal	Direction	Width	Description
m_axis_tdata	output	DW	Packed data.
m_axis_tstrb	output	SW	Contiguous from lane 0. Partial only on the tlast beat.
m_axis_tlast	output	1	Last beat of the packet.
m_axis_tid	output	AXIS_ID_WI DTH	Channel.
m_axis_tdest	output	AXIS_DEST_ WIDTH	Channel.

Signal	Direction	Width	Description
m_axis_tuser	output	AXIS_USER_WIDTH	Zero.
m_axis_tvalid	output	1	Beat valid.
m_axis_tready	input	1	Consumer ready.

21.3.4 Other Ports

The AXI4 read master is that of the [AXI Read Engine](#), with IW-bit IDs and DW-bit data. The debug outputs are dbg_rd_all_complete, dbg_r_beats_rcvd, dbg_sram_writes, dbg_arb_request, dbg_sram_bridge_pending, dbg_sram_bridge_out_valid, dbg_axis_beats_sent and dbg_axis_packets_sent.

21.4 Egress Operation

21.4.1 Reservation and Drain

Unchanged from RAPIDS Beats. A round-robin decision per cycle selects a channel whose available data, less reservations still in flight, reaches cfg_drain_size beats. It may also select one whose fill is finished and has any data left. Each decision is a one-cycle drain_req of up to cfg_drain_size beats. Up to four reservations queue ahead of the drain stage, which loads the next entry on the last beat of the current one.

21.4.2 Pop

A memory beat is popped when all of these hold:

- The drain stage is active on channel c and has a beat.
- The output register is free or being taken.
- No flush is pending.
- The channel's packet record is known.

Each pop counts against the reservation block, so the drain stage advances on pops.

21.4.3 Shift and Hold

With off the head record's offset:

Table 3.7.5: Egress Shift Rules

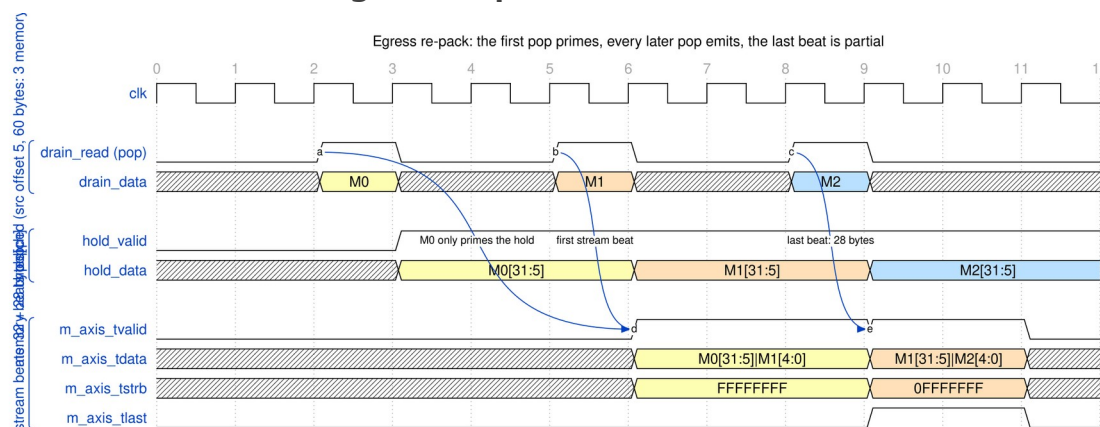
Situation	Action
$off = 0$	Every pop emits a stream beat.
$off > 0$, first pop	The pop only primes the hold with beat $\gg off * 8$. It emits only when the packet fits in that single memory beat.
$off > 0$, later pops	Emit hold $(beat \ll (SW - off) * 8)$. The new hold is beat $\gg off * 8$.
Last memory beat popped, bytes left	The flush emits the hold alone.

The memory beat count of a packet is $(off + bytes + SW - 1) \gg OFF_W$, computed in 33 bits.

21.4.4 Strobe and Last

Each emitted beat carries $n = \min(bytes_left, SW)$ bytes. $tstrb$ is the low n lanes. $tlast$ is set on the beat where $bytes_left$ equals n . That beat also retires the packet record and clears the channel's hold, started flag and counters.

21.4.5 Waveform 3.7: Egress Re-pack



Egress re-pack

Source: [01_egress_repack.json](#)

Offset 5, 60 bytes and a 256-bit beat give three memory beats. The first pop, M0, only primes the hold. The second pop emits the first stream beat with 32 bytes. The third pop emits the last beat with 28 bytes, $tstrb$ 0FFFFFFF and $tlast$.

21.5 Differences from RAPIDS Beats

Table 3.7.6: Egress Delta

Item	RAPIDS Beats	RAPIDS
Stream beat from memory beat	one to one, full lanes	shifted by the offset, hold merged
m_axis_tstrb	all ones	contiguous, partial on the last beat
m_axis_tlast	on the drain block boundary	on the last byte of the descriptor
Drain advance	stream handshakes	memory-beat pops
Packet framing	one per drain block	one per descriptor
Packet record queue	absent	four per channel
Reservation stage, allocation of drain		Unchanged

The beats chapter is [Source Data Path with AXIS](#).

Last Updated: 2026-09-30

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

22 Source SRAM Controller Specification

Module: src_sram_controller_beats.sv **Location:** projects/components/dma-ip/rapids/rtl/macro_beats/ **Status:** Implemented **Last Updated:** 2025-01-10

22.1 Overview

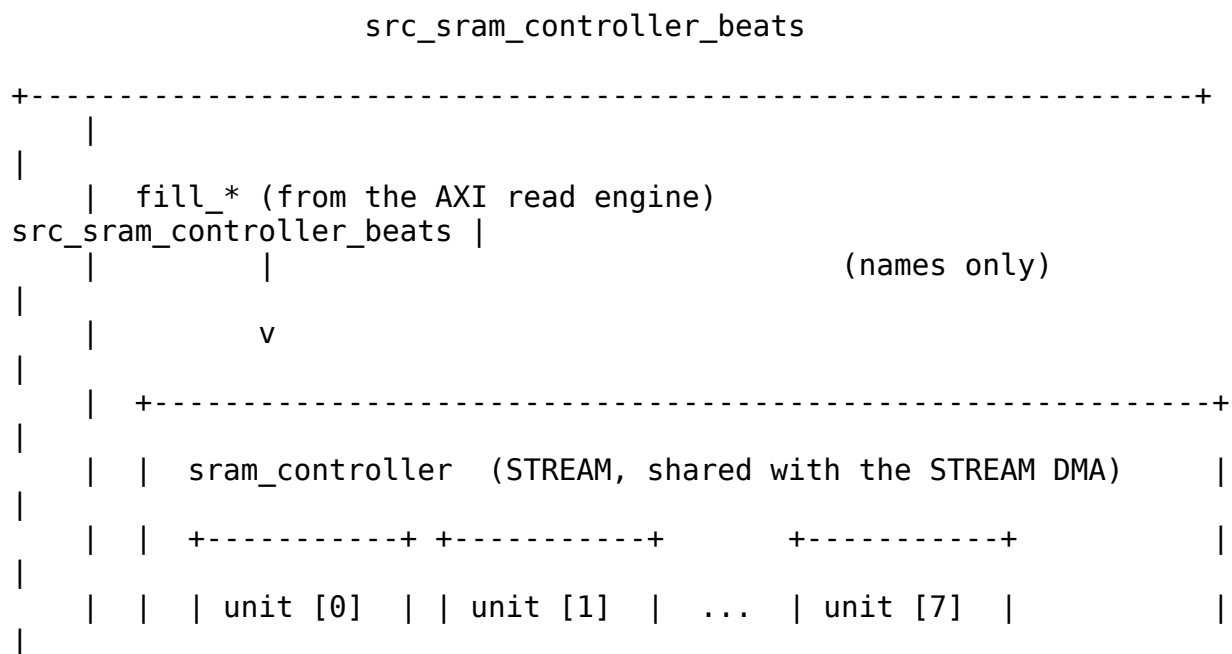
The Source SRAM Controller is a naming wrapper, with no logic of its own, around STREAM's `sram_controller` – the one per-channel SRAM implementation both DMAs share since `bdf4e0dff`. It maps the RAPIDS `fill_*`/`drain_*` names onto STREAM's ports; the eight per-channel units (allocation counter, FIFO, drain counter, latency bridge) live inside STREAM's block. It manages 8 channel units for the source data path, providing per-channel buffering with channel arbitration for data delivery to the network.

22.1.1 Key Features

- **One shared implementation:** instantiates STREAM's `sram_controller` once; it holds 8 `sram_controller_units`, each a `gaxi_fifo_sync` FIFO with `stream_alloc_ctrl`, `stream_drain_ctrl` and `stream_latency_bridge`
- **Channel Arbitration:** none inside the controller – the AXIS egress arbiter picks a channel and presents it on `drain_id`
- **Flow Control Integration:** per-channel `stream_alloc_ctrl` / `stream_drain_ctrl` virtual FIFOs; the drain counter's virtual depth is 2 x `SRAM_DEPTH` (stream BUG-011)
- **Data Availability Tracking:** Reports available data per channel

22.1.2 Block Diagram

22.1.3 Figure 3.8.1: Source SRAM Controller Block Diagram



22.3 Port List

22.3.1 Clock and Reset

Table 3.8.2: Clock and Reset

Signal	Direction	Width	Description
clk	input	1	System clock
rst_n	input	1	Active-low reset

22.3.2 Fill Interface (ID-Selected, from AXI Read Engine)

Table 3.8.3: Fill Interface (ID-Selected)

Signal	Direction	Width	Description
fill_alloc_req	input	1	Allocation request (single, ID-selected)
fill_alloc_size	input	8	Beats to allocate
fill_alloc_id	input	CIW	Transaction ID selects the channel
fill_space_free	output	NC x SCW	Available space per channel
fill_valid	input	1	Fill data valid
fill_ready	output	1	Ready for fill data
fill_id	input	CIW	Transaction ID selects the channel
fill_data	input	DW	Fill data

22.3.3 Arbitrated Drain Interface (to Network)

Table 3.8.4: Arbitrated Drain Interface

Signal	Direction	Width	Description
drain_data_avail	output	NC x SCW	Data available per channel
drain_req	input	NC	Drain reservation request per channel
drain_size	input	NC x 8	Beats to reserve

Signal	Direction	Width	Description
drain_valid	output	NC	Drain data valid per channel
drain_valid_comb	output	NC	Combinational per-channel valid (beat gate)
drain_read	input	1	Consumer read strobe
drain_id	input	CIW	Channel ID select for drain
drain_data	output	DW	Drain data (muxed from selected channel)

22.3.4 Debug Interface

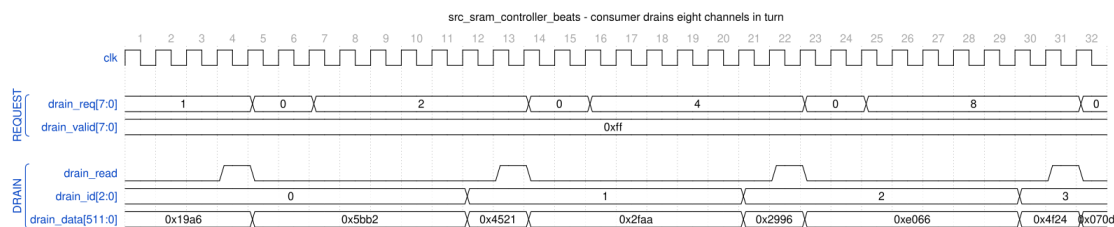
Table 3.8.5: Debug Interface

Signal	Direction	Width	Description
dbg_bridge_p_ending	output	NC	Bridge has a pending beat per channel
dbg_bridge_output_valid	output	NC	Bridge output valid per channel

22.4 Drain Selection

There is no arbiter inside the controller. Every channel unit raises its own `drain_valid[ch]` / `drain_size[ch]` when it has data; the AXIS egress arbiter picks a channel, drives it on `drain_id`, and the controller steers that channel's beats to the single `drain_*` port. Switching channels costs nothing beyond the consumer's own decision, and a channel keeps presenting `drain_valid` until it is empty.

22.4.1 Figure 3.8.2: Drain Selection by the Consumer



src_sram_controller_beats - consumer drains eight channels in turn

Source: [src_sram_controller_drain_select.json](#), captured from `dv/tests/macro_beats/test_src_sram_controller_beats.py` (multi_channel, 8 channels, 512-bit data, 512-entry SRAM, REG_LEVEL=GATE) with WAVES=1.

Reading it: the same selection contract as the sink controller (Figure 3.5.2). With `drain_valid = 0xff` the consumer asks for one channel at a time through `drain_req`; `drain_id` and `drain_data` follow the request on the same cycle and `drain_read` pops the word. The nine-cycle turn per channel is the test's pacing, not the controller's.

22.5 Comparison: Sink vs Source SRAM Controller

Table 3.8.6: Sink vs Source Controller Comparison

Aspect	Sink Controller	Source Controller
Fill Source	Network ingress	AXI read data
Drain Destination	AXI write engine	Network egress
Fill Interface	Per-channel from network	Per-channel from AXI R
Drain Interface	Arbitrated SRAM read	Arbitrated drain to network
Primary Use	Network -> Memory	Memory -> Network

22.6 Integration Example

```
src_sram_controller_beats #(
    .NUM_CHANNELS(8),
    .DATA_WIDTH(512),
    .SRAM_DEPTH(512)
) u_src_sram_ctrl (
    .clk                (clk),
    .rst_n              (rst_n),

    // Per-channel fill interface (from AXI read engine)
    .fill_alloc_req     (fill_alloc_req),
    .fill_alloc_size    (fill_alloc_size),
    .fill_space_free    (fill_space_free),
    .fill_valid         (fill_valid),
    .fill_ready         (fill_ready),
    .fill_data          (fill_data),

    // Arbitrated drain interface (to network)
```

```

        .drain_valid      (src_drain_valid),
        .drain_data      (src_drain_data),
        .drain_id        (src_drain_id),

        // Status
        .drain_data_avail (channel_data_avail),
        .dbg_bridge_pending (channel_empty),
        .dbg_bridge_out_valid (channel_full)
    );

```

Last Updated: 2025-01-10

RTL Design Sherpa · Learning Hardware Design Through Practice · [GitHub](#) · [Documentation Index](#) · [MIT License](#)

23 RAPIDS Core, Sink Half and Source Half

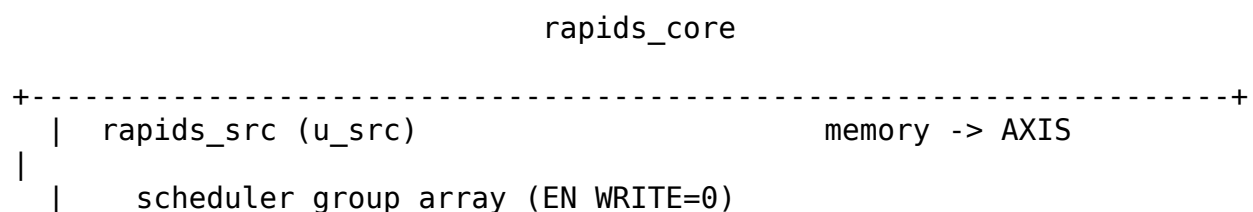
Module: rapids_core.sv, rapids_snk.sv, rapids_src.sv **Location:** projects/components/dma-ip/rapids/rtl/macro/ **Status:** Implemented **Last Updated:** 2026-09-30

23.1 Overview

rapids_core is a thin structural wrapper over two independent halves. rapids_src moves memory to the network. rapids_snk moves the network to memory. The halves share no logic and no signals. The core only presents one boundary and merges the two MonBus streams.

The structure, the port naming convention and the register and monitor plumbing are those of RAPIDS Beats. This chapter records what the byte-granular RTL contains and how it differs. The full port tables are in [RAPIDS Core Beats](#), with the port renames below applied.

23.1.1 Figure 3.9.1: Core Block Diagram



```

|
|   src_data_path_axis  --> m_axis_*      m_axi_rd_* <--
|
|   AXIS egress monitor (agent 0x0A)
|
|   monbus_arbiter
|
|
|
|   rapids_snk (u_snk)                                AXIS -> memory
|
|   scheduler_group_array (EN_READ=0)
|
|   snk_data_path_axis  <-- s_axis_*      m_axi_wr_* -->
|
|   AXIS ingress monitor (agent 0x09)
|
|   monbus_arbiter
|
|
|
|   u_mon_arbiter (2 clients)  --> mon_valid / mon_packet / timestamp
|

```

```

+-----+

```

23.2 Byte-Granular Content

23.2.1 Packet Record Nets

Each half declares the packet record nets of its direction and connects them between its scheduler group array and its AXIS data path:

Table 3.9.1: Packet Record Nets per Half

Half	Nets	Producer	Consumer
rapids_src	sched_rd_pk t_valid, _ready, _bytes, _offset	array read set	src_data_path_axis
rapids_snk	sched_wr_pk t_valid, _ready,	array write set	snk_data_path_axis

Half	Nets	Producer	Consumer
	<code>_bytes,</code> <code>_offset</code>		

The offset nets are $\lceil \log_2(DW/8) \rceil$ bits wide. The unused direction of each array is tied off: valid, bytes and offset are left open and ready is held at all ones so nothing stalls. The record nets are internal, so the core port list has no packet record ports.

23.2.2 Port Renames

The core ports that carry the beat-count unit are renamed. Everything else keeps its Beats name.

Table 3.9.2: Core Port Renames

RAPIDS Beats	RAPIDS
<code>cfg_axi_rd_xfer</code>	<code>cfg_axi_rd_xfer_beats</code>
<code>cfg_axi_wr_xfer</code>	<code>cfg_axi_wr_xfer_beats</code>
<code>src_dbg_r_rcvd</code>	<code>src_dbg_r_beats_rcvd</code>
<code>src_dbg_axis_sent</code>	<code>src_dbg_axis_beats_sent</code>
<code>snk_dbg_axis_received</code>	<code>snk_dbg_axis_beats_received</code>

23.2.3 Data Ports

Table 3.9.3: Data Ports

Interface	Signals	Notes
Source AXIS	<code>m_axis_*</code>	<code>tstrb</code> is contiguous, partial on the tlast beat.
Sink AXIS	<code>s_axis_*</code>	<code>tid</code> selects the channel. Partial <code>tstrb</code> only on tlast.
Source AXI read	<code>m_axi_rd_*</code>	<code>ARADDR</code> is beat-aligned, bursts stop at 4 KB.
Sink AXI write	<code>m_axi_wr_*</code>	<code>wstrb</code> carries the stored byte enables.

23.3 Parameters

Table 3.9.4: Core Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Channels.
ADDR_WIDTH	64	Address width.
DATA_WIDTH	512	Data width in bits. The board uses 256.
AXI_ID_WIDTH	8	AXI ID width.
SRAM_DEPTH	512	Words per channel.
PIPELINE	1	Write engine pipelining.
AR_MAX_OUTSTANDING, AW_MAX_OUTSTANDING	8	Outstanding bursts.
W_PHASE_FIFO_DEPTH, B_PHASE_FIFO_DEPTH	64, 16	Write engine queues.
AXIS_ID_WIDTH, AXIS_DEST_WIDTH, AXIS_USER_WIDTH	8, 4, 1	AXIS side widths.
DESC_MON_BASE_AGENT_ID	16	Descriptor engine agents.
SCHED_MON_BASE_AGENT_ID	48	Scheduler agents.
DESC_AXI_MON_AGENT_ID	8	Descriptor AXI monitor.
SNK_AXIS_MON_AGENT_ID	9	Sink ingress AXIS monitor.
SRC_AXIS_MON_AGENT_ID	10	Source egress AXIS monitor.
ACLK_MHZ	100	Clock in MHz for the AXIS monitors' microsecond tick.
MON_UNIT_ID, MON_MAX_TRANSACTIONS	1, 16	Monitor unit id and table depth.
USE_AXI_MONITORS	1	0 omits the monitor hardware.
USE_ROW_COL_MAJOR_ADDRESSING	1	0 compiles out extended addressing.
GEN_MON	1	0 omits the completion and error emitters.

23.4 Differences from RAPIDS Beats

Table 3.9.5: Core Delta

Item	RAPIDS Beats	RAPIDS
Half structure, monitor merge		Unchanged
Packet record nets inside each half	absent	present
Data path instances	*_axis_beats	byte-granular *_axis
Beat-unit port names	no suffix	_beats suffix, see Table 3.9.2

The Beats chapters for the halves are folded into the core chapter there.

Last Updated: 2026-09-30

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

24 RAPIDS Registers Specification

Module: rapids_regs.sv (generated) **Source:** projects/components/dma-ip/rapids/rtl/macro_beats/rapids_regs.rdl, rapids_mon_regs.rdl **Generated:** projects/components/dma-ip/rapids/regs/generated/rtl/rapids_regs.sv **Status:** Implemented (PeakRDL-generated)

24.1 Overview

rapids_regs is the PeakRDL-generated register block for the RAPIDS Beats top-level. It exposes configuration and status through a single command/response CPU interface and drives the

hardware via a `hwif_out` structure that `rapids_config_block` maps onto the core/monitor `cfg_*` signals.

A single addrmap (one APB slave) contains two regfiles:

- **Base config/status** registers at 0x100-0x3FF.
- A nested **rapids_mon_regs** monitor regfile at base 0x1000, decoded under `hwif_out.MON.*` (all DAXMON / RDMON / WRMON / MON_FIFO and per-monitor performance registers).

Because the monitor regfile lives at 0x1000, the register-block address decode requires at least **13 bits** (`cpuif_addr` is a 13-bit compare in the generated RTL). The APB address bus feeding the block (`APB_ADDR_WIDTH`) must therefore be ≥ 13 to reach the monitor regfile.

24.2 Base Register Map (0x100-0x3FF)

Table 3.10.1: Base Register Map

Offset	Name	Access	Description
0x100	GLOBAL_CTRL	RW	GLOBAL_EN[0], GLOBAL_RST[1] (self-clearing)
0x104	GLOBAL_STATUS	RO	SYSTEM_IDLE[0] (all channels idle)
0x108	VERSION	RO	MINOR[7:0]=0x5A, MAJOR[15:8], NUM_CHANNELS[23:16]=0x08
0x120	CHANNEL_ENABLE	RW	CH_EN[7:0] per-channel enable
0x124	CHANNEL_RESET	RW	CH_RST[7:0] per-channel reset (self-clearing)
0x140	CHANNEL_IDLE	RO	CH_IDLE[7:0] per-channel idle
0x144	DESC_ENGINE_IDLE	RO	Per-channel descriptor-engine idle
0x148	SCHEDULER_IDLE	RO	Per-channel scheduler idle (excludes CH_ERROR)
0x150-0x16C	CH_STATE[0	RO	Per-channel FSM state

Offset	Name	Access	Description
	..7].STATE		(stride 0x4)
0x170	SCHED_ERROR	RO	Per-channel sticky scheduler error
0x174	AXI_RD_COMPLETE	RO	Per-channel read-complete flags
0x178	AXI_WR_COMPLETE	RO	Per-channel write-complete flags
0x200	SCHED_TIMEOUT_CYCLES	RW	TIMEOUT_CYCLES[31:0] write-progress window (reset 1000)
0x204	SCHED_CONFIG	RW	SCHED_EN, TIMEOUT_EN, ERR_EN, COMPL_EN, PERF_EN
0x208	SCHED_TIMEOUT_LIMIT	RW	LIMIT[7:0] consecutive-timeout escalation limit (reset 4, 0 = never)
0x220	DESCENG_CONFIG	RW	DESCENG_EN[0], PREFETCH_EN[1], FIFO_THRESH[5:2] (reset 8)
0x224	DESCENG_ADDR0_BASE	RW	Descriptor address range 0 base [31:0]
0x228	DESCENG_ADDR0_LIMIT	RW	Descriptor address range 0 limit [31:0]
0x22C	DESCENG_ADDR1_BASE	RW	Descriptor address range 1 base [31:0]
0x230	DESCENG_ADDR1_LIMIT	RW	Descriptor address range 1 limit [31:0]
0x240	CTRL_CONFIG	RW	CTRLRD_MAX_TRY[8:0] control-read poll retry budget (0-511, reset 16)
0x2A0	AXI_XFER_CONFIG	RW	AXI transfer sizing configuration
0x2B0	PERF_CONFIG	RW	Performance profiler configuration
0x2C0	OBS_CTRL	RW	Observation mux control

Offset	Name	Access	Description
			(channel/category select)
0x2C4	OBS_FLAGS	RO	Observation flags
0x2C8	OBS_DATA0	RO	Observation data word 0
0x2CC	OBS_DATA1	RO	Observation data word 1
0x35C	PERF_CH_SEL	RW	Performance channel select
0x378	HIST_SEL	RW	Histogram bucket select
0x37C	HIST_DATA	RO	Histogram bucket data
0x380	HIST_TOTAL	RO	Histogram total count

24.2.1 Key Register Fields

SCHED_TIMEOUT_LIMIT (0x208) – LIMIT[7:0] sets the number of consecutive write-progress timeout windows a channel tolerates before the recoverable timeout escalates to a fatal, sticky CH_ERROR. 0 = never escalate (pure soft timeout). Total time to escalate is approximately LIMIT × TIMEOUT_CYCLES. Reset value is 4.

DESCENG_CONFIG (0x220) – PREFETCH_EN[1] and FIFO_THRESH[5:2] control the now-functional descriptor prefetch: prefetch off = on-demand chaining (~1 ahead), prefetch on = buffer up to FIFO_THRESH descriptors ahead.

CTRL_CONFIG (0x240) – CTRLRD_MAX_TRY[8:0] bounds the control-read poll retry budget (0-511, reset 16) fed to every channel's ctrlrd_engine. A CTRL_READ descriptor polls its gate address once per retry; if the masked value never matches within CTRLRD_MAX_TRY attempts the engine raises ctrlrd_error so a never-satisfied gate cannot hang the channel. See the Control-Read / Control-Write Engine specs (Sections 2.8 / 2.9) and the Scheduler control interface (Section 2.1).

24.3 Monitor Register Map (rapids_mon_regs @ 0x1000)

The monitor regfile is instantiated at base 0x1000 under hwif_out.MON.*. It holds the MonBus FIFO status, the three AXI-monitor configuration groups (DAXMON = descriptor monitor, RDMON = read monitor, WRMON = write monitor), and per-monitor performance counters.

Table 3.10.2: Monitor Register Map (base 0x1000)

Offset	Name	Access	Description
0x1000	MON.MON_FIFO_STATUS	RO	MonBus capture/error FIFO status

Offset	Name	Access	Description
0x1004	MON.MON_FIFO_COUNT	RO	MonBus FIFO occupancy
0x10C0	MON.DAXMON_ENABLE	RW	Descriptor-monitor enables (incl. COMPRESS_EN)
0x10C4	MON.DAXMON_TIMEOUT	RW	Descriptor-monitor timeout cycles
0x10C8	MON.DAXMON_LATENCY_THRESHOLD	RW	Descriptor-monitor latency threshold
0x10CC	MON.DAXMON_PKT_MASK	RW	Descriptor-monitor packet mask
0x10D0	MON.DAXMON_ERR_CFG	RW	Descriptor-monitor error select/mask
0x10D4-0x10DC	MON.DAXMON_MASK1/2/3	RW	Descriptor-monitor category masks
0x10E0-0x10FC	MON.RDMON_*	RW	Source-egress AXIS monitor-lite config (same layout as DAXMON; SRC instance, rapids TASK-015)
0x1100-0x111C	MON.WRMON_*	RW	Sink-ingress AXIS monitor-lite config (same layout as DAXMON; SNK instance, rapids TASK-015)
0x1150-0x1178	MON.DAXMON_PERF_*	RO/RW	Descriptor-monitor performance counters
0x1180-0x11A8	MON.RDMON_PERF_*	RO/RW	Read-monitor performance counters
0x11B0-0x11D8	MON.WRMON_PERF_*	RO/RW	Write-monitor performance counters
0x11E0-0x11EC	MON.{RD,WR}MON_PERF_CH_*	RO	Per-channel producer/backpressure/starve/idle
0x11F0	MON.RDMON_PERF_CH_OVERFLOW	RO	Read-monitor per-channel overflow

Offset	Name	Access	Description
0x11F4	MON.WRMON_ PERF_CH_OV_ ERFLOW	RO	Write-monitor per-channel overflow

Each *_ENABLE register carries MON_EN, ERR_EN, COMPL_EN, TIMEOUT_EN (and COMPRESS_EN on the write monitor); each *_PERF_* group provides window, producer, backpressure, starve, idle, beat, byte (lo/hi), and burst counters.

Last Updated: 2026-07-02

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

25 RAPIDS Config Block Specification

Module: rapids_config_block.sv **Location:**

projects/components/dma-ip/rapids/rtl/macro_beats/ **Status:** Implemented

25.1 Overview

rapids_config_block is a combinational adapter that maps the register block's hwif_out structure onto the flat cfg_* signals consumed by rapids_core_beats and the AXI monitors. It contains no state; every output is a direct assignment (with a small number of width expansions and global-enable gates).

The monitor registers arrive under hwif_out.MON.*; base configuration and status arrive at the top level of hwif_out.

25.2 Mapping Groups

25.2.1 Base Configuration

Routes scheduler, channel, and descriptor-engine configuration to the core:

- `cfg_channel_enable = CHANNEL_ENABLE.CH_EN & {8{GLOBAL_CTRL.GLOBAL_EN}}` (per-channel enable gated by the global master enable).
- `cfg_sched_timeout_cycles = SCHED_TIMEOUT_CYCLES.TIMEOUT_CYCLES` (32-bit).
- `cfg_sched_timeout_limit = SCHED_TIMEOUT_LIMIT.LIMIT` (8-bit, direct).
- `cfg_sched_{timeout,err,compl,perf}_enable = SCHED_CONFIG.{TIMEOUT,ERR,COMPL,PERF}_EN`. `COMPL_EN` is the one the scheduler group acts on for its monitor packets: 0 drops the scheduler's and descriptor engine's CORE Completion packets at the group (emitters still acknowledged); `ERR_EN` and `PERF_EN` reach the group but gate nothing today (errors always reported, perf not implemented).
- `cfg_desceng_*` from `DESCENG_CONFIG` (enable / prefetch / FIFO threshold).
- Descriptor address ranges (`cfg_desceng_addr{0,1}_{base,limit}`) are zero-extended from the 32-bit register fields to the 64-bit `ADDR_WIDTH`.

25.2.2 Monitor Configuration (`hwif_out.MON.*`)

Three parallel groups map to the descriptor-read monitor and, since rapids TASK-015, the two AXIS data-path monitor-lites (one per half):

Table 3.11.1: Monitor Register-to-Config Mapping

Register group	<code>cfg_*</code> target
<code>MON.DAXMON_*</code>	<code>cfg_desc_mon_*</code> (descriptor monitor)
<code>MON.RDMON_*</code>	<code>cfg_rdeng_mon_*</code> (source-egress AXIS monitor-lite; wired in the SRC half only)
<code>MON.WRMON_*</code>	<code>cfg_wreng_mon_*</code> (sink-ingress AXIS monitor-lite; wired in the SNK half only)

The `*_ENABLE` bits map onto the AXIS monitor-lite's class enables: `MON_EN` (gated by `GLOBAL_EN`), `ERR_EN` (error, channel-change and TSTRB checks), `COMPL_EN` (one Completion/STREAM_END per packet), `PERF_EN` (stream start/pause/resume and Credit/BACKPRESSURE stall reports), `TIMEOUT_EN`. `*_TIMEOUT` is MICROSECONDS in its low 16 bits (0 = never), `*_LATENCY_THRESH` the stall CYCLES that raise a backpressure report, and `*_PKT_MASK` is a MASK: `bit[type] = 1` masks (drops) that packet type at the monitor-lite and again at the monbus group. It resets to all-masked (0xFFFF); the host clears the bits of the classes it wants to see (rapids BUG-008 corrected the description, which used to read "1 = enable"; the polarity and the reset are as intended).

Within each group the enable/timeout/latency/mask fields map 1:1, e.g. `cfg_desc_mon_enable = MON.DAXMON_ENABLE.MON_EN & GLOBAL_CTRL.GLOBAL_EN`, `cfg_desc_mon_timeout_cycles = MON.DAXMON_TIMEOUT.TIMEOUT_CYCLES`, and the category masks (timeout/compl/thresh/perf/addr/debug) pass through directly. Key enables are gated by `GLOBAL_CTRL.GLOBAL_EN`; masks pass through ungated.

25.2.3 Performance and Observation

`PERF_CONFIG`, `PERF_CH_SEL`, and `OBS_CTRL` map to the profiler and channel-observation-mux `cfg_*` signals.

Last Updated: 2026-07-02

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

26 RAPIDS Top

Module: `rapids_top.sv` **Location:** `projects/components/dma-ip/rapids/rtl/top/` **Status:** Implemented **Last Updated:** 2026-09-30

26.1 Overview

`rapids_top` is the synthesizable top level. It wraps [RAPIDS Core](#) with an APB4 register interface, two config blocks, the register file, per-channel bus meters and the MonBus AXI-Lite group. It has the same hierarchy as RAPIDS Beats. Only the beat-count unit names differ.

The byte-granular behavior is entirely below the top: the packet record nets are internal to the core halves, so the top port list has no new signals. The full port list is in [Top-Level Port List](#) and the top integration description is in [RAPIDS Beats Top](#).

26.1.1 Figure 3.12.1: Top Hierarchy

```

s_apb_*  --> apb4_slave --> peakrdl_to_cmdrsp --> rapids_regs
                                     |
                                     rapids_config_block u_cfg_src <--+-->
rapids_config_block u_cfg_snk
                                     |
                                     +-----> rapids_core (u_core)
                                     |
                                     +----->

```

```

<----> AXI / AXIS

    axi_bus_meter u_rd_bus_meter, u_wr_bus_meter | mon_*
                                           |
                                           v
    (u_monbus_axil_group) monbus_axil4_axil4_group
                                           -->
m_axil_mon_*, mon_irq, s_axil_err_*

```

26.2 APB Address Map

The APB address is 13 bits. Bit 12 selects the half.

Table 3.12.1: APB Address Map

Range	Content
0x0000 to 0x003F	Source staged descriptor addresses.
0x0040	Source kick-off, one single-pulse bit per channel.
0x0040 to 0x0FFF	Source configuration registers.
0x1000 to 0x103F	Sink staged descriptor addresses.
0x1040	Sink kick-off, one single-pulse bit per channel.
0x1040 to 0x1FFF	Sink configuration registers.

The monitor register windows sit at +0x800 of each half's 4 KB space. With USE_MON_REGS clear the windows answer with an APB error, so a host can tell a build without monitors from a zero setting.

26.3 Parameters

Table 3.12.2: Top Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Channels.
DATA_WIDTH	512	Data width in bits. The board

Parameter	Default	Description
		uses 256.
ADDR_WIDTH	64	Address width.
AXI_ID_WIDTH	8	AXI ID width.
SRAM_DEPTH	4096	Words per channel at the top.
APB_ADDR_WIDTH, APB_DATA_WIDTH	13, 32	APB widths.
MON_MAX_TRANSACTIONS	16	Monitor table depth.
SNK_AXIS_MON_AGENT_ID, SRC_AXIS_MON_AGENT_ID	9, 10	AXIS monitor agent ids.
ACLK_MHZ	100	Clock in MHz for the AXIS monitor tick.
USE_AXI_MONITORS	1	0 omits the monitors and the MonBus AXI-Lite group.
USE_ROW_COL_MAJOR_ADD RESSING	1	0 compiles out extended addressing.
GEN_MON	1	0 omits the completion and error emitters.
USE_MON_REGS	USE_AXI_MONITORS != 0	Wire the monitor register windows.
AR_MAX_OUTSTANDING, AW_MAX_OUTSTANDING	8	Outstanding bursts.
PIPELINE	1	Write engine pipelining.
AXIS_ID_WIDTH, AXIS_DEST_WIDTH, AXIS_USER_WIDTH	8, 4, 1	AXIS side widths.

Clock and reset are `aclk` and `aresetn`. `cam_clear` synchronously clears the MonBus group transaction tables.

26.4 Differences from RAPIDS Beats

Table 3.12.3: Top Delta

Item	RAPIDS Beats	RAPIDS
Module	<code>rapids_beats_top</code>	<code>rapids_top</code>

Item	RAPIDS Beats	RAPIDS
Core instance	rapids_core_beats	rapids_core
Register file, APB map, config blocks		Unchanged
Bus meter beat counters	r_rd, r_wr	r_rd_beats, r_wr_beats
Config nets	*_xfer	*_xfer_beats
Port list		Unchanged

The register-file field names RD_XFER_BEATS and WR_XFER_BEATS are the same in both designs. Their meaning is beats in both, since bursts are beat-based.

Last Updated: 2026-09-30

RTL Design Sherpa · Learning Hardware Design Through Practice GitHub · Documentation Index · MIT License

27 AXI4 Interface Specification

Module: axi_read_engine.sv (source AXI read master), axi_write_engine.sv (sink AXI write master) **Location:** projects/components/dma-ip/rapids/rtl/fub/ **Status:** Implemented
Last Updated: 2026-09-30

27.1 Overview

RAPIDS drives two AXI4 masters: a read master on the source half and a write master on the sink half. The channel signals, the outstanding-transaction limits and the response handling are those of RAPIDS Beats and are specified in [AXI4 Interface Specification](#) (shared with RAPIDS Beats). This chapter records only what the byte-granular design adds.

The AXI side stays beat-oriented. Every burst starts at a beat-aligned address and moves whole beats. A descriptor that starts or ends inside a beat is handled by the byte lanes of the data path,

not by narrow AXI transfers. Both engines are specified in [AXI Read Engine](#) and [AXI Write Engine](#).

27.2 Byte Address to Beat Mapping

The scheduler splits a byte address into a beat address and a byte offset. `OFF_W` is $\lceil \log_2(\text{DATA_WIDTH}/8) \rceil$, which is 5 for the 256-bit board build.

Table 4.1.1: Address Split

Quantity	Value
Offset	<code>addr[OFF_W-1:0]</code>
First beat address	<code>addr</code> with the low <code>OFF_W</code> bits cleared
Beats for a descriptor	$(\text{offset} + \text{length} + \text{lanes} - 1) \gg \text{OFF_W}$, zero for zero length

The engines receive the working byte address and the remaining beat count. `ARADDR` and `AWADDR` are driven with the low `OFF_W` bits at zero, so every burst is beat-aligned.

27.3 Burst Rules

Table 4.1.2: Burst Rules

Rule	Read master	Write master
Burst start	beat-aligned	beat-aligned
Burst size	full bus width	full bus width
4 KB boundary	no burst crosses one	no burst crosses one
Burst cap	smaller of the configured burst and the beats left to the 4 KB boundary	same

The number of beats left to the boundary is computed from the beat-aligned address, so a byte offset never lets a burst run past it.

27.4 Write Strobes

The write master carries a byte enable with every beat. The data path stores the strobe next to the data in the sink SRAM, and the write engine drives it onto WSTRB. Interior beats of a packet are fully strobed. The first beat and the last beat of a descriptor that start or end inside a beat carry a partial strobe, so only the bytes of the descriptor are written and neighbouring bytes in memory are untouched.

The read master has no byte enables. It always reads whole beats and the source data path discards the bytes outside the descriptor.

Table 4.1.3: Byte-Granular Write Signal

Signal	Width	Meaning
m_axi_wstrb	DATA_WIDTH/8	Byte lanes written by the beat, from the stored strobe.

27.5 Limitations

- TYPE=EXT descriptors remain beat-aligned in address and length. This is a permanent limitation.
 - A descriptor never issues a narrow AXI transfer. Sub-beat behavior is entirely in the strobes.
-

Last Updated: 2026-09-30

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

28 AXI-Stream Interface Specification

Module: src_data_path_axis.sv (source master), snk_data_path_axis.sv (sink slave)
Location: projects/components/dma-ip/rapids/rtl/macro/ **Status:** Implemented **Last Updated:** 2026-09-30

28.1 Overview

RAPIDS has one AXI-Stream master on the source half and one AXI-Stream slave on the sink half. Both carry packed bytes. Unlike RAPIDS Beats, the stream is no longer a train of full beats: each descriptor is one packet, the last beat of a packet may be partial, and TSTRB says which byte lanes are valid. The handshake, the width parameters and the monitor tap are those of RAPIDS Beats, see [AXI-Stream Interface Specification](#) (shared with RAPIDS Beats). The two data paths are specified in [Source Data Path AXIS](#) and [Sink Data Path AXIS](#).

28.2 Packet Contract

Table 4.2.1: Packet Contract

Property	Rule
Packet	One descriptor is one packet, ending with TLAST.
Packing	Bytes are packed from lane 0 upward. The descriptor start offset in memory does not appear on the stream.
TSTRB	Contiguous from lane 0. All ones on every beat but the last.
Partial beat	Only the TLAST beat may be partial, carrying the remaining bytes.
Packet length	Equals the descriptor length in bytes.
Zero length	A zero-length descriptor produces no packet.

28.3 Source Master Signals

Table 4.2.2: Source Master Signals

Signal	Value
m_axis_tdata	Re-packed bytes from the drain

Signal	Value
	shifter.
m_axis_tstrb	Low min(bytes_left, DATA_WIDTH/8) lanes set.
m_axis_tlast	Set when the remaining byte count equals the bytes in this beat.
m_axis_tid	Channel number.
m_axis_tdest	Channel number.
m_axis_tuser	Zero.

The source path pops memory beats and re-packs them, so a stream beat is not tied one to one to a memory beat. The first pop of an offset descriptor only primes the hold register and produces no output unless the packet fits in that single beat. When the last memory beat leaves bytes in the hold register, one extra flush beat carries them.

28.4 Sink Slave Signals

Table 4.2.3: Sink Slave Signals

Signal	Use
s_axis_tdata	Payload, shifted into the per-channel hold by the descriptor offset.
s_axis_tstrb	Valid byte lanes. Only the TLAST beat may be partial.
s_axis_tid	Channel select, low CIW bits.
s_axis_tdest, s_axis_tuser	Unused.
s_axis_tlast	Ends the packet.
s_axis_tready	Deasserted while the output register is busy, a flush beat is pending, or the selected channel has no packet record.

The sink stalls a packet until the scheduler has issued the record of its descriptor. A packet for a channel whose record queue is empty is not accepted, so no byte enters the SRAM before the destination offset is known. The queue holds four records per channel.

The spill of a shifted beat is held per channel. After TLAST, if the hold register still holds bytes, TREADY drops for one flush beat that writes them to the SRAM.

28.5 Length Check

At TLAST the sink compares the bytes received with the length in the head record of the channel. A mismatch sets the sticky per-channel `r_pkt_len_error` bit, which is ORed with the write engine error into `sched_wr_error`. Nothing is padded or dropped silently. The bytes received are counted from the lanes set in TSTRB on each beat.

Last Updated: 2026-09-30

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

29 MonBus Interface Specification

Last Updated: 2025-01-10

29.1 Overview

The MonBus (Monitor Bus) is a 64-bit packet-based interface for reporting internal events, status, and performance data. All RAPIDS subsystems generate MonBus packets that are aggregated and output through a unified interface.

29.2 MonBus Signal Interface

29.2.1 Signal Table

Table 4.3.1: MonBus Interface Signals

Signal	Direction	Width	Description
monbus_pkt_valid	output	1	Packet valid

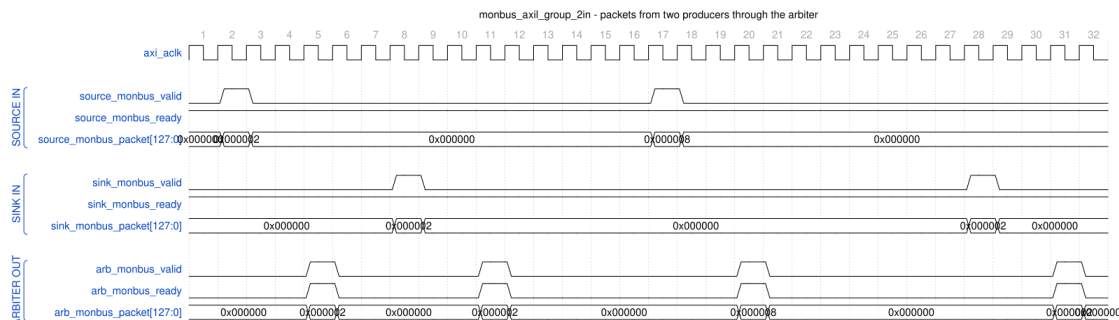
Signal	Direction	Width	Description
monbus_pkt_ready	input	1	Consumer ready
monbus_pkt_data	output	64	Packet data

29.2.2 Handshaking Protocol

Standard valid/ready handshaking: - Data transfers when valid & ready both high - Producer holds valid until ready seen - Consumer asserts ready when able to accept

29.2.3 Timing Diagram

29.2.4 Figure 4.3.1: MonBus Packet Transfer Timing



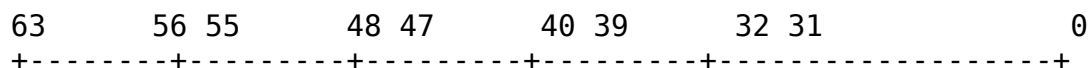
monbus_axil_group_2in - packets from two producers through the arbiter

Source: [monbus_group_packet_transfer.json](#), captured from `dv/tests/macro/test_monbus_axil_group.py` (basic_flow, error FIFO 64 x 32, TEST_LEVEL=full) with WAVES=1. The 128-bit packets are shown as their top 24 bits ([127:104]: packet type nibble, then the protocol field at [108:105], so 0x0000002 is AXIS and 0x0000008 is CORE).

Reading it: a MonBus link is a single-beat valid/ready handshake with the packet presented alongside valid. The group holds ready high on both inputs, so the source and sink packets are accepted the cycle they are offered, and each one reappears on the arbiter output three cycles later as another one-cycle valid/ready pair. Between packets the links idle low; nothing is held across transfers.

29.3 MonBus Packet Format

29.3.1 64-bit Packet Structure



Type	Unit ID	Agent ID	Channel	Payload (32 bits)	
+-----+	+-----+	+-----+	+-----+	+-----+	+-----+

29.3.2 Figure 4.3.2: MonBus Packet Format

Table 4.3.2: MonBus Packet Fields

Field	Bits	Description
Type	[63:56]	Packet type code
Unit ID	[55:48]	Reporting unit identifier
Agent ID	[47:40]	Source agent within unit
Channel	[39:32]	Channel number (0-7)
Payload	[31:0]	Type-specific payload

29.4 Packet Types

29.4.1 Type Code Definitions

Table 4.3.3: MonBus Packet Types

Type Code	Name	Description
0x01	DESC_FETCH	Descriptor fetch started
0x02	DESC_COMPLETE	Descriptor fetch completed
0x03	SCHED_START	Scheduler started transfer
0x04	SCHED_DONE	Scheduler transfer complete
0x10	AXI_RD_START	AXI read burst started
0x11	AXI_RD_DONE	AXI read burst completed
0x12	AXI_WR_START	AXI write burst

Type Code	Name	Description
		started
0x13	AXI_WR_DONE	AXI write burst completed
0x20	SRAM_FILL	SRAM fill event
0x21	SRAM_DRAIN	SRAM drain event
0x30	ERROR	Error event
0x31	TIMEOUT	Timeout event
0xFF	HEARTBEAT	Periodic heartbeat

29.4.2 Payload Formats by Type

29.4.2.1 DESC_FETCH / DESC_COMPLETE (0x01, 0x02)

Table 4.3.4: Descriptor Packet Payload

Bits	Description
[31:16]	Descriptor address [47:32]
[15:0]	Descriptor address [31:16]

29.4.2.2 SCHED_START / SCHED_DONE (0x03, 0x04)

Table 4.3.5: Scheduler Packet Payload

Bits	Description
[31:16]	Transfer ID
[15:0]	Beat count

29.4.2.3 AXI_RD/WR_START/DONE (0x10-0x13)

Table 4.3.6: AXI Event Packet Payload

Bits	Description
[31:24]	AXI ID
[23:16]	Burst length
[15:0]	Reserved

29.4.2.4 ERROR (0x30)

Table 4.3.7: Error Packet Payload

Bits	Description
[31:24]	Error code
[23:16]	Error source
[15:0]	Error details

29.5 MonBus Source Aggregation

29.5.1 RAPIDS Core MonBus Sources

Table 4.3.8: MonBus Source Assignment

Source ID	Agent	Description
0-7	Scheduler[0:7]	Per-channel scheduler events
8-15	DescEngine[0:7]	Per-channel descriptor events
16-23	SnkSRAM[0:7]	Sink SRAM controller events
24-31	SrcSRAM[0:7]	Source SRAM controller events
32	AXI Write Engine	Write transaction events
33	AXI Read Engine	Read transaction events
34	Error Handler	Error events

Scheduler and DescEngine COMPLETION packets are subject to the half's SCHED_CONFIG.COMPL_EN (reset 1): the scheduler group drops them ahead of its arbiter when the bit is clear and acknowledges the emitter, so the data path is unaffected. Their ERROR packets are always forwarded (rapids ISSUE-005).

29.5.2 Aggregation Hierarchy

29.5.3 Figure 4.3.3: MonBus Aggregation Tree

Level 0 (Sources) Level 1 (Groups) Level 2 (Top)
+-----+

```

| Scheduler[0] |-----+
| DescEngine[0] |-----+----> scheduler_group[0] ---+
+-----+
+-----+
| Scheduler[1] |-----+
| DescEngine[1] |-----+----> scheduler_group[1] ---+
+-----+
...
...
+----->
rapids_core
+-----+
monbus_out
| Scheduler[7] |-----+
| DescEngine[7] |-----+----> scheduler_group[7] ---+
+-----+
+-----+
| SnkSRAM[0:7] |-----> snk_sram_group -----+
+-----+
+-----+
| SrcSRAM[0:7] |-----> src_sram_group -----+
+-----+
+-----+
| AXI Engines |-----> axi_engine_group -----+
+-----+

```

29.6 Integration Guidelines

29.6.1 Consumer Requirements

MonBus consumers must: 1. Provide sufficient ready cycles to prevent packet loss 2. Handle bursty traffic (multiple sources may have packets simultaneously) 3. Process or buffer packets at wire rate

29.6.2 Recommended Consumer Patterns

```

// Pattern 1: Direct FIFO buffering
gaxi_fifo_sync #(
    .DATA_WIDTH(64),
    .DEPTH(256)
) u_gaxi_fifo_sync (
    .axi_aclk          (axi_aclk),
    .axi_aresetn       (axi_aresetn),
    .wr_valid          (wr_valid),
    .wr_ready          (wr_ready),
    .wr_data           (wr_data),
    .rd_ready          (rd_ready),
    .count             (count),

```

```

        .rd_valid      (rd_valid),
        .rd_data       (rd_data)
    );

```

29.7 Timing Requirements

Table 4.3.9: MonBus Timing Parameters

Parameter	Value	Description
Min Ready	1 cycle	Must assert ready within N cycles
Max Latency	Configurable	Before packet dropped
Packet Rate	Variable	Depends on activity

Last Updated: 2025-01-10

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

30 Packet Record Interface Specification

Module: scheduler.sv (producer), src_data_path_axis.sv and snk_data_path_axis.sv (consumers) **Location:** projects/components/dma-ip/rapids/rtl/fub/ and rtl/macro/ **Status:** Implemented **Last Updated:** 2026-09-30

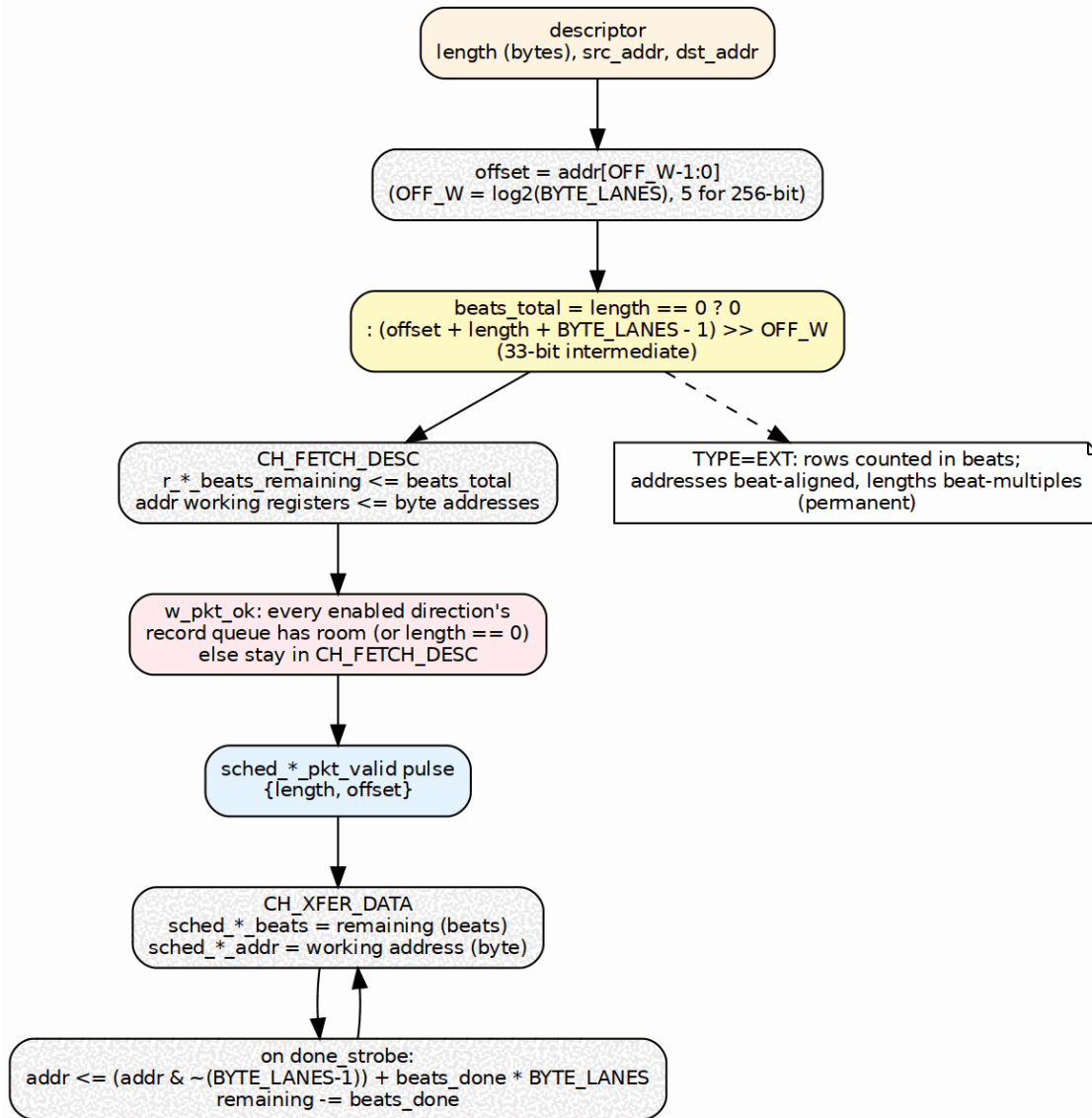
30.1 Overview

The packet record interface is new in the byte-granular design. It tells a data path how many bytes a descriptor covers and where in its first beat the data starts. RAPIDS Beats has no equivalent, because every transfer there is a whole number of aligned beats.

The scheduler is the producer. It emits one record per data descriptor. The source or sink data path is the consumer and holds the records in a small queue per channel. The record interface is internal to each half: it does not appear on the rapids_core or rapids_top ports.

30.1.1 Figure 4.4.1: Scheduler Byte Math

scheduler: bytes in, beats out (per enabled direction)



Scheduler byte math

Source: [assets/graphviz/03_scheduler_byte_math.dot](#)

30.2 Signals

There is one set per direction. The source half uses `sched_rd_pkt_*` and the sink half uses `sched_wr_pkt_*`. Each signal is a vector with one entry per channel.

Table 4.4.1: Packet Record Signals

Signal	Direction (scheduler view)	Width per channel	Description
<code>sched_*_pkt_valid</code>	output	1	One-cycle pulse: a record is presented.
<code>sched_*_pkt_ready</code>	input	1	The consumer queue has room.
<code>sched_*_pkt_bytes</code>	output	32	Descriptor length in bytes.
<code>sched_*_pkt_offset</code>	output	OFF_W	Address offset inside the first beat.

OFF_W is $\lceil \log_2(\text{DATA_WIDTH}/8) \rceil$, 5 for a 256-bit build.

30.3 Protocol

The scheduler raises `valid` for one cycle, in the cycle it leaves the descriptor fetch state. It does not leave that state until every enabled direction can take the record, so `ready` is a pre-condition of the pulse, not a wait state after it. Both directions of a channel see exactly one pulse per data descriptor.

A consumer accepts a record when `valid` and its queue is not full. It drives `ready` low when the queue is full, which is what holds the scheduler in its fetch state.

Table 4.4.2: Record Rules

Rule	Detail
Records per descriptor	One, for data descriptors of nonzero length.
Zero-length descriptors	No record.
Non-data descriptors	No record.
Order	Records are consumed in the order pushed, per channel.

Rule	Detail
Queue depth	Four records per channel in each data path.
Pop	The record pops on the last beat of its packet.

The offset is taken from the source address in the read direction and from the destination address in the write direction. The length is the same for both. When both directions are enabled, each data path receives its own offset.

30.4 Unused Direction

A scheduler group array instantiated for only one direction ties the other direction off inside the half. The unused `valid`, `bytes` and `offset` outputs are left open and the unused `ready` inputs are held at all ones, so the scheduler never waits on a queue that does not exist.

30.5 Consumer Behavior

- The source data path uses the record to compute the number of memory beats, to decide when a pop emits a stream beat, to set `TSTRB` and `TLAST`, and to retire the record.
 - The sink data path uses the record to decide how far to shift each incoming beat, to check the received length at `TLAST`, and it holds off `TREADY` for a channel until the record exists.
-

Last Updated: 2026-09-30

RTL Design Sherpa · Learning Hardware Design Through Practice [GitHub](#) · [Documentation Index](#) · [MIT License](#)

31 RAPIDS Beats Architecture MAS

Version: 0.7 **Date:** 2026-07-13 **Purpose:** Module Architecture Specification for RAPIDS Phase 1 “Beats” Architecture

31.1 Document Organization

Note: All chapters linked below for automated document generation.

31.1.1 Front Matter

- [Document Information](#)

31.1.2 Chapter 1: Overview

- [Architecture Overview](#)
- [Top-Level Port List](#)
- [Clocks and Reset](#)

31.1.3 Chapter 2: FUB (Functional Unit Blocks)

Control Path: - [Scheduler](#) - [Descriptor Engine](#)

AXI Engines: - [AXI Read Engine](#) - [AXI Write Engine](#)

Flow Control: - [Beats Alloc Control](#) - [Beats Drain Control](#) - [Beats Latency Bridge](#)

Control Engines: - [Control-Read Engine](#) - [Control-Write Engine](#)

31.1.4 Chapter 3: Macro (Integration Blocks)

Scheduler Integration: - [Beats Scheduler Group](#) - [Beats Scheduler Group Array](#)

Sink Data Path (Network to Memory): - [Sink Data Path](#) - [Sink Data Path AXIS](#) - [Sink SRAM Controller](#)

Source Data Path (Memory to Network): - [Source Data Path](#) - [Source Data Path AXIS](#) - [Source SRAM Controller](#)

Top-Level Integration: - [RAPIDS Core Beats](#) - [RAPIDS Registers](#) - [RAPIDS Config Block](#) - [RAPIDS Beats Top](#)

31.1.5 Chapter 4: Interfaces

- [Interface Overview](#)
- [AXI4 Interface Specification](#)
- [AXIS Interface Specification](#)
- [MonBus Interface Specification](#)

31.2 Quick Reference

31.2.1 FUB Modules (fub_beats/)

FUB Module Summary

Module	File	Purpose	Status
scheduler_beats	scheduler_beats.sv	Transfer coordinator, descriptor processing	Implemented
descriptor_engine_beats	descriptor_engine_beats.sv	Descriptor fetch/parse (256-bit)	Implemented
axi_read_engine_beats	axi_read_engine_beats.sv	AXI read master, streaming pipeline	Implemented
axi_write_engine_beats	axi_write_engine_beats.sv	AXI write master, streaming pipeline	Implemented
alloc_ctrl_beats	alloc_ctrl_beats.sv	Space allocation tracking (virtual FIFO); kept and tested, no longer in the SRAM path since bdf4e0dff	Implemented
drain_ctrl_beats	drain_ctrl_beats.sv	Data availability tracking (virtual FIFO); kept and tested, no longer in the SRAM path	Implemented
latency_bridge_beats	latency_bridge_beats.sv	Registered-FIFO to valid/ready skid bridge; kept and tested, no longer in the SRAM path	Implemented
ctrlrd_engine	ctrlrd_engine.sv	Control-read consumer gate (poll-until-match, retry budget)	Implemented

Module	File	Purpose	Status
ctrlwr_engine	ctrlwr_engine.sv	Control-write producer doorbell (single-beat write)	Implemented

31.2.2 Macro Modules (macro_beats/)

Macro Module Summary

Module	File	Purpose	Status
scheduler_group_beats	scheduler_group_beats.sv	Scheduler + Descriptor Engine wrapper	Implemented
scheduler_group_array_beats	scheduler_group_array_beats.sv	8-channel scheduler array with arbitration	Implemented
snk_data_path_beats	snk_data_path_beats.sv	Network-to-memory path integration	Implemented
snk_data_path_axis_beats	snk_data_path_axis_beats.sv	AXIS variant of sink path	Implemented
src_data_path_beats	src_data_path_beats.sv	Memory-to-network path integration	Implemented
src_data_path_axis_beats	src_data_path_axis_beats.sv	AXIS variant of source path	Implemented
snk_sram_controller_beats	snk_sram_controller_beats.sv	Sink SRAM: naming wrapper around STREAM's sram_controller (one instance, per-channel FIFOs inside it)	Implemented
src_sram_controller_beats	src_sram_controller_beats.sv	Source SRAM: naming wrapper around STREAM's sram_controller	Implemented
rapids_src_beats / rapids_snk_beats	rapids_src_beats.sv,	One half each: scheduler array +	Implemented

Module	File	Purpose	Status
	rapids_snk_beats.sv	data path + monbus merge	
rapids_core_beats	rapids_core_beats.sv	Both halves + the core monbus arbiter	Implemented
rapids_regs	regs/generated/rtl/rapids_regs.sv (generated from rtl/macro_beats/rapids_regs.rdl)	PeakRDL register block (base + MON regfile @ 0x1000)	Implemented
rapids_config_block	rapids_config_block.sv	Maps register hwif_out to core/monitor cfg_*	Implemented
rapids_beats_top	rapids_beats_top.sv	Top-level: APB slave, kickoff, monitors, MonBus AXI-Lite group	Implemented

31.3 Architecture Comparison: RAPIDS vs STREAM

The “beats” architecture is a Phase 1 implementation of RAPIDS that shares concepts with STREAM but targets network-to-memory (and vice versa) transfers rather than memory-to-memory.

RAPIDS Beats vs STREAM Comparison

Feature	STREAM	RAPIDS Beats
Primary Use	Memory-to-memory DMA	Network-to-memory accelerator
Data Paths	Single bidirectional	Separate sink/source paths
Network Interface	None	AXIS master/slave
SRAM Buffering	Shared buffer	Separate sink/source buffers
Descriptor Format	256-bit	256-bit (compatible)

Feature	STREAM	RAPIDS Beats
Channel Count	8	8
Flow Control	alloc_ctrl/ drain_ctrl	STREAM stream_alloc_ctrl/stream_drain_ctrl inside the shared sram_controller (alloc_ctrl_beats/drain_ctrl_beats kept, out of the path)

31.4 Clock and Reset Summary

31.4.1 Clock Domains

Clock Domains

Clock	Frequency	Usage
aclk	100-500 MHz	Primary - all RAPIDS logic, AXI/AXIS interfaces

31.4.2 Reset Signals

Reset Signals

Reset	Polarity	Type	Usage
aresetn	Active-low	Async assert, sync deassert	Primary - all RAPIDS logic

See: [Clocks and Reset](#) for complete timing specifications

31.5 Interface Summary

31.5.1 External Interfaces

External Interfaces

Interface	Type	Width	Purpose
AXI4 (Descriptor)	Master	256-bit	Descriptor fetch
AXI4 (Sink Write)	Master	512-bit	Sink data write to

Interface	Type	Width	Purpose
		(param)	memory
AXI4 (Source Read)	Master	512-bit	Source data read
		(param)	from memory
AXIS (Sink)	Slave	512-bit	Network data
		(param)	ingress
AXIS (Source)	Master	512-bit	Network data
		(param)	egress
MonBus	Master	64-bit	Monitor packet
			output

31.5.2 Internal Buses

Internal Buses

Interface	Width	Purpose
MonBus	64-bit	Internal monitoring bus
Descriptor Bus	256-bit	Descriptor distribution
SRAM Data Bus	512-bit	SRAM read/write data

31.6 Related Documentation

- [PRD.md](#) - Product requirements and overview
- [CLAUDE.md](#) - AI development guide
- [RAPIDS Beats HAS](#) - Architecture specification (the original pre-beats rapids_spec/ tree was retired)

31.7 Specification Conventions

31.7.1 Signal Naming

- **Clock:** `aclk`
- **Reset:** `aresetn` (active-low)
- **Valid/Ready:** Standard AXI/AXIS handshake

- **Registers:** r_prefix (e.g., r_state, r_counter)
- **Wires:** w_prefix (e.g., w_next_state, w_grant)

31.7.2 Parameter Naming

- **Uppercase:** NUM_CHANNELS, DATA_WIDTH, ADDR_WIDTH
- **Derived:** CHAN_WIDTH = $\lceil \log_2(\text{NUM_CHANNELS}) \rceil$

31.7.3 Figure and Table Numbering

- **Figures:** Figure X.Y.Z: Title where X=chapter, Y=section, Z=figure number
 - **Tables:** Caption line : Table Title after table markdown
-

Last Updated: 2026-07-13 **Maintained By:** RAPIDS Architecture Team